RESEARCH REPORT

Finite partitions of sets of common differences

The partition-regular version of
"2-large is large"

Produced by ChatGPT-6 pro, prompted by Sohail Farhangi

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Abstract

A set of positive integers is r-large if every r-coloring of the positive integers contains arbitrarily long monochromatic arithmetic progressions with common difference in that set. We study whether every set whose every finite partition has a 2-large cell must be large for all finite numbers of colors. We establish uniform finite-interval and cyclic reformulations, prove an affirmative result for automatic sets, and reduce the general question to full difference sets of subsets of a fixed sparse Sidon set. An exact hierarchy determines which bounded lists of partition-cell palettes can be forced, and shows why no fixed bound on the number of cells suffices. Further results concern simultaneous binary tests, quadratic and higher-degree obstructions, ternary winding squares, and multiplicative ultrafilters. Computer-assisted finite lemmas yield five-term and two-test six-term quadratic configurations, together with an exact 462-point extremal bound and its recurrence consequences. Complete proofs, explicit external inputs, and reproducible finite verification programs are included. The unrestricted finite-partition question is left open.

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Chapter 1

Introduction

1.1 The finite-partition question

Throughout, $\mathbb{N} = \{1, 2, \ldots\}$, and all arithmetic progressions in $\mathbb{N}$ have positive common difference. For a coloring $c : X \to C$, where $X \subseteq \mathbb{Z}$, define

$$R_h^X(c) = \{d \in \mathbb{N} : \text{there is } a \text{ with } a, a+d, \ldots, a+(h-1)d \in X \text{ and } c(a) = \cdots = c(a+(h-1)d)\}.$$

We omit the superscript when $X = \mathbb{N}$. A set $D \subseteq \mathbb{N}$ is *r-large* if $D \cap R_h(c) \neq \varnothing$ for every coloring $c : \mathbb{N} \to [r]$ and every $h \geq 2$. It is *large* if it is r-large for every finite r. Colorings using fewer than r colors are permitted. These notions were introduced in the study of restricted common differences in van der Waerden's theorem; see Brown–Graham–Landman [3].

Let $\mathcal{L}$ denote the family of large sets, and let $\mathcal{R}_r$ consist of those D such that every finite partition of D has an r-large cell. Our central question is

$$\boxed{D \in \mathcal{R}_2 \implies D \in \mathcal{L}\ ?} \tag{1.1.1}$$

The partition is a partition of the *allowed differences*, not of the integers being colored. The number of partition cells and the required progression length are both unbounded. The results below do not decide (1.1.1). They give equivalent finite problems, affirmative special cases, exact bounded-parameter separations, and restrictions on possible counterexamples.

Proposition 1.1.1 (Simultaneous tests)**.** *For $r \geq 2$, a set D belongs to $\mathcal{R}_r$ if and only if*

$$D \cap \bigcap_{i=1}^{m} R_h(c_i) \neq \varnothing \tag{1.1.2}$$

for every $m \geq 1$ and $h \geq 2$ and every list of r-colorings $c_1, \ldots, c_m$ of $\mathbb{N}$. The witnessing progressions have a common difference, but their starting points may differ. Moreover, $\mathcal{L} \subseteq \mathcal{R}_r$ for every r.

Proof. If (1.1.2) fails, assign each $d \in D$ to the first index i for which $d \notin R_h(c_i)$. No resulting cell is r-large. Conversely, from a finite partition without an r-large cell choose one obstructing coloring for each cell and increase their obstructing lengths to their maximum. These tests have no common good difference in D. Finally, for $D \in \mathcal{L}$, apply largeness to the product coloring $(c_1, \ldots, c_m)$. □

Lemma 1.1.2 (Finite witnesses)**.** *Fix r, m, h. If D satisfies (1.1.2) for every list of m r-colorings, there exist a finite $F \subseteq D$ and an N such that every such list on $[N]$ has a common good h-term difference in F.*

Proof. The space $([r]^{\mathbb{N}})^m$ is compact in the product topology. For each list, choose its common difference and its finitely many witnessing sites. The lists agreeing on these sites form an open neighborhood on which the same witnesses work. A finite subcover uses only finitely many differences and finitely many sites. Take their union as F and an interval containing all sites as $[N]$. Every coloring of $[N]$ extends to $\mathbb{N}$, so the assertion follows. □

1.2 The main reductions

Theorem 2.2.1 identifies (1.1.1) with a uniform finite simulation problem: for every $r, h \geq 2$, there must exist m, L such that for every N and every $f : [N] \to [r]$ there are binary colorings $c_1, \ldots, c_m$ of $[N]$ satisfying

$$\bigcap_{i=1}^{m} R_L^{[N]}(c_i) \subseteq R_h^{[N]}(f). \tag{1.2.1}$$

The parameters m, L may depend on r, h, but not on N or f. The converse direction of the equivalence is a mixed-radix construction: finite failures of (1.2.1) can be assembled into one infinite counterexample while keeping one obstructing palette and one obstructing progression length fixed. Theorem 2.3.2 gives an equivalent formulation on finite cyclic groups. Any unbounded collection of moduli suffices, including the powers of two or the primes.

A second reduction shows that full difference sets do not make the unrestricted problem easier. For $E \subseteq \mathbb{N}$, write

$$\Delta(E) = \{b - a : a, b \in E,\ a < b\}.$$

Theorem 3.1.1 constructs a single sparse additive Sidon set E_* and a monotone assignment $S \mapsto E_S \subseteq E_*$ such that S and $\Delta(E_S)$ agree on membership in every $\mathcal{R}_r$, on largeness, and on each fixed order of topological and measurable recurrence from order two onward. The construction represents the original set by selected pair differences and makes every cross-pair difference fail one fixed binary test. Consequently, (1.1.1) is equivalent to its restriction to full difference sets of subsets of this one E_*.

The reductions also clarify the required quantifiers for a negative answer. One needs fixed q, h such that, for every m, L, there is a finite set of differences passing every list of m binary tests at length L, while a q-coloring blocks all of those differences at length h. Letting the obstructing number of colors or the obstructing length grow with m, L is insufficient.

1.3 Affirmative cases and exact separations

Automatic sets. For a set D recognized by a finite automaton in a fixed base, Theorem 2.4.1 and Corollary 2.5.5 give

$$\begin{aligned} D \in \mathcal{L} &\iff D \in \mathcal{R}_2 \iff D \text{ is 3-large} \\ &\iff \text{every two-cell partition of } D \text{ has a 2-large cell} \\ &\iff D \cap q\mathbb{N} \text{ is nonarid for every } q \geq 1. \end{aligned}$$

Here aridity is the regular-language condition defined in Section 2.4; for automatic sets it is equivalent to polylogarithmic growth of the counting function. The affirmative direction proves measurable multiple recurrence. If D is automatic and nonlarge, one computable modulus q splits it into the two non-2-large sets $D \cap q\mathbb{N}$ and $D \setminus q\mathbb{N}$. The proof includes a translated digital-sums recurrence criterion and a binary obstruction for bounded signed digit weight along an arbitrary divisibility chain.

An exact hierarchy by the number of cells. Theorem 4.1.1 constructs, for every $k \geq 1$, a set D_k with the following property for every finite list $r_1, \ldots, r_m \geq 2$:

$$[\text{every partition } D_k = A_1 \sqcup \cdots \sqcup A_m \text{ has some } r_i\text{-large } A_i] \iff \sum_{i=1}^{m} (r_i - 1) < k.$$

In particular, for each fixed m, a nonlarge set can pass every partition into at most m cells. Thus no fixed bound on the number of cells can replace "every finite partition" in (1.1.1). Appendix A supplies the underlying localized hierarchy, its torus geometry, and the compact test-family construction.

Fixed progression lengths. For irrational α, put

$$S_{\alpha,\delta} = \{d \in \mathbb{N} : \|\alpha d^2\|_{\mathbb{T}} > \delta\}, \qquad S_\alpha = S_{\alpha,2/5}.$$

Every finite list of binary colorings has a common five-term difference in S_α, even after intersection with any ordinary Bohr neighborhood of zero; the resulting set has positive lower density. The same assertion permits two prescribed six-term binary tests together with any number of five-term tests. Nevertheless, five colors block all S_α-triples, and a finite list of binary tests blocks all S_α-progressions of length seven. These are Theorem 4.2.1, Corollary 4.2.5, and Lemma 3.5.3. For the corresponding five-color triple obstruction, the least common binary simulation length is therefore either six or seven. The case of arbitrary finite lists of six-term binary tests is not decided here.

Theorem 4.3.1 gives an exact palette threshold: for $r \geq 4$, $S_{\alpha,\delta}$ forces a monochromatic triple in every r-coloring precisely when $\delta < 2/r$. In particular, the least number of colors blocking all such triples is $\lceil 2/\delta \rceil$. Corollary 4.3.2 gives the analogous higher-power threshold for sufficiently large palettes. Theorem 2.9.1 supplies further same-length separations at every odd length.

1.4 Structural restrictions and finite extremal bounds

For a ternary coloring, a *winding square* is a parallelogram whose boundary has one stationary color edge and three edges traversing the three colors in a common orientation. Theorem 5.1.1 shows that a ternary h-term obstruction with no winding squares on its allowed directions is covered by two binary obstructions at the common length $\max(h, 3)$. The proof integrates integer lifts of the color increments and then uses two parity colorings. For $h = 3$, one binary obstruction at length five also suffices. A ternary witness against (1.1.1) must therefore have a winding graph of infinite chromatic number. Corollary 5.3.1 strengthens this to a finite-direction, syndetic-start condition for either prescribed orientation.

The polynomial results in Section 5.6 provide a dimension-independent binary simulation for colorings obtained from integer polynomials modulo a fixed integer. The integral lift of an exponent-three model is explicitly blocked at length eight. These arguments isolate why torsion constructions do not automatically give examples on the integers. The ultrafilter argument in Section 2.13 proves $K_r K_s \subseteq K_{rs}$, where K_r consists of ultrafilters all of whose members are r-large. It follows that a multiplicatively idempotent element of K_2 has only large sets as members. A separate example shows that density of products cannot complete this approach.

There is also a computer-assisted finite extremal result:

$$\max\{|X| : X \subseteq \{0, \dots, 461\},\ 42 \notin X + X - 2 \cdot X\} = 126.$$

Theorem 4.6.1 proves this using an exhaustive integer search with independently checked cross-residue masks. It implies the prime-cyclic density bound $3/11$, a strict improvement of that bound for primes greater than eleven, and a Bohr-neighborhood conclusion for triple recurrence of a measurable set of measure greater than $3/11$. The finite problem was proposed in Lev–Matolcsi–Pach–Varga [12]. All three computational inputs to the report are specified in Appendix B; the programs, nine reverse-unit-propagation certificates, and exact outputs are embedded in the PDF.

1.5 Conventions and proof dependencies

An ordinary Bohr neighborhood of zero in $\mathbb{N}$ contains a set of the form

$$\{n \in \mathbb{N} : \|n\theta_j\|_{\mathbb{T}} < \varepsilon \text{ for } 1 \le j \le s\}, \qquad \theta_1, \dots, \theta_s \in \mathbb{T}, \quad \varepsilon > 0.$$

For $A \subseteq \mathbb{N}$, its upper Banach density is

$$d^*(A) = \lim_{N\to\infty} \sup_{M\ge 0} \frac{|A \cap \{M+1, \dots, M+N\}|}{N}.$$

The limit exists by subadditivity. The assertion that a recurrence set contains a Bohr neighborhood up to a null set means that it contains $B \setminus Z$ for some such B and some Z with $d^*(Z) = 0$. It does not imply intersection with every particular zero-density set of allowed differences.

All measure-preserving systems in the proofs are invertible probability-preserving systems; ergodicity is specified when needed. Topological systems are compact metric systems with invertible transformations. A set is topologically k-recurrent when every finite coloring has a monochromatic $(k+1)$-term progression with an allowed difference; this is the standard coloring formulation of topological multiple recurrence [10]. Measurable k-recurrence requires, for every such probability-preserving system and every measurable A of positive measure, some allowed d with $\mu(\bigcap_{j=0}^{k} T^{-jd}A) > 0$.

The finite reductions, digit constructions, winding arguments, and finite verification procedures are proved directly. The analytic arguments use the nilfactor and nilsequence theorems of Bergelson–Host–Kra [4], the rational-IP estimates of Kra–Shalom [13], and polynomial orbit results of Leibman [14]. The use of Moreira–Richter is the corrected, fixed-eigenvalue statement in [15]; only its countable collection of rational-eigenvalue consequences is needed. The exact partition hierarchy uses Qiu's minimal-system lifting theorem [16]. The host-localized construction in Appendix A uses the return-time structure theorem of [8]; its precise linear form is stated there before use. The weighted polynomial argument uses the published Proposition 4.2 of Frantzikinakis–Lesigne–Wierdl [7]. These dependencies are applied with their hypotheses explicitly stated in the corresponding proofs.

Chapter 2

Finite problems, automatic sets, and the obstruction landscape

2.1 The quantifiers

For a coloring $c : X \to C$, where X is either $\mathbb{N}$ or a finite interval, write

$$R_h^X(c) = \{d \geq 1 : \exists a,\ a, a+d, \dots, a+(h-1)d \in X, c(a) = \cdots = c(a+(h-1)d)\}.$$

Omit X when $X = \mathbb{N}$. A set D is r-large if it meets every $R_h(c)$ with $|C| = r$ and $h \geq 2$; it is large if it is r-large for every finite r [3]. Let $\mathcal{L}$ denote the large sets, and let $\mathcal{R}_2$ consist of the D whose every finite partition has a 2-large cell. Then

$$D \in \mathcal{R}_2 \quad \iff \quad D \cap \bigcap_{i=1}^{m} R_L(c_i) \neq \varnothing \quad \text{for all finite lists of binary } c_i \text{ and all } L \geq 2. \tag{2.1.1}$$

Indeed, given a failing list, assign each $d \in D$ to the first test it fails. Conversely, given a partition without a 2-large cell, take one witness for each cell and the maximum of their lengths. The m progressions in (2.1.1) have one common difference; their starting points may differ. A product coloring proves $\mathcal{L} \subseteq \mathcal{R}_2$.

Guide. Theorem 2.4.1 gives the digit-growth criterion; Corollary 2.5.5 gives the stronger two-cell and three-color conclusions. Theorem 2.2.1 is the exact finite target for the unrestricted question. Theorem 2.6.1 treats ternary triples, and Corollary 2.11.1 explains why a zero-density exceptional set cannot be dismissed in a general proof.

2.2 A uniform finite equivalence

Theorem 2.2.1. *The equality $\mathcal{R}_2 = \mathcal{L}$ is equivalent to the following uniform assertion. For every $r, h \geq 2$ there are m, L such that for every N and every $f : [N] \to [r]$ there exist binary $c_1, \dots, c_m : [N] \to \{0,1\}$ with*

$$\bigcap_{i=1}^{m} R_L^{[N]}(c_i) \subseteq R_h^{[N]}(f). \tag{2.2.1}$$

Proof. First suppose (2.2.1) holds. Restrict a fixed coloring $f : \mathbb{N} \to [r]$ to every $[N]$, choose the corresponding tests, and take a subsequence on which all tests converge pointwise. If a difference is good for every limit test, choose their m witnessing progressions. These use a fixed finite interval, so the difference belongs to $R_h^{[N]}(f)$ for all sufficiently large N on the subsequence. In particular it belongs to $R_h(f)$. Apply (2.1.1).

For the other implication, suppose (2.2.1) fails uniformly for fixed r, h. For each $j \geq 2h+1$ choose N_j and $f_j : [N_j] \to \mathbb{Z}/r\mathbb{Z}$ witnessing failure at $m = L = j$. There is a finite nonempty set

$$F_j = \{d \leq (N_j - 1)/(j-1) : d \notin R_h^{[N_j]}(f_j)\}$$

such that any j binary colorings of $[N_j]$ have a common good j-term difference in F_j. Define the periodic extension $g_j(t) = f_j(1 + (t \bmod N_j))$ on $\mathbb{Z}$. No $2h$-term progression of difference $d \in F_j$ is monochromatic in g_j: its span is less than N_j, so it crosses at most one period boundary, and at least h of its terms lie in a single period.

Relabel these packets in increasing order, retaining their original parameters $j \to \infty$. Choose mixed radices $b_j > 4h \max F_j$, place values λ_j starting at 1, and $\lambda_{j+1} = b_j \lambda_j$. Write $n = \sum_j \lambda_j z_j(n)$ with $0 \le z_j(n) < b_j$. Put

$$D = \bigcup_j \lambda_j F_j, \qquad g(n) = \sum_j \left(g_j(z_j(n)) - g_j(0)\right) \pmod r.$$

The sum is finite. On $4h$ terms of step $\lambda_j d$, $d \in F_j$, the digits below j are fixed and the interval of quotients $\lfloor n/\lambda_j \rfloor$ crosses at most one multiple of b_j. Some $2h$ consecutive terms have every digit above j fixed, and their jth digits form a progression of step d. Thus g blocks D at length $4h$.

On the other hand, given m binary tests on $\mathbb{N}$ and a target length L, choose $j \ge \max(m, L)$, restrict $n \mapsto c_i(\lambda_j n)$ to $[N_j]$, and repeat a test to make up the list of j. The packet property, followed by multiplication by λ_j, gives a single difference in D working for all the tests. So $D \in \mathcal{R}_2 \setminus \mathcal{L}$. □

The significance of the mixed-radix step is that a qualitatively affirmative answer would have to furnish bounds independent of the coloring, the interval, and the set of steps. The gluing is an adaptation of the finite-packet method in Farhangi's thesis [6, Ch. 5].

2.3 Uniform cyclic formulations

For a finite cyclic group $C_q = \mathbb{Z}/q\mathbb{Z}$, let

$$R_h^{C_q}(f) = \{d \in C_q \setminus \{0\} : \exists x \in C_q,\ f(x) = f(x+d) = \cdots = f(x+(h-1)d)\}.$$

Progression sites are permitted to repeat when the order of d is small. These difference sets are invariant under $d \mapsto -d$. A binary guard below eliminates problematic wrapping in the transfer back to intervals.

Lemma 2.3.1 (Half-circle guard)**.** *Define $a_q(x) = \mathbf{1}_{[0,1/2)}(x/q \bmod 1)$. If $d \in R_K^{C_q}(a_q)$ with $K \ge 2$, its centered integer representative δ satisfies*

$$0 < |\delta| < \frac{q}{2(K-1)}.$$

Proof. Choose real representatives of the witnessing phases in their common half-open semicircle. Each successive difference is in $(-1/2, 1/2)$ and is congruent to d/q modulo one, so all successive differences are equal to δ/q. The total span is $(K-1)|\delta|/q < 1/2$. □

Theorem 2.3.2 (A uniform cyclic equivalence)**.** *Let $\mathcal{Q}$ be any unbounded set of integers at least two. The following assertions are equivalent.*

(i) $\mathcal{R}_2 = \mathcal{L}$.

(ii) *For every $r, h \ge 2$ there are m, L such that, for every $q \in \mathcal{Q}$ and every $f : C_q \to [r]$, there exist binary $c_1, \ldots, c_m$ on C_q with*

$$\bigcap_{i=1}^{m} R_L^{C_q}(c_i) \subseteq R_h^{C_q}(f). \tag{2.3.1}$$

In particular, one may take $\mathscr{Q} = \{2^j : j \geq 1\}$ or the primes. If the interval formulation for (r, h) has parameters (m, L), the cyclic formulation has parameters $(m + 1, 2L)$. Conversely, cyclic parameters (m, L_0) for $(r, 4h)$ give interval parameters $(m, \max\{L_0, 4h + 1\})$ for (r, h).

Proof. Use the interval equivalence in Theorem 2.2.1. First assume the interval assertion with parameters m, L for (r, h). Given f on C_q, identify its coordinates with $[0, q - 1]$ and choose the corresponding m interval tests. Regard these tests as cyclic colorings and add the guard a_q. Suppose a nonzero d passes all $m + 1$ cyclic tests at length $2L$. By Lemma 2.3.1, its centered representative δ has $(2L - 1)|\delta| < q/2$. Consequently each of the other witnessing cyclic progressions, lifted with increments δ, crosses the endpoint of the interval at most once. At least L consecutive terms lie in a single copy of $[0, q - 1]$. Reverse their order when $\delta < 0$. Thus $|\delta|$ passes all interval tests at length L, with their own starting points. It is a good h-term interval difference for f, hence d is a good cyclic difference, using sign invariance.

For the reverse implication, apply the cyclic assertion to $(r, 4h)$, obtaining m, L_0, and put $L = \max\{L_0, 4h + 1\}$. Fix $f : [N] \to [r]$. Choose $q \in \mathscr{Q}$ with $q \geq 2N$ and define a coloring of its standard representatives by

$$F(x) = f(1 + (x \bmod N)), \qquad 0 \leq x < q.$$

Take the m cyclic tests supplied for F, and restrict them to $[0, N - 1]$. If an integer d passes all restricted tests at length L, then

$$d \leq \frac{N - 1}{L - 1}, \qquad (4h - 1)d < N.$$

It passes their cyclic tests at length L_0, so F has a monochromatic cyclic $4h$-term progression of step d. Lift its sites to an increasing integer progression. Its span is less than N, and $q \geq 2N$. It crosses at most one q-boundary. On either side of that boundary it crosses at most one boundary between successive N-blocks in the definition of F. There are therefore at most four consecutive runs on which the sites are read from one unwrapped copy of an interval inside $[N]$. One run has at least h terms. That run gives an f-monochromatic h-term progression of step d in $[N]$. This proves the uniform interval assertion. □

The result is a reduction, not a resolution of the finite problem. In particular, checking finitely many cyclic orders cannot establish the uniform assertion. The parameters must work for every modulus in the chosen unbounded family.

2.4 The partition question for automatic sets of differences

Fix an integer base $b \geq 2$. A set $D \subset \mathbb{N}$ is b-automatic if a finite automaton recognizes its base-b expansions. Call it *arid* if its expansions form a finite union of languages of the form $u_0 v_1^* u_1 \cdots v_s^* u_s$, with fixed finite words u_i, v_i. Equivalently, for an automatic set, its counting function is $O((\log N)^s)$ for some s [5, Proposition 3.4].

Theorem 2.4.1. *For an automatic set $D \subset \mathbb{N}$, the following are equivalent:*

(i) D is large;
(ii) every finite partition of D has a 2-large cell;
(iii) $D \cap q\mathbb{N}$ is nonarid for every $q \geq 1$.

Under these conditions D is also a set of measurable multiple recurrence.

The main issue is (iii)⇒(i). A branching state in a finite automaton produces an arbitrary number of independent digit choices, but the resulting sets of sums come with translations that vary with their dimension. The following criterion handles such translations.

For $K \geq 2$ let $P_N(K)$ have the uniform distribution on the 2^N sums $\sum_{j=0}^{N-1} \epsilon_j K^j$, $\epsilon_j \in \{0,1\}$, including zero. We use $n \sim P_N(K)$ in averages and $\operatorname{supp} P_N(K)$ for their support. The parameters K and u in the next theorem may depend on q, but they must be fixed as N grows.

Theorem 2.4.2 (Translated digital sums)**.** *Suppose that for each $q \geq 1$ there exist integers $K \geq 2$ and $u \in q\mathbb{N}$ such that, for arbitrarily large N, some $t_N \in q\mathbb{Z}$ satisfies*

$$t_N + u \operatorname{supp} P_N(K) \ \subset \ D. \tag{2.4.1}$$

Then for every ergodic invertible probability preserving system (X, μ, T), every measurable A of positive measure, and every $h \geq 2$, there is $d \in D$ with $\mu(\bigcap_{j=0}^{h-1} T^{-jd} A) > 0$.

Removing the part orthogonal to the nilfactor

Write $T^a f = f \circ T^a$. The Host–Kra factor Z_s of an ergodic system is an inverse limit of ergodic s-step nilsystems; see [4, §4.4.4]. We need a version of Kra and Shalom's bound for finite-sums averages in which the input functions can be translated by amounts depending on N. Here and below the shifts are not assumed to be bounded.

Lemma 2.4.3. *Let $k \geq 2$, let $\ell_1, \dots, \ell_k$ be distinct nonzero integers, and fix $K \geq 2$, $u \geq 1$. If $f_1, \dots, f_k$ are bounded functions and some f_i is orthogonal to Z_{k-1}, then*

$$\lim_{N\to\infty} \sup_{s_1,\dots,s_k \in \mathbb{Z}} \left\| \mathbb{E}_{n\sim P_N(K)} \prod_{i=1}^{k} T^{\ell_i u n + s_i} f_i \right\|_2 = 0.$$

Proof. We verify the uniformity, since the unshifted assertion alone does not imply this lemma. Normalize the functions to have absolute value at most one. Write $V_j(f)$ for the rational IP seminorm $\|f\|_{\mathrm{U}^j_{\mathrm{IP}}}$ of [13, Definition 4.5]. We use two facts from that paper. First, for $j \geq 2$, $f \perp Z_{j-1}$ implies $V_j(f) = 0$ [13, Proposition 4.7 and Lemma 4.10]. Second, the first seminorm dominates the limiting L^2 norm of an orbit average along any IP generated by a sequence with rational spectrum. For higher orders the recursive definition includes the bound

$$\limsup_{M\to\infty} \mathbb{E}_{r,s\sim P_M(K)} V_{j-1}\big((T^{ar} f)\overline{T^{as} f}\big)^{2^{j-1}} \leq V_j(f)^{2^j} \qquad (a \in \mathbb{Z} \setminus \{0\}), \tag{2.4.2}$$

where r, s are independent. Indeed the generator sequence $|a|K^j$, $j \geq 0$, has rational spectrum in the sense of [13, Definition 3.7, Example 3.9 and Lemma 4.8]. Negative a gives the same average, by the symmetry of the difference $r - s$ and invariance of the seminorm under translates and conjugation. For clarity, the rational-spectrum fact here follows directly from the product criterion: for irrational θ the modulus of $\prod_{j=J}^{N-1} \big((1 + e^{2\pi \mathrm{i} |a| K^j \theta})/2\big)$ tends to zero. Otherwise $\| |a| K^j \theta \|_{\mathbb{T}} \to 0$; once small, its successive signed representatives must satisfy $x_{j+1} = K x_j$, forcing eventual zero and contradicting irrationality.

We prove, for all $k \geq 1$, that the limsup with arbitrary shifts is bounded by $\min_i V_k(f_i)$. This is the induction of [13, Proposition 4.9], with the following check at each stage. At $k = 1$ the single shift is an isometry and drops out of the norm. Assume the bound at $k - 1$, choose any coordinate whose seminorm is to bound the average, and, after relabeling, call it coordinate 1. Eliminate a different coordinate, say k. Fix $M < N$ and split n as $r + K^M v$, where $r \sim P_M(K)$ and

$v \sim P_{N-M}(K)$ are independent. By Jensen's inequality the squared norm of the original average is bounded by the mean over v of the squared norm of its inner average over r. Expand this square using independent $r, s \sim P_M(K)$ and write

$$H_i = (T^{\ell_i u r} f_i)\overline{T^{\ell_i u s} f_i}.$$

In the resulting integrals, move the common translate $T^{\ell_k u K^M v}$ off all factors using invariance of μ. The kth factor becomes $T^{s_k} H_k$. Cauchy–Schwarz and $|H_k| \le 1$ bound the result by the average over r, s of

$$\left\| \mathbb{E}_{v \sim P_{N-M}(K)} \prod_{i<k} T^{(\ell_i - \ell_k) u K^M v + s_i} H_i \right\|_2 .$$

For fixed M, r, s, the functions H_i are fixed, as is the new positive multiplier uK^M. The induction bound holds uniformly over every remaining s_i, even if it varies with N. Taking the limsup in N, then applying the power-mean inequality and (2.4.2) as $M \to \infty$, bounds the squared norm by $V_k(f_1)^2$ in the limit. This proves the asserted uniform estimate and hence the lemma. □

Averaging on a connected nilsystem

Lemma 2.4.4. *Let (Y, R) be a totally ergodic nilsystem with Haar probability measure m. For continuous F on Y and $K \ge 2$,*

$$\lim_{N \to \infty} \sup_{y \in Y} \left| \mathbb{E}_{n \sim P_N(K)} F(R^n y) - \int F \, dm \right| = 0.$$

Proof. On a torus rotation this follows by approximating F uniformly by characters: a nonconstant character yields the product in the proof of Lemma 2.4.3, with irrational frequency, times a phase depending on y. We induct on the nilpotence step. In the standard presentation $Y = G/\Gamma$ of an ergodic s-step nilsystem, $s \ge 2$, the last lower central subgroup gives a compact central torus $V = G_s/(G_s \cap \Gamma)$ acting on Y, and Y/V is a nilsystem of step at most $s-1$. Here one can take G generated by its identity component and the translating element, and G_s connected and rational, as in [4, §4.1]. Convolving in this torus action with Fejér kernels uniformly approximates F by finite sums of continuous functions transforming under individual characters of V. The trivial character is handled by passing to the factor Y/V, which inherits total ergodicity. Thus take F of a nontrivial vertical character; then $\int F \, dm = 0$.

Split the first M digits from the rest as above. For $r, s \in \operatorname{supp} P_M(K)$ the function

$$H_{r,s}(y) = F(R^r y)\overline{F(R^s y)}$$

is continuous and V-invariant, since R commutes with V and the two characters cancel. Jensen's inequality and expansion of the square give

$$\left| \mathbb{E}_{n \sim P_N(K)} F(R^n y) \right|^2 \le \mathbb{E}_{r,s \sim P_M(K)} \mathbb{E}_{v \sim P_{N-M}(K)} H_{r,s}(R^{K^M v} y).$$

For fixed M, r, s the inner average converges to $\int H_{r,s} \, dm$ uniformly in y, by induction on Y/V for the totally ergodic rotation induced by R^{K^M}. After taking first the limsup of the supremum in y and then letting $M \to \infty$, the right side is bounded by the limit of

$$\mathbb{E}_{r,s \sim P_M(K)} \int H_{r,s} \, dm = \left\| \mathbb{E}_{r \sim P_M(K)} R^r F \right\|_2^2 .$$

This tends to zero by the spectral theorem: the finite-sums product tends to zero at every irrational frequency, and the spectral measure of F has no rational atoms, by total ergodicity and $\int F\,dm = 0$. This proves uniform convergence on the original nilmanifold, without selecting a continuous section of the torus extension. □

Proof of the criterion

Proof of Theorem 2.4.2. It suffices to treat $h = k + 1 \geq 3$. Write $f = \mathbf{1}_A$, $\delta = \mu(A)$ and $I_f(d) = \int \prod_{j=0}^{k} T^{jd} f\,d\mu$. We first recall a uniform consequence of the finite Szemerédi theorem. There is $c = c(\delta, h) > 0$ such that, for every invertible probability preserving R and every $0 \leq g \leq 1$ of mean δ,

$$\liminf_{L\to\infty} \frac{1}{L} \sum_{a=1}^{L} \int \prod_{j=0}^{k} R^{ja} g\,dm \geq c. \tag{2.4.3}$$

Indeed, $B = \{g \geq \delta/2\}$ has measure at least $\delta/2$. Choose W so that every subset of $[W]$ of density at least $\delta/4$ contains an h-term progression. For any $s \geq 1$, a set of points of measure at least $\delta/4$ has at least $(\delta/4)W$ visits to B at times $s, 2s, \ldots, Ws$. Integrating the union bound over the progressions in $[W]$ gives $W \sum_{e=1}^{W} I_{\mathbf{1}_B,R}(es) \geq \delta/4$. Sum over $s \leq L/W$; each difference occurs at most W times. This gives (2.4.3), for example with $c = (\delta/2)^h \delta/(4W^3)$.

Let $f' = \mathbb{E}(f \mid Z_{k-1})$. Choose a finite-dimensional ergodic nilfactor Y of Z_{k-1} so that $g = \mathbb{E}(f \mid Y)$ satisfies $h\|f' - g\|_1 < c/4$. This choice is made before choosing a modulus or any packets. Let q be the number of connected components of Y. The transformation $R = T^q$ restricts to a totally ergodic nilrotation on each component. Choose u, K, t_N for this q in (2.4.1). For the shifted packet averages of I_f, Lemma 2.4.3 allows us to replace each moving factor by f', by telescoping and pairing the resulting L^2 error with the bounded function at time zero. After replacing the moving factors, their product is Z_{k-1}-measurable, so the time-zero function may be projected exactly. Replacing f' by g in all slots now costs at most $h\|f' - g\|_1$ for every individual difference, by invariance and telescoping. Consequently it suffices to identify the limit on Y; all error estimates also hold along the subsequence of available N.

We claim that the packet average on Y has the same limit as the ordinary average at differences qa, $1 \leq a \leq L$, $L \to \infty$. First suppose g continuous, fix one component Y_0, and let $S = R \times R^2 \times \cdots \times R^k$ act on Y_0^k. For almost every $y \in Y_0$ the nilsystem

$$Y_y = \overline{\{S^n(y, \ldots, y) : n \in \mathbb{Z}\}}$$

is totally ergodic with its Haar measure. Indeed, the *revised* Theorem 7.1 of Moreira–Richter [15] excludes each fixed nonzero rational eigenvalue for almost every y, since it is absent from (Y_0, R); now intersect these countably many full-measure sets. Write $u' = u/q$ and $v_N = t_N/q$. The continuous observable $G = g^{\otimes k}$ in the moving slots is evaluated at

$$(S^{u'})^n S^{v_N}(y, \ldots, y), \qquad n \sim P_N(K).$$

The initial point may change with N, but lies in the fixed space Y_y. By Lemma 2.4.4 for $S^{u'}$ its mean converges to the Haar integral of G on Y_y. The ordinary average of $G(S^a(y, \ldots, y))$ has this same limit by unique ergodicity of the ergodic nilsystem Y_y. Multiplying by $g(y)$, integrating in y using bounded convergence and averaging over the finitely many components proves the claim. For measurable $0 \leq g \leq 1$, approximate it in L^1 by continuous $[0,1]$-valued functions. Telescoping after integration bounds the error in either average by h times this L^1 error, uniformly over all differences and both averaging lengths, so the claim follows.

By (2.4.3) for $R = T^q$ on all of Y, the common limit is at least c. Hence the liminf of the packet averages on X is at least $3c/4$ and some difference in D gives $I_f(d) > 0$. This proves the criterion. □

Bounded variable radices. The translated-sums criterion also holds when the fixed generators K^j are replaced, for each modulus, by a fixed sequence $a_0 = 1$, $a_{j+1}/a_j \in \{2, \dots, B\}$ for some finite B. Indeed $\|a_j\theta\| \to 0$ again forces a rational θ: the successive centered representatives eventually multiply by the indicated integer without any wrap, and hence cannot remain small unless zero. This applies to every normalized tail a_{M+j}/a_M and every nonzero integer multiple. In both inductions above, replace K^M by a_M and use the corresponding tail distribution. The induction statements quantify over every such generator sequence, so the inductions on the functions and on the nilpotence step apply to each fixed tail. For the final spectral bound the prefixes are those of the original fixed sequence, whose irrational-frequency product tends to zero as before.

Reading the criterion from the automaton

Proof of Theorem 2.4.1. Assume (iii), fix q, and take the automaton recognizing $D \cap q\mathbb{N}$. By [5, Proposition 3.4], nonaridity gives an accessible and coaccessible state with two noncommuting return words v_1, v_2. Thus v_1v_2 and v_2v_1 are distinct return words of the same length a. Call them w_0, w_1, ordered by digit value $[w_1]_b > [w_0]_b$. Choose a word p reaching the state and a word z taking it to acceptance. One may use the regular language of canonical expansions, so that every accepted concatenation below represents a positive integer. With $K = b^a$ and $u = b^{|z|}([w_1]_b - [w_0]_b) > 0$, the words

$$p\, w_{\epsilon_{N-1}} \cdots w_{\epsilon_0}\, z, \qquad (\epsilon_0, \dots, \epsilon_{N-1}) \in \{0,1\}^N,$$

represent exactly $t_N + u \operatorname{supp} P_N(K)$, where $t_N = [pw_0^N z]_b$. All these integers are multiples of q, so $q \mid t_N$ and, by subtracting the two words differing only in their lowest free block, $q \mid u$. Theorem 2.4.2 proves the assertion on ergodic systems. For a general invariant measure, apply it to the ergodic components on which the test set has positive measure; countability of D gives a single difference with positive intersection after integration. For a coloring, apply it to a positive-measure color cylinder in an ergodic measure on the forward orbit limit set; any positive-measure cylinder intersection occurs in the original coloring. Thus (i) holds.

Implication (i)⇒(ii) follows by product coloring as in (2.1.1). If (iii) fails, $D \cap q\mathbb{N}$ is arid and has the finite binary cover of Lemma 2.5.1. All the other differences have nonzero residue modulo q and are blocked by a single binary coloring of sufficiently long progressions: color n by the half-circle containing $n/q \pmod 1$. For a step of order $d > 1$ modulo q the orbit consists of d equally spaced phases, which cannot all lie in either half-open semicircle. A full period has length at most q. For $q = 1$ this extra piece is empty. Hence (ii) fails, proving the converse. □

Corollary 2.4.5. *For each fixed automatic vertex coloring f of $\mathbb{N}$ and each fixed h, finitely many binary tests, of some common length, simulate $R_h(f)$.*

Proof. The set $R_h(f)$ is automatic: addition and multiplication by each fixed integer are recognizable relations in base b, and regular relations are closed under Boolean operations and projection. Express the condition by quantifying the initial point of the progression and taking a finite union over its colors. Its complement is therefore an automatic nonlarge set of differences, to which Theorem 2.4.1 applies. □

The bounds here may depend on the automaton. A bound for the finite-interval formulation independent of the number of states would give the unrestricted assertion of Theorem 2.2.1, since every finite word extends to a periodic automatic sequence.

A sparse example with no Schur triple. Let D consist of the integers whose base-four digits are 0 or 1 and whose number of ones is odd. If $x, y, x+y$ all have only these digits, the supports of x and y must be disjoint: an overlap gives a digit 2 and no carry is possible. Thus D is sum-free. Nevertheless D is a set of measurable multiple recurrence. Indeed, write $q = 2^a m$ with m odd, choose J so that $2^a \mid 4^J$, and choose $t \geq 1$ with $4^t \equiv 1 \pmod{m}$. In the positions $J, J+t, J+2t, \ldots$, put m initial ones, then split the remaining positions into consecutive groups of $2m$. Turn each of the next N groups entirely on or entirely off. All 2^N resulting numbers lie in $q\mathbb{N}$ and have an odd number of ones. Their digit lengths are $O_q(N)$, so $D \cap q\mathbb{N}$ is nonarid. Theorem 2.4.1 applies, although D does not contain even the three finite sums $x, y, x+y$.

One computable modulus suffices

The quantifier over q in Theorem 2.4.1 can be tested effectively from the original automaton. Take a deterministic automaton recognizing canonical base-b expansions, read most significant digit first, and write n for its number of states. Call a state *branching* here if it is accessible and coaccessible and admits two distinct return words of equal length. For each such state s, choose one pair $w_{s,0}, w_{s,1}$, ordered by digit value, and put

$$\delta_s = [w_{s,1}]_b - [w_{s,0}]_b > 0.$$

Let M be the least common multiple of the largest divisors of the δ_s coprime to b, with $M = 1$ for an empty list, and set $Q = Mb^{2n+2}$. The list and the word pairs are computable: on the productive part of the transition graph, use the product of two copies, keeping a flag for a discrepancy between their simultaneous digit labels. Reachability from $(s, s, 0)$ to $(s, s, 1)$ detects and produces a pair.

Corollary 2.4.6. *The set D is large if and only if $D \cap Q\mathbb{N}$ is nonarid. Thus largeness and the finite-partition property of an automatic set are decidable.*

Proof. Only sufficiency needs proof. Nonaridity in the product automaton for $D \cap Q\mathbb{N}$ supplies an accepted word W whose path visits a branching state s of the original automaton: project the branching state and its two return words. The positive value of W is divisible by Q, so W ends in at least $2n+2$ zeros.

Fix an arbitrary q and write $q = q_b q_0$, where all prime factors of q_b divide b and $\gcd(q_0, b) = 1$. Choose v so that $q_b \mid b^v$. We may first lengthen the terminal zero run of W, preserving its selected visit to s and acceptance, so that at least v terminal zeros occur after that visit. Indeed, if at least n terminal zeros lie after the visit, a state repeats while reading these zeros, producing a zero loop there. Otherwise the visit lies on the terminal run after at least n zeros have already been read, so s itself lies on a cycle of the deterministic zero transition. In either case repeating the loop multiplies the value by a power of b and in particular preserves divisibility by M. Write the resulting word as pz, with the split at the selected visit to s.

Inserting either chosen return word at that split acts on the prefix residue $x \pmod{q_0}$ by

$$H_i(x) = b^\ell x + [w_{s,i}]_b, \qquad \ell = |w_{s,0}| = |w_{s,1}|.$$

These maps are permutations. Their finite generated semigroup contains their inverses and hence the translation $H_1 H_0^{-1} : x \mapsto x + \delta_s$. After the unchanged suffix z, its effect on the final residue is

addition of $b^{|z|}\delta_s$. The original residue $[pz]_b$ is a multiple of M and hence of $\gcd(q_0, \delta_s)$. As $b^{|z|}$ is a unit modulo q_0, some finite concatenation of the two return words therefore changes the final residue to zero modulo q_0. The terminal zeros guarantee divisibility by q_b throughout.

Finally choose a common positive exponent e such that both H_0^e and H_1^e are the identity modulo q_0. At s insert arbitrarily many independent choices of $w_{s,0}^e$ or $w_{s,1}^e$ in addition to the correcting concatenation. The resulting words are all accepted multiples of q. There are 2^L distinct words after L choices, of length growing linearly with L. Thus $D \cap q\mathbb{N}$ is nonarid for every q. Theorem 2.4.1 completes the proof. For the algorithm, build the product with the residue automaton modulo the explicit Q and apply the finite branching-state test once. □

Corollary 2.4.7 (Effective finite-interval simulation)**.** *Fix a base b, a state bound s, a number of colors r, and a target progression length $h \geq 2$. There are computable m, L such that for every b-automatic coloring $f : \mathbb{N} \to [r]$ produced by a deterministic output automaton with at most s states, one can compute m binary automatic colorings satisfying* (2.2.1) *on* every $[N]$*. The colorings may depend on f, but need not depend on N.*

Proof. Given f, build the synchronous base-b automaton for the relation

$$W(d, a) \quad \Longleftrightarrow \quad a, d \geq 1, \qquad f(a) = f(a + d) = \cdots = f(a + (h - 1)d).$$

This uses the usual digit automaton for addition (carry bounded for each fixed multiplier), product automata for conjunction, projection for existentially quantified variables, and determinization. Equivalently one may first read the digits least significant first and then reverse the regular language. Use most significant digit first in W, padding the shorter coordinate on the left with zeros. Let S be its number of states. If a given d admits any witness, its least witness satisfies $a < b^{S+1}d$. Indeed, if a has more than S excess digits compared with d, inspect the states reached after the first, second, ..., last excess column. Two states coincide; deleting the intervening columns removes only leading d-zeros and preserves the first nonzero a-digit. Thus it gives a shorter accepted witness for the same d, a contradiction.

Project W and complement to recognize $D = \mathbb{N} \setminus R_h(f)$. Since D is not large, Corollary 2.4.6 effectively supplies Q with $D \cap Q\mathbb{N}$ arid. The finite binary cover in Lemma 2.5.1 is effective. To see this directly, in the productive graph of an arid deterministic automaton each strongly connected component containing a cycle is a simple directed cycle, counting digit labels as distinct edges. Otherwise a vertex has two different internal outgoing edges; following each by a path back gives two return words with different first digits and hence a branching state. Enumerate the paths in the finite acyclic graph of components, fixing all entry and exit edges. Each internal cyclic stretch has a fixed partial traversal and an arbitrary number of full turns. This explicitly produces the bounded-language description used in Lemma 2.5.1. Its denominator clearing, choice of radix and search over primes are effective; its digit-square tests are automatic. Take those tests together with the periodic test for $D \setminus Q\mathbb{N}$.

Increase their common length L so that $L \geq b^{S+1} + h$. If d passes every test on $[N]$, it passes them on $\mathbb{N}$, so $d \notin D$. Also $N \geq 1 + (L - 1)d$, and the witness bound gives $a + (h - 1)d \leq N$. Thus $d \in R_h^{[N]}(f)$, which is the required finite-interval assertion. Finally enumerate the finitely many base-b output automata with at most s states and outputs in $[r]$. Take maxima of their computed m, L, repeating a test or introducing a constant test when padding a list. □

What automatic vertex tests can detect

The allowed set D in the next statement need not be automatic.

Corollary 2.4.8. *Fix $b \geq 2$. Every finite-valued b-automatic vertex coloring has a monochromatic h-term progression of difference in D for every $h \geq 2$ if and only if, for each $q \geq 1$, the set $D \cap q\mathbb{N}$ is not contained in any b-arid automatic set.*

Proof. If a b-automatic f fails at length h, then $D \subseteq \mathbb{N} \setminus R_h(f)$. The set on the right is b-automatic and nonlarge, so its intersection with $q\mathbb{N}$ is arid for some q by Theorem 2.4.1. Conversely, suppose $D \cap q\mathbb{N}$ is contained in a b-arid set A. Use the finite family of b-automatic binary tests of Lemma 2.5.1 for A and one periodic test for $\mathbb{N} \setminus q\mathbb{N}$ (omit the latter when $q = 1$). Take their product, still a b-automatic coloring, and the maximum of their obstructing lengths. No monochromatic progression of this length can have a difference in D. □

This gives an explicit limit on approximation by automata, even if their sizes and bases are unrestricted. Take

$$D = \{d \in \mathbb{N} : \|d\sqrt{2}\|_{\mathbb{T}} \geq 1/3\}.$$

For every q, irrational rotation gives $D \cap q\mathbb{N}$ natural density $1/(3q)$. An arid set in any fixed base has polylogarithmic counting function, so by Corollary 2.4.8, every automatic vertex coloring in every base admits arbitrarily long monochromatic progressions with differences in D. Nevertheless, the binary coloring $c(n) = \mathbf{1}_{[0,1/2)}(n\sqrt{2} \bmod 1)$ blocks every triple of such a difference. Indeed, if three consecutive phases belong to one half-open semicircle, their unique representatives in that semicircle have consecutive differences in $(-1/2, 1/2)$. These two differences are congruent modulo one, hence equal. The distance between the first and last representatives would be $2\|d\sqrt{2}\|_{\mathbb{T}} \geq 2/3$, which is impossible in that semicircle.

2.5 A finite binary cover for bounded signed digit weight

Lemma 2.5.1. *Fix a base $b \geq 2$ and integers $p, C \geq 1$. The positive integers of the form $\sum_{\ell=1}^{p} c_\ell b^{n_\ell}$, with $|c_\ell| \leq C$ and $n_\ell \geq 0$, are covered by finitely many non-2-large sets. Consequently every arid automatic set admits such a cover.*

Proof. Choose a large integer K, put $B = b^K$, and allow all alignments $s = 0, \ldots, K-1$. Given one of the prescribed positive integers d, the at most p occupied exponent residues on the circle $\mathbb{Z}/K\mathbb{Z}$ leave a gap of at least $\lfloor K/p \rfloor - 1$ empty residues. Rotate the residues so the gap ends at $K-1$. In this alignment write

$$b^s d = \sum_m v_m B^m, \qquad S = \{m : v_m \neq 0\}, \qquad |S| \leq p, \qquad |v_m| \leq V = pCb^{K - \lfloor K/p \rfloor + 1}.$$

Here the v_m are signed integers, not ordinary base-B digits. The positive integer $\sigma = \sum_m v_m^2$ is at most pV^2. For K sufficiently large, the product of the odd primes up to K^2 exceeds pV^2 (for example, by Chebyshev's estimate for the primorial). Thus some odd prime $\ell \leq K^2$ does not divide σ. We also choose K so large that, with $H = (2p+1)K^4$,

$$HV/(B-1) < 1.$$

This is possible since p, C, b are fixed and $V/(B-1)$ decreases exponentially in K.

Write $Q_B(x)$ for the sum of the squares of the ordinary base-B digits of $x \geq 0$. For each of the finitely many s, ℓ use the bit indicating whether $Q_B(b^s n) \bmod \ell^2$ lies in $\{0, \ldots, \ell - 1\}$. We show that the pair chosen above blocks all H-term progressions of difference d.

For an arbitrary start let $x = b^s n$ have digits x_k. At time t the carry from the part below position m is

$$u_m(t) = \left\lfloor \sum_{k<m} (x_k + tv_k) B^{k-m} \right\rfloor .$$

On $0 \le t < H$ its real argument moves monotonically by less than $HV/(B-1) < 1$. Hence u_m changes at most once. Cut the time interval whenever one of these carries changes, using only $m \in S \cup (S+1)$. There are at most $2p$ cuts, so some run has at least $K^4 \ge \ell^2$ sites. On that run every active digit equals $x_m + tv_m + u_m(t) - Bu_{m+1}(t)$, hence is affine with velocity v_m. Across an inactive gap the integer encoded by its entire block of digits is fixed, since there are no velocities inside and both boundary carries are fixed. The digits above the last active place are fixed for the same reason, and those below the first were unchanged throughout. Thus on the run $Q_B(x + tb^s d)$ is a quadratic with leading coefficient σ.

Write this quadratic as $\sigma t^2 + \beta t + \gamma$. Choose a residue t_0 mod ℓ with $2\sigma t_0 + \beta \not\equiv 0 \bmod \ell$. On ℓ^2 consecutive sites the terms congruent to t_0 cover, in value, an entire class modulo ℓ inside $\mathbb{Z}/\ell^2\mathbb{Z}$: the increment on adding ℓ is $\ell(2\sigma t_0 + \beta) \bmod \ell^2$. Our bit is nonconstant on each such class. This proves the covering assertion.

For the last assertion use the bounded-language description of arid sets [5]: there are finitely many families of words $u_0 v_1^{e_1} u_1 \cdots v_t^{e_t} u_t$, with fixed words and arbitrary nonnegative exponents. Reading a repeated word uses the geometric sum $(b^{e_j|v_j|} - 1)/(b^{|v_j|} - 1)$. After multiplying the represented integers by a common fixed denominator, each becomes a sum of boundedly many bounded signed coefficients times powers of b. Use the first assertion and pull a test back by $n \mapsto Mn$, with M that common denominator. □

2.5.1 One binary obstruction along an arbitrary divisibility chain

The finite-state conclusion for a fixed base has a counterpart when the ratios between consecutive place values can be arbitrarily large. Here the coloring need not be automatic.

Theorem 2.5.2. *Fix $p, C \ge 1$. There exists $H = H(p, C)$ with the following property for every sequence*

$$1 = u_0 \mid u_1 \mid u_2 \mid \cdots , \qquad u_{j+1}/u_j \ge 2.$$

There is a binary coloring of $\mathbb{N}$ with no monochromatic H-term progression of any positive difference $\sum_{r=1}^{p} c_r u_{e_r}$, where $e_r \ge 0$ and $|c_r| \le C$.

Proof. The constants in the proof depend only on p, C. Choose a large integer K, put $L = K^3$ and $H = (2p+1)L$, and group consecutive place values in K different alignments. For alignment $s \in \{0, \dots, K-1\}$ the block starts are 0 and all positive indices congruent to s mod K. If they are denoted $i_0 = 0 < i_1 < \cdots$, the new place values and radices are

$$U_m = u_{i_m}, \qquad B_m = U_{m+1}/U_m.$$

Only the first block can be shorter than K; in particular $B_m \ge 2^K$ for $m \ge 1$, and always $B_m \ge 2$. Every nonnegative integer has a unique finite digit expansion $x = \sum_m x_m U_m$ with $0 \le x_m < B_m$.

A common alphabet for observing all the digit blocks. For a fixed alignment and block m, let $\mathcal{V}_m$ consist of the nonzero sums of at most p terms $c\,u_e/U_m$, with $i_m \le e < i_{m+1}$ and $|c| \le C$.

This set has at most $B_* = (1 + 2CK)^p$ members irrespective of the radices. Choose $\theta_m \in \mathbb{T}$ so that

$$\|jv\theta_m\|_{\mathbb{T}} > \frac{1}{4B_*L} \quad (v \in \mathcal{V}_m,\ 1 \le j \le L).$$

Such a choice exists by Haar measure: jv is a nonzero integer and the union of the excluded sets has measure at most $1/2$. Make these choices separately for every block and alignment. Partition $\mathbb{T}$ into $Q = \lceil 8B_*L \rceil$ half-open equal cells, numbered $0, \dots, Q-1$ with 0 in cell zero; let $c(y)$ be the cell number. Take a prime P strictly between $\max(Q, p+1)$ and $2\max(Q, p+1)$, and in $V = (\mathbb{Z}/P\mathbb{Z})^p$ put

$$w(c) = (c, c^2, \dots, c^p).$$

The sum of these weights identifies any multiset of cell numbers *of specified size* $k \le p$. Indeed the first k power sums determine the elementary symmetric functions by Newton's identities, since $P > k$, hence determine the multiset as the roots of its monic polynomial. In alignment s set

$$F_s(x) = \sum_{m \ge 0} w(c(\theta_m x_m)) \in V, \qquad Z(x) = (F_s(x))_{0 \le s < K}.$$

The sum is finite because $w(c(0)) = 0$.

One alignment gives many states. For a given difference $d = \sum_r c_r u_{e_r} > 0$, write $d = \sum_m v_m U_m$ in each alignment by placing each summand in its original block; these v_m are signed, not ordinary digits. There are at most p nonzero v_m and they belong to their respective menus. The occupied residues e_r mod K leave at least $g = \lfloor K/p \rfloor - 1$ consecutive empty residues on the circle. Place a block boundary immediately after this gap. Then each occupied e_r has at least g indices between itself and the end boundary of its block; in particular $u_{e_r}/U_{m+1} \le 2^{-g}$. This also holds in the possible initial partial block, since the gap condition is imposed on the residues of *all* occupied indices. In this alignment $|v_m| \le pC2^{-g}B_m$.

For a starting $x \ge 0$ the usual carry entering block m at time t is

$$a_m(t) = \left\lfloor \sum_{k<m} (x_k + tv_k)U_k/U_m \right\rfloor.$$

Its argument is $\eta + t\beta$, with $0 \le \eta < 1$. Since $\sum_{k<m} U_{k+1}/U_m < 2$, in the chosen alignment $|\beta| \le 2pC2^{-g}$. For large K this gives $H|\beta| < 1$. Cut the interval $0 \le t < H$ at every change of an entering or leaving carry of a nonzero v_m. There are at most $2p$ cuts, leaving a consecutive run of at least L sites. On the run the ordinary digit of each such block equals

$$x_m + tv_m + a_m(t) - B_m a_{m+1}(t)$$

and has velocity v_m. An uninterrupted group of inactive digits encodes its old block value plus its lower carry, reduced modulo the capacity of the group; hence all its digits are constant on the run. The infinite inactive tail is also constant, and the digits below the first active block never change.

At least one block is active because $d > 0$. Its phase $\theta_m x_m(t)$ visits L distinct cells during the run, by the defining separation of θ_m. Writing $k \le p$ for the number of active blocks, the multiset of their k cells therefore assumes at least L/k different values: each multiset can account for at most k of those distinct cells. The power-sum encoding distinguishes these multisets and the inactive blocks contribute a fixed translate. In particular, every admissible progression has at least L/p different joint states Z.

Counting trajectories without a bound on the radices. In *every* alignment $|v_m| \le pCB_m$, so the slope of every entering carry is bounded by $2pC$ and $|a_m(t)| \le 2pCH + 1$. Each length-H itinerary $\lfloor \eta + t\beta \rfloor$ has $O_{p,C}(H^4)$ possibilities: $O_{p,C}(H^2)$ lines in the (η, β)-plane determine all its floors, with their boundaries included. Prescribe the entering and leaving carries at all the at most p active blocks. Although x_m, v_m, B_m need not be bounded, the cell itinerary of an active digit is obtained from the phase

$$y_0 + ty_1 + a_m(t)y_2 + a_{m+1}(t)y_3 \pmod 1, \qquad y_0, y_1, y_2, y_3 \in [0, 1).$$

Here the four parameters represent $\theta_m x_m, \theta_m v_m, \theta_m, -B_m\theta_m$ modulo one. For fixed carry sequences and fixed t, the real affine expression before reduction ranges over an interval of length $O_{p,C}(H)$. The cell changes only when it equals an integer multiple of $1/Q$. Thus $O_{p,C}(QH^2)$ hyperplanes in a four-dimensional parameter cube determine its entire itinerary over all t. The number of relatively open faces of all dimensions of an arrangement of M hyperplanes in fixed dimension d is $O_d((1 + M)^d)$; this also follows directly by induction on M and d. Consequently each active digit costs only $(QH)^{O_{p,C}(1)}$ possible itineraries.

Every nonempty inactive gap above an active digit begins with a full block of radix $B_m \ge 2^K$, because its index m is positive. Write its initial block value as $e + B_m z$, $0 \le e < B_m$. For large K, its lower carry $a_m(t)$ has absolute value less than $B_m/2$. Write $j(t) = \lfloor (e + a_m(t))/B_m \rfloor \in \{-1, 0, 1\}$. The function $j(t)$ is monotone, since $a_m(t)$ is, and has only $O(H^2)$ possible itineraries. The lowest digit of the gap is $e + a_m(t) - B_m j(t)$. Its phase therefore has the form $z_0 + a_m(t)z_1 + j(t)z_2 \bmod 1$ with three parameters in $[0, 1)$, so its cell itinerary again costs only $(QH)^{O_{p,C}(1)}$ choices. The sum of the cell weights for all remaining digits of the gap has at most three possible values, one for each j: those digits encode $z + j$ modulo the capacity of the remaining gap. Record these three constants in V. The same argument handles the infinite upper tail, with no reduction at its top, since only nonnegative occurring tail values matter. The stationary prefix below the first active digit adds one constant in V.

Listing these data in order and recording which gaps are empty suffices to determine the full F_s trajectory. No absolute digit locations, radices or gap lengths are required. Since p, C are fixed, $|V| = P^p = K^{O_{p,C}(1)}$, and Q, H are polynomial in K, there are $K^{O_{p,C}(1)}$ possibilities in each alignment. The number of trajectories of the joint state Z is thus at most $\exp(O_{p,C}(K \log K))$.

Assign independent fair bits to the finite space V^K. A trajectory having at least L/p distinct states is monochromatic with probability at most $2^{1-L/p}$. As $L = K^3$, the union bound over all possible trajectories is less than one for large K. The bit of $Z(n)$ from a successful assignment is the claimed coloring. □

Corollary 2.5.3. *Every arid b-automatic set is blocked at some finite length by a single b-automatic binary coloring.*

Proof. Clear denominators as in Lemma 2.5.1 and apply Theorem 2.5.2 to $u_e = b^e$. In each alignment the menus for all full blocks are identical, so the same angle θ can be used for all of them; there may be one different angle for the initial partial block. All these finitely many angles can be rational, since the avoidance conditions are finitely many strict inequalities on a nonempty open set. In this case F_s is computed by grouping base-b digits read from the right, first into the initial block and then into blocks of size K. Each block makes one of finitely many prescribed additions to a finite group; the final partial block is padded on the left with zeros. The Boolean function of these simultaneous finite states is therefore b-automatic. Reversing a regular language permits reading from the other end; pulling the coloring back by the fixed denominator also preserves automaticity. □

Lemma 2.5.4 (Adding the nonmultiples of a modulus)**.** *If $E \subset \mathbb{N}$ is not 2-large and $q \geq 1$, then $E \cup (\mathbb{N} \setminus q\mathbb{N})$ is not 3-large.*

Proof. The case $q = 1$ is immediate. For $q \geq 2$, let $f : \mathbb{N} \to \{0, 1\}$ block the steps in E at length H. Let $a(n)$ be 1 or 2 according as $\{n/q\}$ belongs to $[0, 1/2)$ or $[1/2, 1)$. Order the two elements of $\{1, 2, 3\} \setminus \{a(n)\}$ increasingly. Let $v(n)$ be the first or the second of them according as $\{n/q^2\}$ belongs to the first or the second half-circle. Color n by $a(n)$ if $f(n) = 0$, and by $v(n)$ if $f(n) = 1$.

If $q \mid d$, the progression stays in a single class modulo q, so $a(n)$ is constant there and the color determines the old bit $f(n)$. If $q \nmid d$, consider one complete orbit of addition by d modulo q^2. Its reduction modulo q has at least two equally spaced points and thus meets both half-circles. Above each visited residue modulo q, the orbit has $\operatorname{ord}_{q^2}(d)/\operatorname{ord}_q(d) \geq 2$ equally spaced points on the circle $n/q^2 \bmod 1$, where ord_m means additive order modulo m. Indeed, for some prime $\ell \mid q$ we have $v_\ell(d) < v_\ell(q)$, and this quotient of orders is divisible by $\ell^{v_\ell(q)}$. Consequently, both possible values of $v(n)$ occur above that residue. The intersection of the permitted palettes $\{a(n), v(n)\}$ over this fibre is the singleton consisting of its value of $a(n)$. As both values of $a(n)$ occur on the base orbit, the intersection of the palettes over the full orbit is empty. No progression of length $\max(H, q^2)$ with an allowed difference can therefore be monochromatic. □

Corollary 2.5.5. *For an automatic set D of positive integers, each of the following is equivalent to largeness: D is 3-large; every partition of D into two cells has a 2-large cell. If D is not large there is a modulus q for which both $D \cap q\mathbb{N}$ and $D \setminus q\mathbb{N}$ are not 2-large.*

Proof. If D is automatic and not large, Theorem 2.4.1 gives q for which $E = D \cap q\mathbb{N}$ is arid. Apply Theorem 2.5.3 and Lemma 2.5.4, observing that $D \subset E \cup (\mathbb{N} \setminus q\mathbb{N})$. The first of the two cells has the single arid obstruction; the second is blocked by the half-circle coloring modulo q from the proof of Theorem 2.4.1. A large set satisfies both conditions by definition and the product-coloring argument. □

For arbitrary D, Lemma 2.5.4 also gives a useful implication independent of automaticity: if D is 3-large, then $D \cap q\mathbb{N}$ is 2-large for every q. Otherwise D would be a subset of the non-3-large set in the lemma.

The obstruction is effective. Enumerate binary digit automata and candidate lengths H until there is no monochromatic H-term progression with difference in the specified arid set. For any candidate, the question is emptiness of a regular relation, by the automaton constructions in Corollary 2.4.7. Termination follows from Corollary 2.5.3. The periodic palette modification is automatic as well. In particular, in Corollary 2.4.7 one can always take *exactly two* binary tests, irrespective of the state bound, original palette and target length: use one arid test and the periodic test, duplicating a test if needed. The length still depends on the automaton.

Proposition 2.5.6 (A single rotation window)**.** *For arbitrary $D \subseteq \mathbb{N}$, if D is 3-large then $D \cap \{d : \|d\alpha\|_{\mathbb{T}} < \epsilon\}$ is 2-large for every $\alpha \in \mathbb{T}$ and $\epsilon > 0$.*

Proof. Suppose a binary f blocks the indicated intersection at length H. Choose a real representative of α and put $y(n) = n\alpha/2 \pmod 1$. On $\mathbb{T}$ ban one of three labels according to which half-open third contains the point. On each half-open sixth assign the two remaining labels bijectively to the bits of f. Pair each sixth with its antipodal sixth: the two bans differ, so the palettes have exactly one label in common; assign this label to the same bit in both. This gives a three-coloring g of the integers.

Choose $\tau > 0$ with $2\tau < \epsilon$ and $2H\tau < 1/6$. If $\delta = d\alpha/2$ lies within τ of $\theta = 0$ or $1/2$, then $\|d\alpha\| < \epsilon$. On $2H$ sites the corrected phases $y(a) + t(\delta - \theta)$ cross at most one boundary between sixths. Some H consecutive corrected sites are in one sixth. On that run the actual phases either stay in one sixth or alternate between its antipodal pair. A constant output g therefore decodes to a constant f, impossible if $d \in D$.

Outside those two neighborhoods there is a uniform length such that every circle progression visits all three half-open thirds. Otherwise pass to a convergent subsequence of starts and steps for increasingly long failed progressions, all missing the same third. The limiting orbit misses its open interior, impossible for an irrational step or for a rational step of order at least four. Orders one and two were excluded. For a limiting step $p/3$ of exact order three, correct each of the first six approximating phases by subtracting $tp/3$. The corrections advance monotonically over an arc of length less than $1/3$, hence some three consecutive corrected phases lie in one half-open third. Adding back $tp/3$ puts their actual phases in all three thirds, again a contradiction. For these steps a long run cannot be monochromatic since every label is banned somewhere. Taking the larger of the two lengths proves the contrapositive. □

2.6 Ternary triples: an ordinary Bohr neighborhood up to a null set

A Bohr neighborhood of zero in $\mathbb{Z}$ contains a set $\{n : \max_{j \le t} \|n\theta_j\|_{\mathbb{T}} < \varepsilon\}$. Upper Banach density refers to the supremum over intervals of a given length, followed by the limit as the length tends to infinity.

Theorem 2.6.1. *For every three-coloring f of $\mathbb{N}$, there are a Bohr neighborhood B of zero and a set $E \subseteq \mathbb{N}$ of zero upper Banach density such that*

$$(B \cap \mathbb{N}) \setminus E \;\subseteq\; R_3(f).$$

Here the factor defining B is a rotation on a compact abelian group. The usual nil-Bohr-minus-null result for triples in one positive-density set permits a two-step nilsystem [11]. The restriction to three colors is what lets us use the smaller factor.

Lemma 2.6.2. *There exists K_0 such that every three-coloring of $[K_0]$ has monochromatic i, j, k with $i - 2j + k = 6$. Repetitions are permitted.*

Proof. By compactness and translation invariance it suffices to work on $\mathbb{Z}$. Suppose there is no such triple. No two sites distance 6 apart can have the same color (use $i = j$), and no two sites distance 3 apart can have the same color (use $i = k$). Consequently every three consecutive sites of each progression of step 3 have three different colors, so $f(n + 9) = f(n)$. Fix one color. Modulo 9 it occurs at three representatives a_0, a_1, a_2 with $a_j \equiv j \bmod 3$. Writing $S = a_0 + a_1 + a_2$, the three numbers $S - 3a_j$ run over $0, 3, 6 \bmod 9$. Use the a_j giving 6 as the middle term and the other two as the outer terms. Moving an outer term by a multiple of 9 gives the required equality over $\mathbb{Z}$. □

Lemma 2.6.3. *There exists an absolute $\eta > 0$ with the following property. Let U be a compact connected abelian Lie group with normalized Haar measure. Let $g_1, g_2, g_3 : U \to [0, 1]$ be measurable with sum 1. For every $t \in U$,*

$$\int_{U^2} \sum_{l=1}^{3} g_l(z) g_l(z + v) g_l(z + 2v + t)\, dz\, dv \;\ge \eta.$$

Proof. Color each point by an index attaining $\max_l g_l$. At that index the weight is at least $1/3$. Choose $s \in U$ with $6s = t$. For independent Haar x, w form the grid $x + aw + bs$, with $b \in [K_0]$ and $a \in [W]$, where $W = W(3^{K_0}, 3)$. Color a by its entire column of K_0 labels. Van der Waerden supplies $a, a+e, a+2e$ with identical columns, $e > 0$. Lemma 2.6.2 supplies indices i, j, k in that common column of one color and with $i - 2j + k = 6$. Select the corresponding three sites, one in each column. Their second difference is t. For any fixed choice of the indices, the first two sites are independent Haar points: the linear map $(x, w) \mapsto (x + aw, x + (a + e)w)$ is onto because multiplication by e on U is onto. Their shifts by is, js do not change this law. The third site is determined by the first two and t. There are at most $W^2K_0^3$ choices, and for every (x, w) some choice succeeds. The union bound gives the assertion with $\eta = (27W^2K_0^3)^{-1}$. The same proof applies to a compact connected abelian group without the Lie assumption. □

Proof of Theorem 2.6.1. Choose an ergodic invariant probability measure on a forward orbit limit system of the symbolic coloring; each nonempty cylinder meeting its support occurs at arbitrarily late positions in f. Let h_l be the indicators of the three colors and write $I_h(d) = \int h(x)h(T^d x)h(T^{2d}x)\, d\mu(x)$. By Bergelson–Host–Kra (BHK), Corollary 4.6 [4], replacing the h_l by their expectations on Z_2 changes the correlations by sequences tending to zero in uniform density. This means that, for any prescribed positive tolerance, the set of errors exceeding the tolerance has zero upper Banach density. The factor Z_2 is an inverse limit of ergodic two-step nilsystems. Choose one such finite-dimensional factor Y sufficiently far along the inverse limit, and take the three conditional expectations g_l on Y. These take values in $[0, 1]$ and sum to one. The additional error in the sum of the three correlations is at most $3\sum_l \|\mathbb{E}(h_l \mid Z_2) - g_l\|_1$, uniformly in d. Choose this bound smaller than $\eta/8$. It matters here that η in Lemma 2.6.3 is independent of Y and the three weights.

Write $Y = G/\Gamma$ with niltranslation a, in the reduced presentation of BHK, Section 4.1: G is generated by G° and a, and Γ contains no nontrivial subgroup normal in G. Then $U = [G, G]$ is a compact connected central torus (possibly trivial) and $U \cap \Gamma$ is trivial. By BHK, Proposition 6.5 and Sections 6.4, 7.2, the correlation $I_g(d)$ on Y differs, by a sequence tending to zero in uniform density, from

$$J_g(d) = \int_Y \int_U g(x)\, g(a^d u x)\, g(a^{2d} u^2 x)\, du\, dx.$$

Indeed the measure in BHK's formula for $k = 2$ is the law of (x, ux, u^2x) for independent Haar x, u. Thus it suffices to bound $\sum_l J_{g_l}(d)$ from below near the identity in the horizontal rotation $H = G/(U\Gamma)$.

Consider any sequence for which $a^d \to 1$ in H. After passing to a subsequence, $a^d\Gamma$ converges to $b\Gamma$ for some $b \in U$. Using local lifts to G, write $a^d = b_d\gamma_d$ with $b_d \to b$ and $\gamma_d \in \Gamma$. Take a further subsequence on which $t_d = [\gamma_d, b_d] \to t \in U$, using $[g, h] = ghg^{-1}h^{-1}$. For $x = \xi\Gamma$ write $v_d(x) = [\gamma_d, \xi] \in U$. This is well defined because $[\Gamma, \Gamma] \subset U \cap \Gamma$ is trivial. The two moving points in J_g are

$$b_d u v_d(x) x, \qquad b_d^2 t_d u^2 v_d(x)^2 x.$$

Absorb $v_d(x)$ into the Haar variable u; this is done at each fixed x and does not require the commutators to be small. Taking limits and absorbing b into u now gives

$$J_g(d) \;\longrightarrow\; \int_Y \int_U g(x) g(ux) g(tu^2 x)\, du\, dx.$$

For continuous g this follows from compactness. For bounded measurable g approximate in L^1: each of the three marginals is Haar, both before and after taking limits. Finally disintegrate x over

H. On every relevant fiber its conditional law is Haar on a translate of U; on that fiber the last display is precisely the expression of Lemma 2.6.3, after summing over $g = g_l$.

Every subsequential limit along a horizontal return is therefore at least η for the sum. By compactness there is a neighborhood of the identity in H whose positive return times satisfy $\sum_l J_{g_l}(d) > \eta/2$ (equivalently it suffices to take a smaller such neighborhood). These times contain a Bohr neighborhood B. Removing the uniform-density exceptional sets, with the combined remaining errors less than $\eta/4$, gives $\sum_l I_{h_l}(d) > 0$. A positive cylinder probability gives a monochromatic triple in the original coloring. □

Consequently a set meeting *every ordinary Bohr neighborhood* in positive upper Banach density forces a monochromatic triple in every ternary coloring. This assertion concerns triples only. For a ternary triple obstruction D, its part inside the Bohr set of Theorem 2.6.1 has zero upper Banach density. Its part outside can be covered by finitely many binary triple obstructions: write the Bohr conditions as $\|d\theta_j\| < \varepsilon$; if one fails, some $2^k d\theta_j$ with k in a fixed finite range has norm at least $1/4$. The half-circle coloring at frequency $2^k\theta_j$ blocks triples there. In particular the unresolved sparse part is a real condition, not an error term one can discard for arbitrary D.

2.7 An exponent-three counterexample and its obstruction over the integers

Write $G = \bigoplus_{i\geq 1} \mathbb{F}_3 e_i$, and let $D = \{e_i - e_j : i < j\}$. In this section a progression is a list $x + td$. At length at least three, monochromaticity means that the full three-point orbit is one color; longer lists repeat. No claim about arbitrarily many distinct points in one such orbit is intended.

Proposition 2.7.1. *Every finite partition of D has a cell such that every binary coloring of G has a monochromatic full orbit in a direction in that cell. A single ternary coloring of G has no such orbit in any direction of D.*

Proof. For the upper obstruction color by $Q(x) = \sum_i x_i^2 \pmod 3$. The second difference in direction $d = e_i - e_j$ equals $2Q(d) = 1$ in $\mathbb{F}_3$, which excludes a constant triple.

For the lower assertion, Ramsey's theorem first gives an infinite set of indices all of whose pair differences belong to one cell. On this set thin again so that for each of the four words $1, 22, 112, 211$ the color of a vector with those ordered nonzero coefficients is independent of the indices chosen. Write their colors as A, B, C, D' respectively. Choose $i < j$ with spare indices before, between and after them. All the following triples are full orbits of the same direction $e_i - e_j$:

- If $A = B$, use $(1, 0), (2, 2), (0, 1)$ on (i, j).
- If $B = C$, use $(2, 0), (0, 2), (1, 1)$ with a further coefficient 2 after j. If $B = D'$, put that coefficient before i instead.
- Otherwise $A = C = D'$, since there are only two colors. Use $(0, 0), (1, 2), (2, 1)$ with a further coefficient 1 strictly between i and j.

Thus the cell works for every binary coloring. Alternatively one can thin for a finite vector of binary colors at once; each test chooses its own row of the argument, with the same i, j. □

On the free abelian group $\bigoplus_i \mathbb{Z} e_i$, the cubic bit from Theorem 2.10.1 already blocks the same directions at length *four*, with no wraps to discuss. Thus the passage from a three-cycle to its torsion-free lift fails at the very next length.

2.8 A positive result at all lengths inside rotation windows

A short consequence of the known nilfactor theorems is useful for comparison. Say that D is *open for nilsystem return times* if it is a union of sets $\{n \geq 1 : T^n y \in V\}$ with T a nilrotation and V open (it need not contain y). For such a D, nil-Bohr recurrence, 2-largeness and largeness are equivalent. Indeed, for every basic nil-Bohr neighborhood B of zero, nil-Bohr recurrence gives $d \in B \cap D$. Take a single open visiting set in D containing d. Its intersection with B contains all the visits to a nonempty relatively open set in the joint nilsystem orbit closure. This orbit closure is minimal, so those visits are syndetic and have positive lower density. By Huang–Shao–Ye [11, Theorem 2.2.4], for any positive-measure set in an ergodic system and any fixed length, its good differences contain a basic nil-Bohr neighborhood of zero less a zero-upper-Banach-density set. The intersection argument therefore gives measurable multiple recurrence and hence largeness. The converse implications are largeness $\Rightarrow$ 2-largeness $\Rightarrow$ nil-Bohr recurrence, the latter by Alweiss–Farhangi [1]. Arbitrary sets that merely intersect every neighborhood need not have this positive-density intersection.

The following result goes in a different direction: the allowed differences are completely arbitrary, and the rotation windows restrict the *locations of the progressions*.

Proposition 2.8.1. *Let $D \in \mathcal{R}_2$. For $i = 1, \dots, m$ let W_i be the positive visiting times to a nonempty relatively open set on the orbit closure of a rotation on a compact abelian Lie group. Give each W_i any binary coloring c_i. For every h there is a common difference $d \in D$ such that every W_i contains a c_i-monochromatic h-term progression of difference d.*

Proof. Realize each rotation as $n \mapsto x_i + n\alpha_i$ in a torus, including rational coordinates if necessary. Its forward orbit is dense in its orbit closure Y_i. If V_i is the target open set, finitely many sets $V_i - t\alpha_i$, $t \geq 0$, cover Y_i. Tile the ambient torus by sufficiently small half-open grid rectangles so that, for every tile C meeting Y_i, there is a fixed $t_C \geq 0$ with $(C \cap Y_i) + t_C\alpha_i \subset V_i$. This follows by the Lebesgue number lemma. Encode a binary coloring of all of $\mathbb{N}$ by $g_i(n) = c_i(n + t_C)$ on the times in C.

Let k be the largest ambient torus dimension and $L = (k+1)h$. Choose a Bohr condition on d that makes the total travel in *each* coordinate along L sites smaller than one grid spacing; it suffices that $(L-1)\|d\alpha_{i,j}\|$ be that small for all i, j. This condition can be imposed along with the tests g_i in (2.1.1): its complement is covered by finitely many binary triple blockers, using the dyadic half-circle construction following Theorem 2.6.1. Increase the common test length to at least three if necessary. On each witnessing progression, each coordinate crosses at most one grid boundary. There are at most $k+1$ consecutive runs in single tiles, one of length at least h. Its correction t_C is constant, giving a progression in W_i of the required color and unchanged difference. □

In particular a ternary witness against $D \in \mathcal{R}_2$ cannot omit a color throughout a rotation window. In a minimal symbolic witness, every fiber of the maximal equicontinuous factor must meet every color cylinder: a fiber missing a clopen color has an open neighborhood over which that color is absent, containing a window from a finite-dimensional rotation factor. This conclusion uses the independent-test quantifier and does not require a density hypothesis.

2.9 Over the integers, equal test lengths do not suffice

Theorem 2.9.1. *Let $L \geq 3$ be odd, put $s = L - 1$, choose irrational α, and set*

$$D_s = \{d \in \mathbb{N} : \|\alpha d^s\|_{\mathbb{T}} > 1/4\}.$$

For every finite list of binary colorings, D_s contains a common good L-term difference. Yet a single finite coloring has no monochromatic L-term progression of difference in D_s. In particular, every finite partition of D_s has a cell that works for every binary test of this fixed length.

These are the polynomial-phase examples used, up to endpoints, by Frantzikinakis, Lesigne and Wierdl (FLW) to separate orders of recurrence [7]. The extra binary term in Theorem 2.9.1 comes from a parity cancellation.

Proof. For the obstructing coloring, partition $\mathbb{T}$ into 2^{s+2} equal half-open intervals and color n by the interval containing $\alpha n^s/s! \pmod 1$. On $s+1$ sites of one color, lift the phases into that interval. Their sth finite difference is congruent to αd^s modulo one. The binomial coefficients sum to zero, and the sum of their absolute values is 2^s. Subtracting the interval's left endpoint therefore gives $\|\alpha d^s\|_{\mathbb{T}} < 1/4$.

For the positive assertion, write the binary colors as signs. For signs $u_0, \dots, u_{L-1}$,

$$\mathbf{1}_{u_0=\cdots=u_{L-1}} = 2^{1-L} \sum_{\substack{J\subseteq\{0,\dots,L-1\}\\ |J| \text{ even}}} \prod_{j\in J} u_j. \tag{2.9.1}$$

This follows by adding $\prod_j(1+u_j)$ and $\prod_j(1-u_j)$ and dividing by 2^L. Since L is odd, every summand uses at most s sites.

For each binary coloring take a shift-invariant probability measure μ_i on its forward orbit limit set in the two-sided shift (extend the coloring arbitrarily to negative indices). One may take a weak limit of empirical measures of its positive shifts. Write T_i for the shift and u_i for the sign at coordinate zero. Let $M_i(d)$ be the probability that $u_i(x), \dots, u_i(T_i^{(L-1)d}x)$ are all equal, and put $M(d) = \prod_i M_i(d)$. If $M(d) > 0$, each coloring has a witnessing progression in its original positive sequence.

Here is the point of using *independent* initial points. After expanding (2.9.1), each term of $M(d)$ is a product over i of integrals indexed by subsets J_i with $|J_i| \le s$. In each nonempty group translate the integration variable by its first time. One factor now occurs at time zero. Place the other at most $s-1$ factors in slots $1, \dots, s-1$, padding with constant functions. On the product probability space the term has the form

$$\int F_0 \prod_{v=1}^{s-1} S_v^d F_v \, d\mu,$$

where in coordinate i, S_v is the appropriate power of T_i, and all S_v commute. Empty groups contribute 1. FLW, Proposition 4.2, applies to these $s-1$ transformations and the degree-s polynomial αd^s. With $g(t) = \mathbf{1}_{\|t\|>1/4}$, whose integral is $1/2$, it gives

$$\lim_{N\to\infty} \frac{1}{N} \sum_{d\le N} (\mathbf{1}_{D_s}(d) - \tfrac{1}{2}) M(d) = 0. \tag{2.9.2}$$

The number of transformations is independent of the number of tests.

For completeness, the unweighted mean of $M(d)$ has positive lower limit using just finite van der Waerden. Choose $W = W(2^m, L)$, where m is the number of tests. For each $q \ge 1$ and each product point, color $[W]$ by the tuples $(u_i(T_i^{kq}x_i))_i$ for $k \in [W]$. There is a monochromatic L-term progression, with index difference $e \le W$. Integrating the union bound and using invariance gives $1 \le W \sum_{e\le W} M(eq)$. Sum over $q \le N/W$; each d is counted at most W times, so $\sum_{d\le N} M(d) \ge \lfloor N/W \rfloor / W^2$. Now (2.9.2) gives some $d \in D_s$ with $M(d) > 0$. The assertion about partitions follows by assigning one failing test to each cell, just as in (2.1.1). □

For triples the calculation simplifies. With the autocorrelation $\rho_i(d) = \int u_i \cdot T_i^d u_i \, d\mu_i$, we have

$$4M_i(d) = 1 + 2\rho_i(d) + \rho_i(2d).$$

Each ρ_i is positive definite. Thus M is the Fourier transform of a positive measure on $\mathbb{T}$ with an atom of mass at least 4^{-m} at zero. The Fourier transform of the normalized counting measure on $D_2 \cap [N]$ tends to zero at each nonzero frequency, by Weyl's theorem applied to $\ell\alpha d^2 + \theta d$ and trigonometric approximation. Integrating gives a positive limiting mean of M on D_2. For this case one may use just *eight* colors in the upper obstruction: if three lifts x_0, x_1, x_2 lie in an interval of length $1/8$, then $|x_0 - 2x_1 + x_2| < 1/4$.

The length qualification is essential. Each D_s misses a polynomial nil-Bohr neighborhood of zero. By Alweiss–Farhangi [1] a longer binary test already blocks it. The example varies with L. If desired, for triples one can instead take $\|\alpha d^2\| > 2/5$. The same positive argument works and five equal arcs suffice for the obstructing coloring.

2.10 Four colors versus arbitrarily many binary triple tests

The polynomial example above supplies one set working for all finite lists. For triples, the following finite sets give a different tradeoff: the obstructing coloring needs only four colors. The sets themselves depend on the number of binary tests.

Theorem 2.10.1. *For every $m \geq 1$ there exist a finite set $D \subset \mathbb{N}$ and a periodic coloring $f : \mathbb{N} \to \mathbb{Z}/4\mathbb{Z}$ such that*

$$D \cap R_3(f) = \varnothing, \qquad D \cap \bigcap_{i=1}^{m} R_3(c_i) \neq \varnothing \quad \text{for every list of binary colorings } c_1, \dots, c_m.$$

Nevertheless, D admits a single periodic binary coloring with no monochromatic 10*-term progression of difference in D, uniformly in m.*

Proof. We first record a quantitative feature of finite difference sets. Take any $q > 4^m$ distinct integers $b_1, \dots, b_q$, and put $D = \{|b_j - b_i| : i < j\}$. For any given binary tests let $M(d)$ be the product of their monochromatic triple probabilities, using shift-invariant measures as in the preceding section. The identity $4M_i(d) = 1 + 2\rho_i(d) + \rho_i(2d)$ shows that $M(d) = \int_{\mathbb{T}} \mathrm{e}^{2\pi \mathrm{i} d\theta} \, d\nu(\theta)$, where ν is positive and $\nu(\{0\}) \geq 4^{-m}$. Hence

$$\sum_{i,j=1}^{q} M(b_j - b_i) = \int_{\mathbb{T}} \left| \sum_{j=1}^{q} \mathrm{e}^{2\pi \mathrm{i} b_j \theta} \right|^2 d\nu(\theta) \geq 4^{-m} q^2 > q.$$

The diagonal contributes exactly q, since $M(0) = 1$. Some off-diagonal term is positive, so its absolute difference belongs to all $R_3(c_i)$.

It remains to construct the b_i and the four-color obstruction. On the four possible increments $0, 1, 2, 3 \in \mathbb{Z}/4\mathbb{Z}$, allow the arrows

$$0 \longrightarrow 1, \qquad 1 \longrightarrow 2, \qquad 2 \longrightarrow 3, \qquad 3 \longrightarrow 0, \qquad 1 \longrightarrow 3.$$

No arrow and its reverse are both allowed. There is a closed walk from 0 of length three, with successive labels $0, 1, 3$, and one of length eight, with labels $0, 1, 2, 3, 0, 1, 2, 3$. The sum of the

labels is zero modulo four in both walks. Every integer $p \ge 14$ has the form $3a + 8b$ with $a, b \ge 0$: this holds for $14, 15, 16$ and persists on adding 3. Concatenation therefore gives a cyclic word $u_p : \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ such that consecutive labels follow allowed arrows and $\sum_t u_p(t) = 0$. The latter condition lets us choose v_p with $v_p(t+1) - v_p(t) = u_p(t)$ at every site, including across the end of the word.

Choose distinct primes $p_i > 27$ and identify $\mathbb{Z}/P\mathbb{Z}$ with $\prod_{i=1}^{q} \mathbb{Z}/p_i\mathbb{Z}$ by the Chinese remainder theorem, where $P = \prod_i p_i$. Take $b_i \in [0, P)$ representing the ith coordinate unit vector e_i, and define

$$f(x) = \sum_{i=1}^{q} v_{p_i}(x_i) \pmod 4.$$

For a step $e_j - e_i$, the first two changes of f are

$$u_{p_j}(x_j) - u_{p_i}(x_i - 1), \qquad u_{p_j}(x_j + 1) - u_{p_i}(x_i - 2).$$

If both vanished, the arrow from $u_{p_j}(x_j)$ to $u_{p_j}(x_j + 1)$ would be the reverse of the arrow from $u_{p_i}(x_i - 2)$ to $u_{p_i}(x_i - 1)$. This is impossible. Reversing a triple handles the opposite step, so the pullback of f to the integers blocks all of D.

For the binary upper bound take representatives $0 \le x_k < p_k$ and use

$$C(x) = \sum_{u<v} \binom{x_u}{2} x_v \pmod 2.$$

In ten consecutive sites only two coordinates move, each at speed one in absolute value. Each can wrap at most once. The resulting at most three runs include one of at least four sites without a wrap. On this run $C(x + td)$ is the reduction of an integer-valued cubic; its third forward difference is $3d_i^2 d_j$ if $i < j$, since only that pair contributes to the cubic coefficient. This is odd. Modulo two the third difference of four equal bits is zero, a contradiction. □

The four-color example exploits an oriented graph with closed walks of two coprime lengths. A loopless oriented graph on three vertices has no closed walk unless it contains the directed triangle, in which case every closed walk has length divisible by three. This explains why simply reducing the number of increment labels to three fails in the construction.

2.11 How simple a negative witness could be

The reduction in Theorem 2.2.1 can be made explicit on both the difference and vertex sides. In particular, entropy or mere computability cannot be the missing hypothesis.

Corollary 2.11.1. *If the finite-partition assertion fails, fix any computable nondecreasing function $H : \mathbb{N} \to \mathbb{N}$ tending to infinity. There is a computable $D \in \mathcal{R}_2 \setminus \mathcal{L}$ with*

$$|D \cap [1, x]| \le H(x) \quad (x \ge 1), \qquad \lim_{k \to \infty} (d_{k+1} - d_k) = \infty$$

for its increasing enumeration $d_1, d_2, \ldots$. Its obstructing coloring g can be chosen computable, with a minimal and uniquely ergodic symbolic orbit closure of zero topological entropy. More precisely, if $p_g(n)$ counts its distinct consecutive n-letter words, then $p_g(\lambda) \le r^2 \lambda$ along an unbounded sequence of powers of two, for some fixed palette size r.

There is also a dynamical obstruction on a finite cyclic extension of the dyadic odometer: the cocycle has at most one discontinuity and the observable is simply the cyclic coordinate.

Proof. Fix the failed pair (r, h) in Theorem 2.2.1. The finite condition on (N_j, f_j) there is decidable by finite exhaustive search, so we can compute successive obstructing packets F_j. We can compute the next packet before choosing its place value. Use only powers of two for the radices and start above an arbitrarily long initial dummy block with zero coloring. Choose successive place values λ_j so large that

$$H(\lambda_j) \geq \sum_{i \leq j} |F_i|, \qquad \lambda_j > 2\lambda_{j-1} \max F_{j-1} + j.$$

We can simultaneously impose the radix lower bounds of Theorem 2.2.1. These choices give the counting bound: if $\lambda_j \leq x < \lambda_{j+1}$ only the first j packets contribute. Gaps within packet j are at least λ_j, and gaps between consecutive packets tend to infinity. Membership in D and the digit-sum coloring g are computable by finite searches terminated by increasing place values.

We also arrange exact balance of the digit contributions. Embed the original packet palette in a cyclic group of power-of-two order, now denoted by r, and use these new labels for the periodic packet colorings g_j. Choose the radix $b_j = rR_j$, where R_j is a power of two with $R_j > 4h \max F_j$. It may be enlarged arbitrarily to meet all of the preceding spacing and counting requirements. On the jth digit, use the contribution

$$G_j(t) = g_j(t \bmod R_j) + \lfloor t/R_j \rfloor - g_j(0) \pmod r, \qquad 0 \leq t < b_j.$$

For fixed $t \bmod R_j$, varying the quotient gives each label exactly once, so every group element occurs exactly R_j times. The dummy initial digits still contribute zero. For $d \in F_j$, the $4h$ consecutive quotients at place λ_j along a progression of step $\lambda_j d$ span less than R_j. They cross at most one multiple of R_j, including any possible radix boundary. On a consecutive run of at least $2h$ sites, the higher digits and the palette offset are therefore fixed, and $t \bmod R_j$ advances by d without a wrap. The periodic g_j obstructs such a run, since its $2h$-term obstruction holds at every start. This preserves the original obstruction; no divisibility of R_j by the packet period is needed.

For each j, all aligned blocks of g of length λ_j are translates, in the cyclic color group, of the same word of that length: all higher digits contribute one constant across each block. An arbitrary word of length λ_j crosses at most two such blocks, so its offset and two cyclic constants determine it. This proves the complexity bound and, by subadditivity of $\log p_g(n)$, zero entropy.

To check strict ergodicity, index positions from zero and let $W_j = g[0, \lambda_j)$. Every aligned $(j+1)$st-level block is a concatenation of the same number R_j of copies of each of the r color translates of W_j, in an order that can depend on its own translate. This holds for every actual packet level, including the first one above the uncolored dummy block. Every word occurring in g occurs wholly inside W_J for some J, simply by taking λ_J beyond the end of one occurrence. Since every aligned block at level $J+1$ contains an unshifted copy of W_J, that word recurs with bounded gaps. Thus the two-sided subshift defined by the language of g is minimal.

For a word w of length $l \leq \lambda_j$, count its occurrences lying wholly inside each translate of W_j, divide by λ_j, and average the r resulting frequencies to get $A_j(w)$. In every aligned block of length λ_{j+1} these within-child starts have frequency exactly $A_j(w)$ by balance. Starts crossing a boundary between children contribute at most $(l-1)/\lambda_j$ in frequency, counting also the right boundary of the aligned block. For an arbitrary interval of possible starts, removing its two partial level-$(j+1)$ boundary blocks costs at most $2\lambda_{j+1}$ starts. Hence the difference between the limiting upper and lower frequencies, uniformly over starting positions, is at most $(l-1)/\lambda_j$. Letting j grow proves uniform word frequencies and therefore unique ergodicity.

For the last statement take the dyadic integers $\mathbb{Z}_2$ with Haar probability and their mixed-radix digits at the same powers-of-two place values. Denote by G_j the contribution of digit j to g. Outside

$x = -1$ define

$$\eta(x) = \sum_j (G_j(z_j(x+1)) - G_j(z_j(x))) \pmod r.$$

Only finitely many digits change, and η is locally constant there. Give it any value at -1. Then $(x, a) \mapsto (x + 1, a + \eta(x))$ preserves Haar probability on $\mathbb{Z}_2 \times \mathbb{Z}/r\mathbb{Z}$. For every nonnegative integer n, the sum of the cocycle on k steps is $g(n + k) - g(n)$, defining $g(0)$ by the same digit formula. The obstruction proof in Theorem 2.2.1 applies also at $n = 0$. For a fixed $d \in D$, the cocycle sums on $d, 2d, \ldots, (4h - 1)d$ steps are locally constant away from finitely many negative integers. If they all vanished at any such point, they would vanish on an open set, hence at a nonnegative integer. This contradicts the $4h$-term obstruction. Thus every individual cyclic fiber has zero multiple intersection at every difference in D. □

Conversely, a restriction that asks only for D itself to be syndetic does not weaken the unresolved question. Given any counterexample D and $q \geq 2$, the syndetic set

$$D' = (\mathbb{N} \setminus q\mathbb{N}) \cup qD$$

is again a counterexample. The common-step test property for qD follows by pulling each test back under $n \mapsto qn$. If a finite f blocks D, color n by $(n \bmod q, f(1 + \lfloor n/q \rfloor))$: a monochromatic pair forces $q \mid d$, and a progression of step d divisible by q projects to one of step d/q. Starting with the sparse D of Corollary 2.11.1 gives such D' of natural density exactly $1 - 1/q$. These constructions are conditional; neither gives a counterexample without the missing finite packets.

Lemma 2.11.2. *In the balanced realization of Corollary 2.11.1, the skew product with product Haar measure is ergodic. Its Kronecker factor, and indeed every Host–Kra factor Z_s, $s \geq 1$, is exactly the dyadic odometer. Consequently for any vertical color $a_0 \in \mathbb{Z}/r\mathbb{Z}$ and any $L \geq 2$, if $A = \mathbb{Z}_2 \times \{a_0\}$, then*

$$d \longmapsto \mu\Big(A \cap T^{-d}A \cap \cdots \cap T^{-(L-1)d}A\Big) - r^{-L}$$

converges to zero in uniform density. In particular, for $L = 4h$ every $d \in D$ lies in an exceptional set of zero upper Banach density, whereas the structured part of this correlation is the same positive constant at every d.

Proof. Discard the countable orbit of -1 in the base, which has Haar measure zero. Fix j at or above the first actual digit level and start at any remaining base point x and any fiber label a. Partition its integer time axis into segments of length λ_{j+1} beginning when the base is $0 \pmod{\lambda_{j+1}}$. Within each such segment the higher digits stay fixed; summing the cocycle along its successive steps shows that the vertical word is a cyclic color translate of W_{j+1}. This uses only finitely many lower digit contributions: there is no need to assign a value to an infinite digit sum at x. For every fixed residue $v \pmod{\lambda_j}$ and vertical label e, the pair (v, e) occurs exactly R_j times in that segment. Indeed, the lower digits are fixed at v, and digit j runs through its $b_j = rR_j$ possible values, with every label repeated R_j times. The two partial segments at the ends of an interval cost fewer than $2\lambda_{j+1}$ time indices. Thus the limiting frequency of each cylinder pair is $1/(r\lambda_j)$, for every such x and every a. These cylinders generate the product Borel σ-algebra. In particular an invariant probability measure projects to Haar measure on the uniquely ergodic odometer, hence gives the discarded orbit no mass. Integrating the cylinder frequencies shows that this invariant measure must be product Haar, which proves ergodicity.

Let $F \neq 0$ be an eigenfunction of the skew product, with eigenvalue ξ. Ergodicity lets us normalize $|F| = 1$ and makes its eigenspace one-dimensional. The vertical translation $(x, a) \mapsto (x, a + 1)$ commutes with T and therefore multiplies F by an rth root of unity. It follows that F^r descends to

the base, where it has eigenvalue ξ^r. The eigenvalues of the dyadic odometer are exactly the roots of unity of power-of-two order (its characters are exhausted by those of its finite cyclic quotients). Since r is a power of two, ξ is itself such a root and hence is already an eigenvalue on the base. Its one-dimensional eigenspace is therefore spanned by a base character. This proves the Kronecker assertion.

Every finite-dimensional ergodic nilfactor $Y = G/\Gamma$ of this skew product is a finite cycle. To see this, use the standard presentation $G = \langle G^0, t\rangle$, where t induces its rotation. Its horizontal factor $G/G_2\Gamma$ is an ergodic rotation on a compact abelian Lie group [4, Theorem 4.1]. A positive-dimensional such group has a character onto the circle, yielding an irrational eigenvalue by ergodicity. Since the eigenvalues of a factor lift, the horizontal factor must be finite. The connected G^0 therefore has trivial image in $G/G_2\Gamma$ and, projecting to G/G_2, has image contained in the discrete image of Γ. Thus $G^0 \subseteq G_2$. Conversely G_2 is connected in this presentation, and $G_2 = [G, G^0]$ [4, Theorem 4.1 and its proof]. It follows that $G_2 = G^0$, and then $G_3 = [G, G_2] = G_2$. Nilpotence forces $G_2 = G^0$ to be trivial, so Y is a finite cycle as asserted. Every such factor lies in the Kronecker factor. By the Host–Kra inverse-limit theorem [4, §4.4.4], every Z_s is therefore contained in the odometer; the reverse inclusion follows from $Z_1 \subseteq Z_s$.

Finally, $\mathbb{E}(\mathbf{1}_A \mid Z_{L-1})$ is the constant $1/r$, by the product Haar measure. The uniform-density comparison in [4, Corollary 4.6] gives the claimed correlation formula. For $L = 4h$ the skew-product obstruction proved above makes the correlation zero on every difference in D. □

Why local binary recodings of a witness are insufficient

The next example is unconditional. There is a uniformly recurrent ternary word with no monochromatic triple of any permitted difference, although every finite list of binary colorings obtained by local recoding of that word, even with additional periodic input, has arbitrarily long monochromatic progressions of a common permitted difference. An external binary coloring blocks all the permitted triples.

To construct it, take $\lambda_1 = 1$ and, for $j \geq 1$, let R_j be even, divisible by $j!$, and at least $10j$. Set $b_j = 3R_j$, $\lambda_{j+1} = b_j\lambda_j$, and $D = \{\lambda_j : j \geq 1\}$. If $z_j(n)$ is the corresponding digit, define

$$G_j(t) = \big((t \bmod R_j) \bmod 2\big) + \lfloor t/R_j \rfloor \pmod 3, \qquad g(n) = \sum_{j\geq 1} G_j(z_j(n)) \pmod 3.$$

The sum is finite because $G_j(0) = 0$. Each digit has exactly R_j occurrences of each label: the three slab offsets cycle every fixed position through the three labels. The preceding balanced-block argument gives a minimal, uniquely ergodic symbolic orbit closure. Among three sites of a progression of difference λ_j, the quotients at place λ_j are three successive integers, hence cross at most one R_j-slab boundary. Two consecutive sites stay in the same slab with the same higher digits, and their two low-order parities in the definition of G_j differ. Thus no such triple is monochromatic for g.

Now take finitely many binary recodings, each of the form

$$c_i(n) = F_i\big(n \bmod q,\, g(n-k), \dots, g(n+k)\big),$$

for all $n > k$, using a common period q and window radius k after enlarging them if needed. Fix a length L. Choose j so large that $q \mid \lambda_j$, $\lambda_j > 2k+2$, and $R_j \geq L$. Fix v with $k < v < \lambda_j - k$. At the L sites of the progression starting at $n = v + aR_j\lambda_j$ of step λ_j, with $a \in \{0,1,2\}$, the periodic input always equals $v \bmod q$. The lower-digit window is fixed, since every $v-k, \dots, v+k$ lies in the same lower block. The higher digits vanish, and at time $t = 0, \dots, L-1$ the whole window

is translated by the single ternary label $a + (t \bmod 2)$. For each i this defines a map from three possible translations to two bits. It identifies some pair, necessarily $\{a, a+1\} \pmod 3$ for some a. Choosing this slab separately for each i gives a monochromatic L-term progression for every recoding, with the same difference λ_j.

In the ambient integers take instead

$$\alpha = \sum_{j \geq 1} \frac{b_j/2}{\lambda_{j+1}}.$$

For every j, the terms before j in $\lambda_j \alpha$ are integers and the term at j equals $1/2$. The remaining positive tail is at most $\frac{1}{2} \sum_{s \geq 1} 30^{-s} < 1/10$, since all $b_j \geq 30$. It follows that $\|\lambda_j \alpha\|_{\mathbb{T}} > 2/5$. The half-circle coloring of $n\alpha$ therefore blocks every permitted triple: three representatives in one half-open semicircle would have equal successive differences of absolute value exceeding $2/5$ and total span exceeding $1/2$. Thus even all finite-window binary recodings of a minimal witness, together with arbitrary periodic observations, need not find a binary obstruction that exists on $\mathbb{N}$.

In fact this last example defeats *all measurable binary partitions of its own skew product*, in a stronger quantitative sense. Use the preceding radices $b_j = 3R_j$ and the same ternary digit contributions, take their inverse-limit odometer with Haar measure, and form the cyclic skew product with vertical group $\mathbb{Z}/3\mathbb{Z}$ by the same finite-carry cocycle. Give it product Haar probability μ. It is ergodic by the aligned-block balance proof in Lemma 2.11.2: that proof of ergodicity uses divisibility and balance, not the power-of-two assumption. For any measurable $c : X \to \{0, 1\}$ and any fixed $L \geq 2$, we have

$$\liminf_{j \to \infty} \mu\{x : c(x) = c(T^{\lambda_j} x) = \cdots = c(T^{(L-1)\lambda_j} x)\} \;\geq\; \tfrac{1}{3}.$$

First take c depending only on a fixed finite base residue modulo λ_k and on the vertical label. For $j \geq k$ the residue stays constant throughout. Except for a proportion at most $(L-1)/R_j$ of the initial positions of digit j, no slab boundary is crossed. On the remaining positions the vertical coordinates alternate between a and $a+1$, or between a and $a-1$. Conditional on the base, the uniform initial a gives each of the three unordered pairs with probability $1/3$. Every binary map on three points is constant on at least one pair. The required lower bound follows for cylinder tests. For any measurable c, choose a binary cylinder test disagreeing on a set of measure less than ϵ; invariance bounds the error on the L-site event by $L\epsilon$ uniformly in j. Let $\epsilon \to 0$. For the original ternary vertical observable the monochromatic-triple probability is nevertheless zero for every λ_j, by the two-adjacent-sites argument in each slab. This distinction between tests inside one system and universal tests persists even under a uniform positive-measure hypothesis on the binary side.

The constant $1/3$ is sharp. For the three singleton-versus-complement recodings of a ternary coloring, a progression using at least two colors is monochromatic in at most one of the three recodings. Their monochromatic probabilities therefore sum to at most one at each obstructed difference. Even adding independent copies of the skew product does not remove the example. For a cylinder observable on a finite product of s copies, away from slab boundaries the vertical vector alternates between a and $a + v$ with $v \in \{1, -1\}^s$, while the observed low base residues remain fixed. Conditional on the base vector, a is uniform on $(\mathbb{Z}/3\mathbb{Z})^s$. The cosets of $\langle v \rangle$ are triangles, and any binary coloring is constant on at least one of their three edges. The same lower bound follows, with boundary loss at most $s(L-1)/R_j$. Cylinder approximation extends this to every binary observable on the countable independent product.

2.12 An elementary ternary obstruction that one binary test suffices to block

Proposition 2.12.1. *Let $f : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$ block monochromatic h-term progressions at all $d \in D$. Suppose that for every $d \in D$ the set $\{f(n+d) - f(n) : n \in \mathbb{Z}\}$ omits a residue of $\mathbb{Z}/3\mathbb{Z}$. There exists a single $\theta \in \mathbb{T}$ with*

$$\|d\theta\|_{\mathbb{T}} \geq \frac{1}{3(h-1)} \qquad (d \in D).$$

In particular a single binary rotation coloring blocks D at length $\lceil 3(h-1)/2 \rceil + 1$. For a three-color obstruction to pairs the hypothesis is automatic, and a binary coloring blocks triples.

Proof. Let P be the subgroup generated by D. For each d choose two consecutive integer representatives for the realized residues, using one of $\{-1, 0\}, \{0, 1\}, \{1, 2\}$. Lift $f(n+d) - f(n)$ to the resulting integer $a_d(n)$. The two paths around a d, e square give the same lifted increment: their difference is a multiple of three and has absolute value at most two. Give a reversed edge the negative of its forward increment.

On the Cayley graph of P, the sum for a closed word in the generators is a multiple of three since the residues telescope. It is independent of the starting point in P: move the path by a generator, sum the square identities along the path, and cancel its two endpoint edges. Take a translation-invariant mean on P and put $\beta_d = \mathbb{E}_{n \in P} a_d(n)/3$. For every signed relation $\sum_d k_d d = 0$, averaging its closed path gives $\sum k_d \beta_d \in \mathbb{Z}$. Thus $\beta_d \bmod 1$ defines a character on P. Every character of $P = g\mathbb{Z}$ extends to $\mathbb{Z}$, so write it as $d \mapsto d\theta$.

If the lifts lie in $\{1, 2\}$ the norm of the mean divided by three is at least $1/3$. Otherwise the nonzero lifts have one sign and every run of $h-1$ successive d-edges contains at least one: a run of zeros would make h colors equal. Averaging gives $|\mathbb{E} a_d| \geq 1/(h-1)$. This proves the separation. If k points of a rotation progression all lie in the same half-open semicircle, their representatives in that semicircle have successive differences equal to the same centered increment (the antipodal case cannot have even a pair). Hence $(k-1)\|d\theta\| < 1/2$. The stated length contradicts this inequality. □

The pair case is the winding-number criterion for three-colored abelian Cayley graphs. The proof above allows zero increments, provided each direction omits some increment. For a coloring originally given only on $\mathbb{N}$, one may work with a two-sided forward orbit limit: every forbidden finite pattern stays forbidden there.

The analogous second-difference identity does not follow from the first one. For example, take the period-nine ternary coloring $(0, 0, 2, 0, 0, 1, 1, 2, 2)$. At steps 1 and 3 its triples all use exactly two colors. Let $b_d(n) \in \{1, 2\}$ be the nonzero representative of $\Delta_d^2 f(n) \bmod 3$. Directly

$$(b_1(0), b_1(3), b_1(6)) = (2, 1, 2), \qquad (b_3(0), b_3(1), b_3(2)) = (1, 2, 2).$$

Thus $\Delta_3^2 b_1(0) - \Delta_1^2 b_3(0) = 3$, not zero. This shows exactly why lifting a commuting pair of second differences modulo three need not give commuting integer differences.

2.13 A multiplicative case and the remaining target

For an ultrafilter p on $\mathbb{N}$ write $p \in K_r$ when every member of p is r-large. Equivalently, p contains $R_h(c)$ for every r-coloring and h. Equation (2.1.1) and the ultrafilter lemma show that $D \in \mathcal{R}_2$ precisely when $D \in p$ for some $p \in K_2$. The question is therefore whether $K_2 = \bigcap_r K_r$.

Proposition 2.13.1. *For the usual multiplication on $\beta\mathbb{N}$, $K_rK_s \subseteq K_{rs}$. Every multiplicatively idempotent member of K_2 has only large sets as its members.*

Proof. Let $p \in K_r$, $q \in K_s$, $D \in pq$ and set $A = \{e : e^{-1}D \in q\} \in p$. Here $e^{-1}D = \{b : eb \in D\}$. For prescribed h, by r-largeness and compactness choose a finite $F \subset A$ and N such that every r-coloring of $[N]$ has an h-term progression of difference in F. The set $\bigcap_{e\in F} e^{-1}D$ belongs to q. Given a product coloring with s and r coordinates, take an N-term progression of difference b in this intersection monochromatic in the first coordinate. Use the choice of F on the second coordinates of its terms. The final difference eb belongs to D. Iteration gives the idempotent assertion. This extends the set-product argument of Brown–Graham–Landman [3]. □

This multiplication argument cannot prove the conjecture by making products dense. In fact the closure of K_2K_2 misses a relatively open piece even of $K_\infty := \bigcap_r K_r$. Choose $D = \bigcup_j [X_j, X_j + j] \cap \mathbb{N}$ with $X_{j+1}/X_j \to \infty$, say $X_j = 2^{2^j}$. For distinct fixed $a, b \geq 1$, the intersection $a^{-1}D \cap b^{-1}D$ is finite: the ratios of two large numbers in one interval tend to one, and those from different intervals cannot remain at a fixed ratio greater than one. Thus for a nonprincipal $q \in K_2$ there is at most one a with $a^{-1}D \in q$. No pq with $p, q \in K_2$ contains D; members of K_2 are nonprincipal, as singleton sets are not 2-large. On the other hand D is thick and hence large. For completeness, a thick set contains all nonempty finite sums of an arbitrarily long list of positive integers, by successively fitting each translate of the old finite family into one of its intervals. For a given vertex coloring apply the finite Hales–Jewett theorem to the coloring of words $w \in \{0, \dots, h-1\}^n$ by the color of $1 + \sum w_i u_i$. A combinatorial line gives a monochromatic h-term progression with difference one of the nonempty finite sums. The product-coloring argument implies some $p \in K_\infty$ contains D (use the finite intersection property with all the $R_h(c)$). The clopen set of ultrafilters containing D proves the claimed failure of density.

For an affirmative answer one still needs to cover the *whole* bad-step set of an arbitrary finite coloring by finitely many sets with binary obstructions. A null exceptional set can contain every step of a proposed example. For a negative answer, Theorem 2.2.1 specifies the task: keep the obstructing palette and length fixed while the number of binary tests *and their length* grow together. Neither repeating a three-cycle nor choosing a new polynomial degree for each length accomplishes this.

A concrete candidate for the missing finite packets. Here is one version of the negative direction, motivated by the torus obstructions of Alweiss [2]. Fix $\epsilon = 10^{-3}$. Given $\alpha \in \mathbb{T}^M$, take the positive steps d for which there is an i with

$$\|d\alpha_i - \tfrac{1}{2}\|_{\mathbb{T}} < \epsilon, \qquad \sum_{j\neq i} \|d\alpha_j\|_{\mathbb{T}}^2 < \epsilon.$$

Call this set $S(\alpha)$. A *fixed finite palette* blocks every such step at length three, uniformly in M, α. Indeed set $F(x) = \sum_j \exp(2\pi i x_j)$. Each term of $\Delta_v^2 F(x)$ has modulus $4\sin^2(\pi v_j)$; for $v = d\alpha$ the distinguished term has modulus at least $4\cos^2(\pi\epsilon)$ while the sum of the others' moduli is at most $4\pi^2\epsilon$. Thus $3 < |\Delta_v^2 F(x)| < 5$. Color n by the square of side $1/4$ containing $F(n\alpha)$ modulo $16(\mathbb{Z} + i\mathbb{Z})$. For three points in one square the centered second difference has modulus less than one, and a nonzero lattice correction would have modulus greater than five. This excludes a monochromatic triple.

The unproved lower assertion is that for all m, L some M, α, N make $S(\alpha)$ meet $\bigcap_{i=1}^m R_L^{[N]}(c_i)$ for *every* list of m binary tests on $[N]$. If so, Theorem 2.2.1 gives the desired integer counterexample.

Nothing in the upper argument gives this lower assertion. One reason the half-turn deserves testing is a two-periodic identity: for any binary coloring c of an abelian group and any offset v, some x and $b \in \{0, v\}$ make $c(x + jb + (j \bmod 2)v)$ constant in j. If $c(x) = c(x + v)$ somewhere use $b = 0$; otherwise translation by v flips every bit, and $b = v$ works. The open difficulty is to arrange compatible corrections in the other coordinates for arbitrary tests.

Chapter 3

Full difference sets and finite-length examples

3.1 The reduction

Throughout, $\mathbb{N} = \{1, 2, \ldots\}$, $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and $\|x\|_{\mathbb{T}}$ is the distance to the nearest integer. For a coloring c write

$$R_h(c) = \{d \in \mathbb{N} : \exists a \in \mathbb{N},\ c(a) = c(a+d) = \cdots = c(a+(h-1)d)\}.$$

A set S is r-large if it meets $R_h(c)$ for every r-coloring c and every $h \geq 2$. It is large if this holds for every finite r [3]. Let $\mathcal{L}$ denote the large sets and put

$$\Delta(E) = \{b - a : a, b \in E,\ b > a\}.$$

A Sidon set means an additive B_2 set. Equivalently, its positive differences have unique representations by ordered pairs of distinct elements.

Theorem 3.1.1. *There is one Sidon set $E_* \subset \mathbb{N}$ and a monotone assignment $S \mapsto E_S \subseteq E_*$ with the following properties for every $S \subseteq \mathbb{N}$.*

(i) *For every $r \geq 2$, $S \in \mathcal{R}_r$ if and only if $\Delta(E_S) \in \mathcal{R}_r$.*

(ii) *S is large if and only if $\Delta(E_S)$ is large.*

(iii) *For every $k \geq 2$, S and $\Delta(E_S)$ agree on topological k-recurrence and on measurable k-recurrence.*

The elements of E_* come in pairs. Both sequences of endpoints can be required to grow faster than any prescribed rate from one pair to the next. A single binary coloring blocks all differences between distinct pairs at a fixed length.

We use the coloring formulation of topological k-recurrence. It requires a monochromatic $(k+1)$-term progression with an allowed difference in every finite coloring [10]. Measurable k-recurrence means that for every invertible probability-preserving T and every set A of positive measure, some allowed d satisfies $\mu(\bigcap_{t=0}^{k} T^{-td} A) > 0$. The restriction $k \geq 2$ matters. Every infinite full difference set is a set of measurable and topological one-recurrence.

Corollary 3.1.2. *The finite-partition implication in the question is equivalent to the same implication restricted to full positive difference sets $\Delta(E)$ of subsets of the single set E_*.*

If $E = \{e_1, e_2, \ldots\}$ is infinite and μ is a finite positive measure on $\mathbb{T}$, then

$$\frac{1}{n^2} \sum_{i,j=1}^{n} \widehat{\mu}(e_i - e_j) = \int \left| \frac{1}{n} \sum_{i=1}^{n} \exp(2\pi i e_i t) \right|^2 d\mu(t) \geq \mu(\{0\}).$$

For each fixed K, at most $(2K+1)n$ pairs have $|e_i - e_j| \leq K$. Consequently,

$$\limsup_{\substack{d \to \infty \\ d \in \Delta(E)}} \operatorname{Re} \widehat{\mu}(d) \geq \mu(\{0\}).$$

In particular, a negative answer to the question would persist even after requiring this nice van der Corput inequality of D.

3.2 Choosing the pairs

We use the elementary equivalence

$$S \in \mathcal{R}_r \quad \Longleftrightarrow \quad S \cap \bigcap_{i=1}^{m} R_h(c_i) \neq \varnothing \quad \text{for every } m, h \text{ and every list of } r\text{-colorings } c_i. \tag{3.2.1}$$

If the intersection is empty, assign each $d \in S$ to its first failed test. Conversely, take an obstructing coloring for each cell of a failed finite partition, then take the maximum of their obstructing lengths. For each fixed m, h, compactness of $([r]^{\mathbb{N}})^m$ gives a finite $F \subset S$ that works for every list whenever the right-hand side holds. Indeed, any particular solution uses only finitely many color values. The same finite-witness observation holds for one finite coloring with a fixed number of colors. The starting points in (3.2.1) can depend on i. The use of separated scales below builds on the finite-packet method in Farhangi's thesis [6, Chapter 5].

Fix irrationals $\beta, \tau \in (0,1)$. There are H and $\eta > 0$ such that, whenever $\|a - \tau\|_{\mathbb{T}} < \eta$, the points

$$at^2 + bt + c \pmod 1, \qquad 0 \leq t < H, \tag{3.2.2}$$

meet the interiors of both half-circles for every $b, c \in \mathbb{T}$. For $a = \tau$ this follows pointwise in (b, c) from Weyl equidistribution. The two visits are open conditions. A finite subcover of the compact parameter space $\{\tau\} \times \mathbb{T}^2$ gives H and then an open tube about that space. Shrink η so that this tube of leading coefficients stays a positive distance from 0. In particular the bit $n \mapsto 1_{[0,1/2)}(\beta n^2 \bmod 1)$ blocks all positive steps u with $\|\beta u^2 - \tau\|_{\mathbb{T}} < \eta$.

Choose place values λ_j with $\lambda_j \mid \lambda_{j+1}$. At stage j introduce the j offsets

$$t_{j,s} = \lambda_j s, \qquad 1 \leq s \leq j.$$

Enumerate the pairs of indices (j, s) in stage order, writing t_u for the corresponding offset. Its two endpoints will be B_u and $B_u + t_u$. We choose λ_1 freely and require

$$b_j = \lambda_{j+1}/\lambda_j > 2^{j+4}(j+1)^2. \tag{3.2.3}$$

In particular all offsets are distinct. For a fixed $0 < \varepsilon < \min(\eta, \|\tau\|_{\mathbb{T}})/40$ impose throughout

$$\begin{aligned} \|\beta t_u t_v\|_{\mathbb{T}} &< \varepsilon \quad \text{for all } u, v, \\ \|\beta B_u t_v\|_{\mathbb{T}} &< \varepsilon \quad \text{for all } u, v, \\ \|\beta B_u B_v\|_{\mathbb{T}} &< \varepsilon \quad \text{for } u \neq v, \\ \|\beta B_u^2 - \tau/2\|_{\mathbb{T}} &< \varepsilon \quad \text{for all } u. \end{aligned}$$

To choose λ_j, take an arbitrarily large multiple of its predecessor with $\|\beta\lambda_j\|_{\mathbb{T}}$ so small that multiplication by s times any previous base or offset still gives an error below ε. Also require $\|\beta\lambda_j^2\|_{\mathbb{T}} < \varepsilon/j^2$ to handle products of two offsets at this stage. These two requirements follow from joint equidistribution of $(\beta n, \beta n^2)$ on any progression of multiples. After fixing the offsets of this stage, choose the bases one at a time. Use the same equidistribution to make $\|\beta B_u\|_{\mathbb{T}}$ sufficiently small for all currently relevant products and βB_u^2 sufficiently close to $\tau/2$. Each choice can exceed any lower bound imposed by earlier choices.

Write $E_* = \bigcup_u \{B_u, B_u + t_u\}$. Every difference d between different pairs satisfies

$$\|\beta d^2 - \tau\|_{\mathbb{T}} < 16\varepsilon. \tag{3.2.4}$$

Indeed d, up to sign, is $B_u + et_u - B_v - e't_v$ with $u \neq v$ and $e, e' \in \{0, 1\}$. The squared bases contribute τ up to error 2ε. Every other monomial is controlled by the preceding bounds. The sum of the absolute coefficients in the square is at most 16. Consequently the binary coloring in (3.2.2) blocks all the cross differences. These differences are disjoint from the matching offsets, since $\|\beta t_u^2\|_{\mathbb{T}} < \varepsilon$.

To ensure Sidonicity, require at each insertion that $B_u > 10(\max E_{\text{old}} + t_u + 1)$, taking the maximum of the empty set to be zero. The offset t_u is different from all old matching offsets and all old cross differences. The new differences from old points are larger than all old differences and larger than t_u. A collision between $B_u - x$ and $B_u + t_u - y$ would give $t_u = y - x$ in the old difference set. Within either of these two lists the differences are distinct. Induction gives the asserted Sidon property. Extra growth conditions on the bases can be imposed here.

For $S \subseteq \mathbb{N}$, retain exactly the pairs at indices (j, s) with $s \in S$. Call their union E_S. The matching part of its full difference set is precisely

$$M(S) = \bigcup_{j \geq 1} \lambda_j (S \cap [1, j]). \tag{3.2.5}$$

The remaining differences all satisfy (3.2.4).

3.3 Binary tests and finite colorings

If S satisfies (3.2.1), choose a finite witnessing F at the desired m, h and take $j \geq \max F$. Pull the given colorings back by $n \mapsto \lambda_j n$. A common difference in F for the pulled-back tests gives a common difference in $\lambda_j F \subset M(S)$ for the original tests. The same argument, with one finite palette, transfers largeness and each fixed order of topological recurrence in the forward direction.

For the reverse implication, suppose a coloring $c : \mathbb{N} \to \mathbb{Z}/r\mathbb{Z}$ blocks length h for all steps in $A \subseteq S$. Use $c_0(t) = c(t + 1)$ on nonnegative integers. Include an initial dummy radix λ_1 below the listed place values, and write $z_j(n) \in [0, b_j - 1]$ for the digit at place λ_j. Color n by

$$g_c(n) = \sum_{j \geq 1} \big(c_0(z_j(n)) - c_0(0)\big) \pmod{r}. \tag{3.3.1}$$

Only finitely many summands are nonzero. For a step $\lambda_j s$ with $s \leq j$, a run of $2h$ sites crosses at most one jth digit boundary as soon as $b_j > 2hj$. At least h consecutive sites avoid that boundary. On them every digit except the jth is fixed, and the jth digit increases by s at each step. Hence g_c blocks length $2h$ for the corresponding matching steps whenever $s \in A$. There are only finitely many offsets at the exceptional early stages. Their union admits a periodic binary obstruction, for example by using one test for each offset.

If a finite partition of S has no r-large cell, apply (3.3.1) to the obstructing coloring for each cell. Together with the early-step tests and the cross-difference bit, this gives a finite partition of $\Delta(E_S)$ with no r-large cell. This proves part (i). If S fails largeness, one finite coloring blocks it. The product of the matching and cross tests gives one finite coloring blocking the full difference set, proving part (ii).

Preserving the exact length

Let $h = k + 1 \geq 3$ be fixed. Two copies of (3.3.1) allow us to avoid a carry for the whole progression. Regard an ordinary integer as a point of the mixed-radix odometer. Take $y = 0$ in the first copy

and take y with jth digit $\lfloor b_j/2 \rfloor$ in the second copy, setting the dummy digit to zero. The two observations are

$$g_{c,y}(n) = \sum_{j \geq 1} \left(c_0(z_j(y+n)) - c_0(z_j(y)) \right) \pmod{r}.$$

Adding a fixed integer to either of these odometer points changes only finitely many digits. The two starting jth digits differ, modulo b_j, by $\lfloor b_j/2 \rfloor$ or $\lfloor b_j/2 \rfloor + 1$, owing to the possible carry from below. The arc of starting digits that would wrap on an h-term progression of step $\lambda_j s$ has length at most $(h-1)j$. When $b_j > 4hj + 4$, at least one chart avoids this arc. Its h output colors cannot all be equal. The finitely many early steps can be blocked even at length two by the finite coloring $n \bmod Q$ with Q larger than those steps.

Finally a fixed finite coloring blocks all cross differences at length three. Color n by a sufficiently short interval containing $\beta n^2/2 \bmod 1$. If three sites lie in one interval of width w, the second difference of their real representatives has absolute value less than $2w$. Modulo one it equals βd^2, which stays a fixed distance from zero by (3.2.4). Taking the product of these finite colorings proves the reverse direction of the topological assertion in part (iii).

3.4 Measurable recurrence

For the forward direction at a fixed order k, let $\delta > 0$. If S is measurably k-recurrent, some finite $F \subseteq S$ works for all systems and all sets of measure at least δ. To see this, code visits to the set by a shift-invariant probability on $\{0,1\}^{\mathbb{Z}}$. The invariant probabilities giving the symbol 1 measure at least δ form a compact space. For each d, positivity of the measure of the $(k+1)$-site all-ones cylinder is an open condition. Recurrence supplies an open cover, hence a finite subcover. Applying it to T^{λ_j} for $j \geq \max F$ proves recurrence for $M(S)$.

For the reverse direction let (X, μ, T) and A witness failure of measurable k-recurrence for S. Let Y be the preceding odometer with Haar probability and addition by one. Define an integer cocycle

$$\rho(y) = \sum_{j \geq 1} \left(z_j(y+1) - z_j(y) \right).$$

The sum is finite almost everywhere. Give it any value at the single point with all digits maximal. The transformation $(y, x) \mapsto (y+1, T^{\rho(y)}x)$ preserves product measure and is invertible modulo null sets. For each fixed positive time its cocycle sum is the sum of the changes in the digits, by telescoping outside a null set.

Choose J so large that $kj/b_j < 1/2$ for $j \geq J$ and define

$$C = \{ y : y = 0 \bmod \lambda_J, \quad z_j(y) \leq b_j - 1 - kj \text{ for every } j \geq J \}.$$

The measure of C is $\lambda_J^{-1} \prod_{j \geq J} (1 - kj/b_j) > 0$ by (3.2.3). On C, for $j \geq J$, $s \leq j$ and $0 \leq t \leq k$, adding $t\lambda_j s$ changes only the jth digit, by ts. Thus $C \times A$ has zero $(k+1)$-fold intersection for every matching step $\lambda_j s$ with $s \in S$. For $j < J$ the positive matching steps are smaller than λ_J and the residue condition on C already blocks pairs.

To deal with cross differences, take the product with the Haar-preserving skew shift $(u, v) \mapsto (u + \beta, v + u)$ on $\mathbb{T}^2$. On three sites separated by d, the second difference of v is βd^2 modulo one. Requiring v to lie in a short interval blocks every cross difference, just as in the preceding section. This product gives a positive-measure witness for $\Delta(E_S)$ at the original order k, proving Theorem 3.1.1.

Corollary 3.4.1. *If the question has a negative answer, then for every sufficiently large k there is a counterexample of the form $\Delta(E)$, $E \subset E_*$, that is measurably k-recurrent and fails topological $(k+1)$-recurrence.*

Proof. Let $D_0 \in \mathcal{R}_2$ have a finite coloring obstructing length h_0, and take $k \geq \max(2, h_0 - 2)$. For irrational α, the set $A_k = \{d : \|\alpha d^{k+1}\|_{\mathbb{T}} \geq 1/4\}$ is measurably k-recurrent and fails topological $(k+1)$-recurrence by Frantzikinakis, Lesigne and Wierdl [7, Theorem A and the following remark]. The union $D_0 \cup A_k$ belongs to $\mathcal{R}_2$ and is measurably k-recurrent by upward closure. The product of the two obstructing colorings blocks length $k+2$ on the union, since $k+2 \geq h_0$. Now apply Theorem 3.1.1. □

3.5 A five-color obstruction that survives every binary four-term test

The following example concerns ordinary integers. Its hypotheses on the number of tests are unrestricted. The binary progression length is four.

Theorem 3.5.1. *For every finite list of binary colorings $c_1, \dots, c_m$ there are an ordinary Bohr neighborhood B of zero and a set Z of zero upper Banach density with*

$$(B \cap \mathbb{N}) \setminus Z \subseteq \bigcap_{i=1}^{m} R_4(c_i).$$

Consequently, for every irrational α, the set

$$S = \{d \geq 1 : \|\alpha d^2\|_{\mathbb{T}} > 2/5\}$$

has the following two properties. Every finite partition of S has one cell that meets $R_4(c)$ for every binary c. There is a five-coloring f of $\mathbb{N}$ for which $R_3(f) = \mathbb{N} \setminus S$.

Proof. First fix one binary coloring and take a stationary sign process $s \in \{-1, 1\}$ from a shift-invariant probability on a forward orbit-limit subshift of the coloring. Any cylinder of positive measure supplies a finite pattern occurring in the original coloring. Write $s_j = s \circ T^{jd}$ and $\rho(kd) = \mathbb{E}(s_0 s_k)$. Let $M_4(d)$ be the probability that four consecutive signs at step d agree. Expanding the indicator gives

$$8M_4(d) = 1 + 3\rho(d) + 2\rho(2d) + \rho(3d) + \gamma(d), \qquad \gamma(d) = \mathbb{E}(s_0 s_1 s_2 s_3).$$

We apply at fixed d the overlapping-quadruple parity observation used by Wolf [18, Section 2] for averages over finite cyclic groups. The two signs $V = s_0 s_1 s_2 s_3$ and $W = s_1 s_2 s_3 s_4$ have the same mean $\gamma(d)$, and $VW = s_0 s_4$. Since $(1+V)(1+W) \geq 0$, we have $1 + 2\gamma(d) + \rho(4d) \geq 0$. Therefore

$$8M_4(d) \;\geq\; \frac{1 - \rho(4d)}{2} + 3\rho(d) + 2\rho(2d) + \rho(3d). \tag{3.5.1}$$

Let a be the total mass of the atomic part of the spectral probability measure of s. For each $1 \leq k \leq 4$ the continuous part has kdth Fourier coefficient smaller than $1/100$ off a set of zero upper Banach density. For clarity, the mean square over any interval of N consecutive d is bounded by

$$\iint \left| N^{-1} \sum_{j=1}^{N} \mathrm{e}^{2\pi i k j (x-y)} \right| d\sigma_{\mathrm{c}}(x)\, d\sigma_{\mathrm{c}}(y) \;\longrightarrow\; 0.$$

The bound is independent of the interval's starting point and the exceptional set $k(x-y)=0$ has product measure zero. Approximating the atomic part in total variation by finitely many atoms gives a Bohr neighborhood on which all four atomic Fourier coefficients are within $1/100$ of a. Thus in (3.5.1) the right side is within $6.5 \cdot 2/100$ of $1/2 + (11/2)a$ off the stated exceptional set. It is strictly positive there. Intersect the Bohr neighborhoods and unite the exceptional sets for the finitely many tests.

The set S has positive lower density inside every ordinary Bohr neighborhood. Indeed $d \mapsto (d\theta_1, \dots, d\theta_t, \alpha d^2)$ equidistributes on the product of the linear orbit closure with $\mathbb{T}$. For a character using the last coordinate this follows from Weyl's theorem for an irrational quadratic leading coefficient, and the other characters give ordinary rotation equidistribution. So S meets $(B \cap \mathbb{N}) \setminus Z$. The fixed-length form of (3.2.1) proves the partition assertion. Finally color n by the fifth of the circle containing $\alpha n^2/2$. For three points in one half-open fifth, the second difference of the representatives has absolute value strictly less than $2/5$. Modulo one this second difference is αd^2. Conversely, let δ be the centered representative of αd^2 with $|\delta| < 2/5$. The real triple $1/10 + \delta/4,\ 1/10 - \delta/4,\ 1/10 + \delta/4$ lies strictly in the first fifth and has second difference δ. The initial value and first increment $(\alpha n^2/2, \alpha dn + \alpha d^2/2)$ are jointly equidistributed modulo one as n varies, by Weyl's criterion. Approximating the initial value and increment of this triple proves $d \in R_3(f)$. Equality at $2/5$ cannot occur for irrational α. □

Corollary 3.5.2. *There is an infinite Sidon set E such that every finite partition of $\Delta(E)$ has a cell forcing a monochromatic four-term progression in every binary coloring, while $\Delta(E)$ fails topological two-recurrence.*

Proof. Apply the construction to the set S of Theorem 3.5.1. The forward finite-witness argument from Section 3.3 also works with the binary length fixed at four. Failure of topological two-recurrence transfers with the triple length preserved by Theorem 3.1.1(iii). □

The use of stationarity matters in (3.5.1). The product of four consecutive signs is not generally a positive-definite correlation sequence. The inequality replaces that product by an ordinary two-point correlation at distance $4d$.

Pair correlations alone cannot give the analogous pointwise lower bound for five sites. To see this, take independent fair signs X_j, $j \in \mathbb{Z}$, and one further independent fair sign σ. Make successive two-site blocks

$$(X_j,\ \sigma(-1)^j X_j)$$

and randomize the parity of the block origin uniformly. The resulting process is stationary, since a shift by two changes the law of neither the X_j nor σ. Distinct signs in one block have zero correlation by averaging over σ. Signs from distinct blocks have zero correlation by independence of their X_j. Yet every five-site interval contains two complete consecutive blocks, whose sign products are opposite. So one of the two blocks changes sign, and a five-term constant run of step one never occurs.

Lemma 3.5.3. *For every $\varepsilon > 0$ and every real α, finitely many binary colorings jointly block the set $\{d : \|\alpha d^2\|_{\mathbb{T}} \geq \varepsilon\}$ at length seven.*

Proof. We first find an open interval J around $a = 1/14$ such that seven values $at^2 + bt + c$, $0 \leq t \leq 6$, cannot all lie in one closed semicircle, for any b, c. If they did so at $a = 1/14$, lift them to a real interval of length $1/2$ and let w_t be the six successive real increments. They lie in $[-1/2, 1/2]$ and $w_t = w_0 + t/7 \pmod 1$. If $w_t \bmod 1$ belongs to $(5/28, 10/28)$, the centered lifts have $w_t + w_{t+1} > 1/2$. If it belongs to $(14/28, 19/28)$, they have $w_t + w_{t+1} < -1/2$. Either

inequality contradicts the common interval. The translates of these two open arcs by $-t/7$ for $0 \le t \le 4$ cover the circle. Indeed their unions before reducing modulo one are $(-11/28, 10/28)$ and $(-2/28, 19/28)$. This proves the assertion at $1/14$, and compactness in b, c and the position of the semicircle gives the interval J.

Choose K such that every finite subgroup of the circle of order greater than K meets J, and let $Q = K!$. If $\|Qx\|_{\mathbb{T}} \ge \varepsilon$, the positive multiples of x meet J, since x is irrational or has order greater than K. The compact set of these x is therefore covered by finitely many open conditions $mx \in J$. Use the corresponding binary colorings $n \mapsto 1_{[0,1/2)}(m\alpha n^2/Q \bmod 1)$. For each allowed d, apply the cover to $x = \alpha d^2/Q$. One of the tests then has leading coefficient in J on every progression of step d. □

Thus for this particular f, the least L for which some finite family of binary colorings satisfies $\bigcap_i R_L(c_i) \subset R_3(f)$ lies between five and seven, inclusive.

3.6 An affirmative result at every progression length

The rotation-window argument in the earlier finite-partition note gives the following consequence for extensions. A single small fiber suffices.

Theorem 3.6.1. *Let $D \in \mathcal{R}_r$ with $r \ge 2$. For each i in a finite list, let (X_i, T_i) be a minimal compact metric system with a rotation factor $\pi_i : X_i \to Y_i$ on a compact abelian group. Suppose that some fiber of each π_i has at most r points. For any nonempty open $U_i \subseteq X_i$ and any h, there is one $d \in D$ and points $x_i \in X_i$ such that $T_i^{jd} x_i \in U_i$ for every i and $0 \le j < h$.*

Proof. For each point of the small fiber choose a nonnegative time sending it into U_i. The corresponding at most r inverse translates of U_i cover that fiber. By compactness they cover the full inverse image of a neighborhood V_i of the base point. Characters separate the points of a compact abelian group, so after shrinking the window it contains the pullback of a relatively open neighborhood from a closed subgroup $K_i \subseteq \mathbb{T}^{s_i}$. Write its rotation vector as θ_i. For an orbit in X_i, color each visit to this smaller base window by one of the at most r selected times sending that orbit point into U_i.

To encode these partially defined colorings on all of $\mathbb{N}$, take a fine rectangular grid on each ambient torus. Finitely many nonnegative inverse translates of the open window cover K_i by minimality. By the Lebesgue number lemma, the grid can be chosen so that for each cell C meeting K_i there is a fixed $t_C \ge 0$ sending $C \cap K_i$ into the open window. At time n in cell C, give n the visit color at $n + t_C$.

Take $L = (1 + \max_i s_i)h$. In addition to these encoded colorings, impose an ordinary Bohr condition on d requiring $L\|d\theta_{i,j}\|_{\mathbb{T}}$ to be less than one grid spacing in each coordinate. It can be imposed in (3.2.1) with finitely many additional binary tests. Indeed, if a circle progression of length K with increment $d\theta$ lies in one half-open semicircle, its successive differences in that semicircle all have the same centered representative. It follows that $(K-1)\|d\theta\|_{\mathbb{T}} < 1/2$. Use K large enough in each auxiliary half-circle test and increase the common target length if necessary.

On the first L sites of each resulting progression, every coordinate of the corresponding rotation crosses at most one grid boundary. Consequently some run of h consecutive sites lies in one cell. On this run the first time correction t_C is constant, and the common color makes the second time correction constant as well. Applying both corrections gives the required progression in U_i, with its difference unchanged. □

For pairs there is a further consequence. Every finite family of three-color pair tests has a common good difference in $D \in \mathcal{R}_2$. Indeed, for a three-coloring of $\mathbb{N}$, its set of good pair differences contains $\{d : \|d\alpha\|_{\mathbb{T}} < 1/3\}$ for some α by Proposition 2.12.1 with $h = 2$, applied to the complement of $R_2(f)$. Intersect these neighborhoods over the family and use the auxiliary half-circle tests above. This argument relies on the three-color pair criterion and does not extend it to triples.

3.7 Testing the half-turn construction

Here is an independent partial result for the high-dimensional candidate. It applies to arbitrary finite lists of binary triple tests, even to spectral tests with a fixed atom at zero. All dimension bounds below are independent of the particular tests.

Lemma 3.7.1. *Fix $r > 0$ and $\delta > 0$. For all sufficiently large M, no positive measure ν on $\mathbb{Z}^M$ of mass at most one and with $\nu(\{0\}) \geq \delta$ has Fourier transform vanishing throughout*

$$O_M = \bigcup_{i=1}^{M} \{v \in \mathbb{T}^M : \operatorname{dist}_2(v, e_i/2) < r\}.$$

Here dist_2 *is the quotient Euclidean distance.*

Proof. If there were counterexamples along $M \to \infty$, let I be a uniform random coordinate and let W have independent centered real Gaussian coordinates of variance t^2/M, independently of I. For $0 < t < r$, the probability that $\|W\|_2 \geq r$ tends to zero. The Fourier transform has modulus at most one. Its expectation at $e_I/2 + W$ must therefore tend to zero. Integrating each character gives

$$\int a_M(k) \exp(-2\pi^2 t^2 u_M(k))\, d\nu(k) \longrightarrow 0, \quad a_M(k) = \frac{1}{M}\sum_i (-1)^{k_i}, \qquad u_M(k) = \frac{1}{M}\sum_i k_i^2.$$

Pass to a weak limit σ of the image measures on the compact space $[-1, 1] \times [0, \infty]$. For each $s > 0$ extend $\exp(-su)$ by zero at infinity. The function $\int a \exp(-su)\, d\sigma(a, u)$ vanishes on a nonempty open interval, hence on $(0, \infty)$ by analyticity of the Laplace transform. Letting $s \to \infty$ gives $\int_{u=0} a\, d\sigma = 0$. But $1 - a_M \leq 2u_M$, so $a = 1$ wherever $u = 0$ on the support of the limit. The atom at $(1, 0)$ has mass at least δ by the closed-set inequality for weak convergence. This is a contradiction. □

Corollary 3.7.2. *Choose $r = 10^{-3}$. For every m, take M sufficiently large and let $1, \theta_1, \ldots, \theta_M$ be rationally independent. The set $D_M = \{d \geq 1 : d\theta \in O_M\}$ meets every intersection of m binary triple tests. A finite palette of size independent of m and M blocks every D_M-triple.*

Proof. For the stationary sign process from a binary coloring the monochromatic triple probability is

$$M_3(d) = \big(1 + 2\rho(d) + \rho(2d)\big)/4.$$

It is the Fourier transform of a positive probability on $\mathbb{T}$ whose atom at zero is at least $1/4$. Taking the product of m such probabilities gives $\widehat{\mu}(d)$ for a probability μ with $\mu(\{0\}) \geq 4^{-m}$, by convolution. We use the convention $\widehat{\mu}(d) = \int \exp(2\pi i d t)\, d\mu(t)$. If there were no common good difference in D_M, then $\widehat{\mu}(d) = 0$ for every $d \in D_M$. Define a positive measure on $\mathbb{Z}^M$ by $\nu(\{k\}) = \mu(\{k \cdot \theta\})$. Independence ensures distinct atoms for distinct k. For every continuous g supported in O_M one has

$$0 = \lim_{N\to\infty} \frac{1}{N} \sum_{d=1}^{N} \widehat{\mu}(d)\, g(d\theta) = \int_{\mathbb{T}^M} \widehat{\nu}(v) g(v)\, dv.$$

The identity follows first for the unconditioned average against a character by geometric summation and dominated convergence, then for continuous g by uniform approximation. Thus $\widehat{\nu}$ vanishes on O_M, contradicting Lemma 3.7.1 with $\delta = 4^{-m}$.

For the obstructing coloring put $P(x) = \sum_{j=1}^{M} \exp(2\pi i x_j)$. For $v \in O_M$ the distinguished summand in $P(x+2v) - 2P(x+v) + P(x)$ has modulus at least $4\cos^2(\pi r)$. The sum of the absolute values of the other summands is at most $4\pi^2 r^2$. Consequently the second difference has modulus between 3 and 5. Color n by the square of side $1/4$ containing $P(n\theta)$ modulo $16(\mathbb{Z} + i\mathbb{Z})$. If three colors agreed, the second difference would be within 1 of the lattice $16(\mathbb{Z} + i\mathbb{Z})$, contradicting those bounds. This last calculation follows the half-turn construction of Alweiss and Green [2, 9]. □

Theorem 3.7.3. *Fix $q > L \geq 2$ and $M \geq q + 1 + 2q(4L + 4)$. In $(\mathbb{Z}/2q\mathbb{Z})^M$ take the direction*

$$v = (q, \underbrace{1, \ldots, 1}_{q}, 0, \ldots, 0).$$

Every permutation-invariant binary coloring has a monochromatic L-term progression of this same direction, with the starting point allowed to depend on the coloring. In $\mathbb{T}^M$ the scaled direction $v/(2q)$ lies at distance $1/(2\sqrt{q})$ from the first coordinate half-turn.

Proof. Such a coloring depends only on the histogram H of the $2q$ symbols. Write e_s for the unit count at symbol s, $Z_s = e_{s+q} - e_s$, and $W = \sum_{s \text{ odd}} e_s - \sum_{s \text{ even}} e_s$. Suppose the coloring has no monochromatic progression of the indicated length. Put $U_s^{\pm} = Z_s \pm W$.

For any histogram with at least two copies of each symbol, place s at the first coordinate. Fill the q moving coordinates with one copy of each even symbol. All other coordinates are fixed. The progression alternates between the starting histogram and the histogram obtained by adding U_s^{+}. Thus that move flips the bit. Starting with all odd symbols proves the same for U_s^{-}. The inverse moves also flip the bit whenever both endpoints have at least two copies of each symbol.

Now start the marked coordinate at zero and the q moving coordinates at $0, 1, \ldots, q-1$. Freeze at least $4L + 4$ copies of each symbol, placing any extra coordinates at zero. At time $0 \leq t < L$, telescoping the counts shows that the displacement from H_0 is

$$H_t - H_0 = \sum_{s=0}^{t-1} Z_s + (t \bmod 2) Z_0.$$

This displacement can be obtained in exactly $2t$ bit flips. First add U_s^{+} for $s = 0, \ldots, t-1$. Then repeat the pair of moves $-U_0^{+}$, U_0^{-} a total of $\lfloor t/2 \rfloor$ times, canceling $2W$ per pair. For odd t add one final U_0^{-}. Each move changes any one count by at most two, so the intermediate histograms still have at least four copies of every symbol. The total number of flips is even. Hence every H_t has the color of H_0, a contradiction. □

The finite-palette obstruction in Corollary 3.7.2 is itself permutation-invariant and applies to this grid for sufficiently large q. The binary assertion here is exact on the grid. Arbitrary colorings pulled back from $\mathbb{N}$ need not respect coordinate permutations.

3.8 Where the unrestricted problem remains

The desired implication is equivalent to a statement about an arbitrary finite coloring f and length h. There must be a finite binary list $c_1, \ldots, c_m$ and a length L such that $\bigcap_i R_L(c_i) \subset R_h(f)$. Indeed,

if this fails, $\mathbb{N} \setminus R_h(f)$ itself lies in $\mathcal{R}_2$ by (3.2.1) and fails largeness. Conversely, the inclusion and (3.2.1) give an allowed good difference for f. The full difference-set reduction permits all progression lengths, but assumes an initial counterexample in its negative application. Theorem 3.5.1 gives an actual fixed set on the integers and allows any number of binary tests, but covers length four. Lemma 3.5.3 gives a finite list that already fails at length seven. The theorem of Alweiss and Farhangi [1] also rules out 2-largeness for this set because it misses a nil-Bohr neighborhood. Corollary 3.7.2 verifies the triple case for a different construction that leaves the long-progression question available as the dimension grows.

The finite-packet criterion underlying Theorem 2.2.1 specifies exactly what that construction would need to achieve. A negative answer is equivalent to the existence of fixed integers q, h such that for every m, L there is a finite $F \subset \mathbb{N}$ satisfying both conditions below.

(a) Every list of m binary colorings admits a common difference in F at length L, with starting points allowed to depend on the coloring.

(b) Some q-coloring blocks all of F at length h.

Indeed, a counterexample supplies the finite sets F by compactness in (3.2.1). Conversely, take packets F_j for $m = L = j$ and corresponding blocking colorings c_j. Choose divisible place values λ_j with $b_j > 2h \max F_j$, increasing the radices as needed. Their union $D = \bigcup_j \lambda_j F_j$ satisfies (3.2.1) by pullback. In (3.3.1) use c_j on the jth digit. On any progression of $2h$ sites at step $\lambda_j s$ with $s \in F_j$, at least h consecutive sites avoid a carry and are blocked by c_j. This one q-coloring witnesses failure of largeness. The unproved requirement is that q, h stay fixed as both m and L grow.

Chapter 4

The exact partition hierarchy and quadratic thresholds

4.1 An exact hierarchy by number of cells

For a coloring c of a set of positive integers let $R_h(c)$ consist of the positive differences of its monochromatic arithmetic progressions of length h. A set D is r-large if it meets $R_h(c)$ for every r-coloring of $\mathbb{N}$ and every $h \geq 2$. It is large if it is r-large for every finite r. A coloring with fewer than r colors is allowed.

Theorem 4.1.1. *For every $k \geq 1$ there is a set $D_k \subset \mathbb{N}$ such that for any finite list $r_1, \dots, r_m \geq 2$ the following are equivalent.*

(i) *Every partition $D_k = A_1 \sqcup \cdots \sqcup A_m$ has an r_i-large cell A_i for some i.*

(ii) $\sum_{i=1}^{m}(r_i - 1) < k$.

For example, with $k = m + 1$ the set passes every m-cell binary test at every length and fails an $(m+1)$-cell binary test. It is also $(m+1)$-large and fails $(m+2)$-largeness. Thus fixing the number of partition cells in the question, however large, gives a false implication.

We use the elementary equivalence between (i) and

$$D_k \cap \bigcap_{i=1}^{m} R_h(c_i) \neq \varnothing \quad \text{for every } h \text{ and every list of colorings } c_i : \mathbb{N} \to [r_i]. \tag{4.1.1}$$

Indeed, a failure of (4.1.1) assigns each d to an index whose test it fails. Conversely, choose one obstructing coloring for each cell and take the maximum of their forbidden lengths. The starting points for distinct tests are independent.

We strengthen the single-palette construction of Appendix A. Two changes are needed. In the positive direction the polynomials in Qiu's theorem give independent starting points with a common difference. In the negative direction we keep the partition into layers and construct an obstruction for each union of layers.

For a niltranslation T on a compact finite-dimensional nilmanifold Y, a point y and a neighborhood V of y, put $N(y, V) = \{n \geq 1 : T^n y \in V\}$. For each k we prove the following two assertions simultaneously.

(a) If $\sum_i (r_i - 1) \leq k - 1$, arbitrary r_i-colorings of $N(y, V)$ have monochromatic h-term progressions there with a common difference in D_k. With no colorings prescribed, an uncolored progression in the window is required.

(b) There is a partition $D_k = C_1 \sqcup \cdots \sqcup C_k$ and a finite H_k such that, for every nonempty $J \subseteq \{1, \dots, k\}$, some $(|J| + 1)$-coloring blocks all H_k-term progressions with difference in $\bigcup_{j \in J} C_j$.

A one-color coordinate in (a) may always be dropped, since a progression in the window already meets it. If no other tests remain we use the uncolored case.

The finite-dimensional input

We use the finite-stage construction, torus geometry, and palette estimates proved in Sections A.3–A.5. Given k, put $Q = 100(k+1)$, $\delta = Q^{-1}$ and $\eta = (10^6 Q^2)^{-1}$. Choose $\alpha \in \mathbb{T}^M$ with $1, \alpha_1, \dots, \alpha_M$ rationally independent. Let O_M consist of vectors with centered coordinates θ_j for which some i satisfies

$$|\theta_i + \delta| < \eta, \qquad |\theta_j| < \eta \ (j \neq i), \qquad \sum_{j \neq i} |\theta_j| < 2\delta.$$

Write $G_M = \{d : d\alpha \in O_M\}$ and $\phi(n) = \sum_j \mathrm{Re}(\exp(2\pi i n \alpha_j) - 1)$. We use two facts.

(1) For fixed h, C, k and all sufficiently large M, every pointed nilsystem of dimension at most C has an $m \in G_M$ with $T^{tm} y \in V$ for $0 \leq t < h$. The bound on M is independent of V and of the translation.

(2) For $2 \leq q \leq k+1$, the integer clock $J_q(n) = \lfloor q\phi(n) \rfloor$ visits all residues modulo q on each $2Q$-term progression of difference in G_M. If instead $\sum_j \|d\alpha_j\|_{\mathbb{T}} < 1/(4\pi(k+1)H)$, the first H clock values on a progression of difference d belong to two consecutive integers.

For (1), the joint orbit closure of $(y, \dots, y, 0)$ under $(1, T, \dots, T^{h-1}, R_\alpha)$ has a connected component whose pure torus directions form a subtorus of codimension at most hC. Lemma A.4.1, applied as in Lemma A.4.2 makes such a subtorus meet O_M. For (2), the ℓ^1 bound on the coordinate increments gives $|q\phi(n+d) - q\phi(n)| < 1$ on G_M. The distinguished coordinate turns through approximately one revolution in Q sites. Its second difference over a half turn, at a suitable starting site, has real part greater than 3.9 after subtracting the errors of the other coordinates. The error estimate uses $\sum_{j \neq i} \theta_j^2 \leq \eta \sum_{j \neq i} |\theta_j|$. Thus the range of ϕ on $2Q$ sites exceeds 1.95. The integer clock takes at least q consecutive values. For small torus motion the range of $q\phi$ on H sites is less than $1/2$ by the triangle inequality. These estimates also explain why the same auxiliary torus works for all the smaller palettes in (b).

Independent starting points

We use Qiu's Theorem B [16] in the following form. Let $\pi : X \to Z$ be the maximal infinite-step pronilfactor of a minimal compact metric system. If a finite collection of open sets $V_j \subset X$ has images with intersection containing a nonempty open subset of Z, and p_j are distinct integer polynomials vanishing at zero including $p_0 = 0$, then some $x \in X$ and $n \in \mathbb{Z}$ satisfy $T^{p_j(n)} x \in V_j$ for all j. Indeed, in Qiu's almost one-to-one modification choose a singleton factor fiber over a point of the open intersection. The inverse images of the V_j then have a common point in their modified factor images, as required by Theorem B.

Suppose first that $k \geq 2$ and that $B = D_{k-1}$ has already been constructed, with old blocking length $H = H_{k-1}$. At a finite stage use

$$B_M = \{d \in B : \sum_j \|d\alpha_j\|_{\mathbb{T}} < 1/(4\pi(k+1)H)\}, \qquad S_M = B_M \cup G_M.$$

At $k = 1$ take $B_M = \varnothing$. Fix a window of dimension at most C, a length h, and a list of colorings in (a). Extend them arbitrarily off the window to two-sided words, then take a minimal subsystem X of the forward orbit limit of their joint names, the given nilsystem point, and the auxiliary torus point 0. Write $\pi : X \to Z$ for the maximal pronilfactor. The joint projection on the source nilsystem orbit and the auxiliary torus is a nilsystem factor, so the source and torus projections factor through maps $\rho : Z \to Y$ and $\gamma : Z \to \mathbb{T}^M$. Replace Y by the orbit closure of y. Set $W = \rho^{-1}(V)$ and let E_i be the clopen cylinder of the first color of test i.

If $W \subseteq \pi(E_i)$ for every i, select $m \in G_M$ by (1). Choose z_0 over y. Take open torus neighborhoods $P_0 \ni \gamma(z_0)$ and $P_1 \ni \gamma(z_0) + m\alpha$ with $P_1 - P_0 \subset O_M$, and set $P_t = \mathbb{T}^M$ for $t \geq 2$. Put

$$U_{i,t} = E_i \cap \pi^{-1}(W \cap \gamma^{-1}(P_t)) \qquad (1 \leq i \leq m_0, \quad 0 \leq t < h),$$

where m_0 is the number of tests. The images of $T^{-tm}U_{i,t}$ contain a common open neighborhood of z_0. Apply Qiu's result to all these sets at once with

$$p_{i,t}(n) = \big((i-1)h + t\big)n^2.$$

All polynomials are distinct, with exactly one zero. For test i this gives starting point $T^{(i-1)hn^2}x$ and positive difference $d = m + n^2$. The first two visits give $d\alpha \in P_1 - P_0 \subset O_M$. Each test has the required progression over the original window. In the uncolored case one may simply use (1) and translate the finite configuration along the orbit if a positive starting time is needed.

Otherwise $W \setminus \pi(E_i)$ is nonempty and open for some i. It contains the full inverse image of a nonempty window in a finite nilsystem factor. Choose a point of X over this window and consider the times of its visits to the window. On these times test i loses one color and the other tests retain their palettes. Adjoin to this finite factor an auxiliary torus window about zero whose difference set satisfies the defining smallness bound of B_M. The remaining palette excess is at most $k-2$. Apply (a) for B on the product window. The common difference belongs to B_M, and the loss of one color makes every recorded progression valid for its original test. All windows and all finitely many color assignments are open conditions, so the configurations obtained in X occur in the original forward orbit at positive starting times. If the starts in X differ by some fixed times, one common forward translate suffices. This proves the positive assertion at the finite stage.

Blocking each union of layers

Give the old part of S_M its inherited $k-1$ labels and give the new part label k, resolving any overlap in favor of the new part. A union omitting the new label is already blocked by its old coloring. Consider a union of t labels including the new one. Its old portion has a blocking input coloring χ with $u = t$ colors, or the constant input when $t = 1$. We need $q = t + 1$ output colors. For an integer J and input $i \in \mathbb{Z}/u\mathbb{Z}$, choose the unique $s \in [J-u, J-1] \cap \mathbb{Z}$ congruent to i modulo u, and set

$$\iota_J(i) = s - 1 \pmod{q}.$$

This injective map omits $J-1$ modulo q. When J increases by one, a single input moves into the previously missing output, so an output present in two adjacent charts has the same input in both. Use the output coloring $n \mapsto \iota_{J_q(n)}(\chi(n))$. On an old step the first H sites use two adjacent charts, and a constant output would give a constant input. On a new step all missing outputs are visited, by (2). Thus $L = \max(H, 2Q)$ is a uniform blocking length at this stage for all unions. The base case $k = 1$ uses just the one new layer and the constant input.

For clarity, the passage from a stage to D_k uses the countable compact test list for pointed nilsystems in Lemma A.7.1, proved in Section A.12. Each entry consists of a connected nilmanifold, an identity base point, an open window, a component period m, and a compact family of translations or affine return maps depending continuously on the parameter. After passing to multiples of m its windows refine those of every given pointed nilsystem represented by that entry. There are only finitely many ordered palette lists of excess at most $k-1$ once redundant one-color coordinates are removed.

Choose successively place values λ_j, starting at one. At stage j retain those of the first j entries whose component periods divide λ_j, together with the trivial system. Use length $j+2$ and choose

a single M_j large enough for their dimensions. Use the (λ_j/m)th power of each component return map. For each retained family and palette list the product of the parameter space with all two-sided coloring spaces is compact. Requiring specified starts and one specified difference in S_{M_j} uses only finitely many open window tests and color equalities. The preceding argument gives an open cover, hence a finite subcover. Taking their union over the lists and families gives a finite packet $F_j \subset S_{M_j}$ that works for all of them. Choose $b_j > 2L \max F_j$ divisible by $(j+1)!$ and set $\lambda_{j+1} = b_j \lambda_j$. Set

$$D_k = \bigcup_{j \geq 1} \lambda_j F_j.$$

The packets are disjoint, so their labels define the layers $C_1, \dots, C_k$. Every fixed nilsystem test eventually qualifies since $j! \mid \lambda_j$. Pulling all its colorings back to times $\lambda_j n$ and using the packet proves (a).

Fix any nonempty set J of labels, put $q = |J| + 1$, and let κ_j be its local q-color obstruction at stage j. With $z_j(n)$ the digit at place λ_j, the single coloring

$$n \longmapsto \sum_{j \geq 1} \big(\kappa_j(1 + z_j(n)) - \kappa_j(1)\big) \pmod q$$

blocks the selected union at length $2L$. Indeed, for $d = \lambda_j e$ from the union the lower digits are fixed, and a progression of $2L$ sites has span less than λ_{j+1}. There is a run of at least L sites on which all higher digits are fixed and the jth digit increases by e. The local obstruction excludes a constant run. This proves (b) with $H_k = 2L$ and completes the induction.

Condition (ii) of Theorem 4.1.1 implies (i) by the one-point window in (a) and (4.1.1). If the sum is at least k, distribute the k layers among the cells so that cell i gets at most $r_i - 1$ layers. Part (b) supplies the obstruction for each cell. The layers are infinite. Indeed, if one were finite, choose a finite cyclic window requiring a common difference divisible by a modulus larger than every element of that layer. Apply (a) to the $k-1$ binary blockers for the other layers and take a length at least H_k. No resulting difference is possible. At $k = 1$ the same argument uses the uncolored case. When the convention for partitions requires additional nonempty cells, subdivide an infinite layer as necessary. □

4.2 Five binary sites for a quadratic set

For an irrational α put

$$S_\alpha = \{d \geq 1 : \|\alpha d^2\|_{\mathbb{T}} > 2/5\}.$$

Theorem 3.5.1 shows that every finite list of binary colorings has a common four-term progression difference in S_α, whereas Lemma 3.5.3 gives a finite list of binary tests excluding every such difference at length seven. Here we increase the lower bound to five.

Theorem 4.2.1. *Given any finite list $c_1, \dots, c_m$ of binary colorings of $\mathbb{N}$ and any ordinary Bohr neighborhood U of zero, the set*

$$S_\alpha \cap U \cap \bigcap_{i=1}^{m} R_5(c_i)$$

has positive lower density. The coloring $f(n) = \lfloor 5\{\alpha n^2/2\} \rfloor$ satisfies $R_3(f) = \mathbb{N} \setminus S_\alpha$.

Consequently every finite partition of S_α has a cell forcing a five-term monochromatic progression for every binary coloring. For this f the least binary length L for which some finite list satisfies $\bigcap_i R_L(c_i) \subseteq R_3(f)$ is either six or seven.

A finite configuration

Lemma 4.2.2. *Every binary coloring of* $\{0, 1, \dots, 122\}$ *is constant on a set*

$$\{a + bt + 10t^2 : 0 \le t \le 4\}$$

for some integers a, b*, with all five values in the interval. Repeated values are permitted.*

The exhaustive verifier printed in Section B.3 proves the lemma; its search rules and output are given in Section B.1. For each $-70 \le b \le -10$ form $V_b = \{bt + 10t^2 : 0 \le t \le 4\}$ and enumerate all translates contained in $[0, 122]$. This range of slopes suffices since the difference between the last and first entries is $4b + 160$, of absolute value at most 122. Use the distinct points in each edge and discard edges containing another edge. A partial binary coloring is rejected if a fully colored edge is monochromatic, and the last uncolored vertex of an edge with only one assigned color is forced to take the opposite color. After propagating all forced choices, branch on an uncolored vertex in an unsatisfied edge, trying both colors. There are 1,397 inclusion-minimal edges. Fixing the color of vertex zero by symmetry, the exhaustive search finds no coloring avoiding them all. The code checks the final Boolean result as well as the number of edges.

The coefficient 10 has an independent arithmetic interpretation. A nonzero integer C forces a monochromatic $\{a + bt + Ct^2 : 0 \le t \le 4\}$ in every binary coloring of $\mathbb{Z}$ if and only if $10 \mid C$. For multiples of ten, apply Lemma 4.2.2 on the progression of step $C/10$. If $5 \nmid C$, reduce modulo five and multiply the argument of the coloring by C^{-1}. The residue set of $t^2 + bt$ is a translate of $\{0, 1, 4\}$, and the cyclic word 00101 splits all its translates. If C is odd, the same scaling modulo eight reduces to $C = 1$. The cyclic word 00001111 splits all translates of $\{t^2 + bt : 0 \le t \le 4\}$ for each of the eight slopes b. These last assertions can also be checked by enumerating the residues.

The finite lemma makes a point about radial colorings of the Euclidean plane. For a red-blue coloring depending only on distance from the origin, color $i \in [0, 122]$ by the color at radius $\sqrt{100 + i/10}$. If $i_t = a + bt + 10t^2$ is monochromatic, the radii squared are $100 + a/10 + (b/10)t + t^2$. These are the squared norms of $x + te_1$ for $0 \le t \le 4$, by choosing $x_1 = b/20$ and $x_2^2 = 100 + a/10 - b^2/400 \ge 0$. Thus every such radial coloring contains a monochromatic unit five-term progression.

The structured five-point correlation

We detail the transfer because an exact finite coloring statement on a torus need not by itself give a good difference for a coloring of $\mathbb{N}$. For one binary coloring take an ergodic invariant probability supported on a minimal subshift in its forward orbit limit. Positive probability of a finite pattern there implies its occurrence in the original coloring. For $0 \le f \le 1$ put

$$I_f(n) = \int \prod_{t=0}^{4} f(T^{tn}x)\, d\mu(x).$$

For the symbol indicator f, let $M_f = I_f + I_{1-f}$. Bergelson–Host–Kra [4, Corollary 4.6, Propositions 6.5 and 7.2] give the following specific approximation. For every $\varepsilon > 0$ there are a finite-dimensional ergodic nilsystem $X = G/\Lambda$ of step at most four, a measurable $F : X \to [0, 1]$ and a continuous nilsequence $n \mapsto \Phi(s^n\Gamma)$ for which

$$M_f(n) = \Phi(s^n\Gamma) + u(n) + e(n), \qquad |e(n)| < \varepsilon, \qquad \mathrm{UDlim}\, |u(n)| = 0. \tag{4.2.1}$$

The last condition means that, for each positive threshold, the exceptional set has zero upper Banach density. First project f onto the order-four Host–Kra factor and approximate that projection

in L^1 by its conditional expectation on a finite nilfactor. The same factor treats $1-f$. The product telescoping bound controls this finite-factor approximation uniformly in n. The explicit representation of Φ from [4, Sections 6–7] is important below.

Use their reduced presentation, in which all lower central groups G_j for $j \geq 2$ are closed and connected, and write $G_1 = G$. In the following groups the index t is an integer and evaluation at $t = 0, \ldots, 4$ identifies each sequence with five elements of G. The Hall–Petresco group A consists of polynomial sequences

$$h(t) = g_0 g_1^t g_2^{\binom{t}{2}} g_3^{\binom{t}{3}} g_4^{\binom{t}{4}}, \qquad g_0 \in G, \quad g_j \in G_j \ (j \geq 1).$$

Its normal subgroup B uses $g_0 \in G$ and $g_j \in G_{j+1}$ for $j \geq 1$. Write $\Lambda_A = A \cap \Lambda^5$, $\Lambda_B = B \cap \Lambda^5$, $K = A/B$, and $\Gamma = \Lambda_A B/B$. The integration space $\bar{X} = B/\Lambda_B$ has Haar probability $\bar{\mu}$. For the translation element a of X, let $s \in K$ be the image of $t \mapsto a^t$. For any representative $p \in A$ of $k \in K$,

$$\Phi(k\Gamma) = \int_{\bar{X}} \left(\prod_{t=0}^{4} F(p(t)h(t)\Lambda) + \prod_{t=0}^{4} (1 - F(p(t)h(t)\Lambda)) \right) d\bar{\mu}(h). \tag{4.2.2}$$

The Haar measure in this formula already averages linear slopes from G_2. This allows different binary tests to have independent starting points even after their external progression parameters are tied together.

Lemma 4.2.3. *There is an absolute $\delta_0 > 0$ with the following property. For every $Z \in \mathrm{Lie}(G_2)$ the polynomial $p_Z(t) = \exp(t^2 Z)$ belongs to A, and the corresponding value of (4.2.2) is at least δ_0. The bound is independent of G, F, Z.*

Proof. The membership follows from $t^2 = t + 2\binom{t}{2}$ along a one-parameter subgroup. Let $N = 123$, take J large enough that every 2^N-coloring of $[0, J-1]$ has a monochromatic five-term progression, and put $\tau = \exp(Z/10)$. Draw h with law $\bar{\mu}$. Color the N-by-J array by thresholding $F(\tau^i h(j)\Lambda)$ at $1/2$. Some five columns $a' + kt$, $0 \leq t \leq 4$, $k > 0$, are identical as binary columns. Lemma 4.2.2 supplies indices $a + bt + 10t^2$ of one color in that common column. There are only a fixed finite number P of possible tuples (a, b, a', k). For at least one tuple, the sum of the two products of the five F or $1 - F$ values is at least 2^{-5}.

For each fixed tuple that sum has expectation (4.2.2) with $p = p_Z$. Indeed the substitution $h(t) \mapsto h(a' + kt)$ respects B and Λ_B. Polynomial sequences for a descending filtration are closed under integer affine substitution, by collecting binomial coefficients and using $[G_i, G_j] \subseteq G_{i+j}$. An element of Λ_B is lattice-valued at every integer, by the triangular binomial formula. The substitution maps B onto B. On its Lie algebra the substitution is invertible triangular on the polynomial coefficients, and every component of B has a constant polynomial representative since all its nonconstant coefficient groups are connected. It fixes these representatives. Consequently its pushforward on B/Λ_B is B-invariant probability, hence Haar. Left multiplication by $t \mapsto \tau^{a+bt}$ preserves the same measure because this polynomial belongs to B. Multiplication on the left by $p_Z(t)$ now gives the desired array entries. Summing the expectations over the P possible tuples gives the bound $\delta_0 = 2^{-5}/P$. □

A common quadratic parameter

For the extra coordinate tracking αn^2, use the two-step nilpotent Lie group $G_\alpha = \mathbb{Z} \times \mathbb{T} \times \mathbb{T}$ with multiplication

$$(k, x, y)(l, u, v) = (k + l, x + u, y + v + ku - lx).$$

Its quotient by the lattice $\mathbb{Z} \times \{0\} \times \{0\}$ has torus coordinates $(x, y + kx)$. The orbit of the identity under $s_\alpha = (1, \alpha, 0)$ is $(n\alpha, n^2\alpha)$ and is dense by Weyl's theorem. The endomorphism $(k, x, y) \mapsto (2k, 2x, 4y)$ preserves the lattice and sends s_α to s_α^2.

Apply (4.2.1) separately to the finite list of binary tests with $\varepsilon < \delta_0/8$. Take the product of their parameter spaces K_i/Γ_i, the auxiliary nilmanifold above, and the finite torus rotation defining a basic neighborhood inside U. Let s be the tuple of their translations and e the identity lattice point. Substitution $h(t) \mapsto h(2t)$ on A_i descends to an endomorphism on K_i preserving its lattice and sending s_i to s_i^2. Together with the above auxiliary endomorphism and doubling on the final rotation this defines a smooth map Δ satisfying $\Delta(s^n e) = s^{2n} e$ and $\Delta(e) = e$.

The orbit closure is a finite union of connected compact subnilmanifolds, cyclically permuted by s, and the orbit equidistributes on each component on the corresponding residue class [14]. Write its identity component as He, with H connected, and let $\mathfrak{h} = \mathrm{Lie}(H)$. The ambient lattice is discrete, so $\mathfrak{h}$ is the tangent space there. The map Δ sends He into itself, giving $d\Delta(\mathfrak{h}) \subseteq \mathfrak{h}$. The projection of He onto the auxiliary torus is all of $\mathbb{T}^2$. Indeed this projection is a closed connected subgroup since the identity component of G_α is $\mathbb{T}^2$. If it were proper, all its finitely many translates by powers of s_α would have zero area, contrary to surjectivity of the projection of the full orbit closure. Thus $\mathfrak{h}$ projects onto both real auxiliary coordinates.

In $\mathrm{Lie}(A_i)$ one may use ordinary polynomial coefficients X_j of t^j, with $X_0 \in \mathrm{Lie}(G)$ and $X_j \in \mathrm{Lie}(G_j)$ for $j \geq 1$. This follows by differentiating the binomial parametrization at the identity and changing triangularly to the power basis. For $\mathrm{Lie}(B_i)$ replace G_j by G_{j+1} for $j \geq 1$. Thus on $\mathrm{Lie}(K_i)$ the derivative of doubling acts by 2^j on the degree-j quotient, for $1 \leq j \leq 4$. On the auxiliary coordinates it acts by 2 and 4, and it acts by 2 on the Bohr rotation. Take a vector of $\mathfrak{h}$ with auxiliary y-coordinate $1/2$. Apply the polynomial in $d\Delta$ taking value one at 4 and zero at $2, 8, 16$. The resulting vector still belongs to $\mathfrak{h}$ and has only degree-two components. Its exponential at the identity point therefore belongs to He, has auxiliary coordinates $(0, 1/2)$, projects to zero on the Bohr rotation, and in each test has representative $t \mapsto \exp(t^2 Z_i)$ with $Z_i \in \mathrm{Lie}(G_{i,2})$.

By Lemma 4.2.3 and continuity there is a relatively open neighborhood of this point in He where every $\Phi_i > \delta_0/2$, the quadratic weight has distance greater than $2/5$ from zero, and the linear Bohr condition holds. Equidistribution gives its visits positive lower density, even if one restricts to an open subset with boundary of measure zero inside it. Removing the finitely many exceptional sets $|u_i| \geq \delta_0/8$ costs zero upper Banach density. At every remaining visit all the original monochromatic five-point probabilities are positive. This proves the density assertion in Theorem 4.2.1.

Finally, three arguments in one half-open fifth of the circle have second difference of norm less than $2/5$. For the phase $\alpha n^2/2$ this second difference is αd^2. If its centered representative v has $|v| < 2/5$, the three interior points $1/10 + v/4, 1/10 - v/4, 1/10 + v/4$ in the first fifth have second difference v. The initial argument and first increment of the quadratic phase on progressions of step d are jointly equidistributed as the start varies, by Weyl's criterion. Thus a monochromatic triple of that step exists. Equality at $2/5$ cannot occur for irrational α. □

One and two six-term binary tests

A direct spectral argument accommodates one six-term test. A separate finite certificate accommodates two.

Corollary 4.2.4. *For any binary colorings $c, b_1, \dots, b_m$ and ordinary Bohr neighborhood U of zero,*

$$S_\alpha \cap U \cap R_6(c) \cap \bigcap_{i=1}^{m} R_5(b_i)$$

has positive lower density.

Proof. First, every binary coloring of $\mathbb{Z}$ has a monochromatic triple $x, x+d, x+3d$ with $d > 0$ and $d \equiv 3 \pmod 7$. Indeed use a stationary sign process for the coloring and let σ be its spectral probability measure. Four times the probability that these three signs agree is

$$1 + \widehat{\sigma}(d) + \widehat{\sigma}(2d) + \widehat{\sigma}(3d).$$

This follows by expanding the binary monochromatic indicator into its constant and three pair terms. Average over $d = 7n + 3$. By geometric summation and dominated convergence the limiting integrand, after taking real parts, is

$$1 + \sum_{j=1}^{3} 1_{\{7j\theta=0\}} \cos(2\pi \cdot 3j\theta).$$

At $\theta = 0$ this is four. At the other seventh roots it is $1/2$, since the three cosines sum to $-1/2$. Off the seventh roots at most one summand survives: simultaneous survival for $j = 2, 3$ would imply $7\theta = 0$. A surviving summand is the real part of a seventh root, and hence the whole integrand is at least $1 + \cos(6\pi/7) > 0$ there. This proves the claim. By compactness there is a fixed finite integer interval on which every binary coloring already has such a triple contained in the interval.

Scale this interval by $2/7$ and write $C = d/7$. The three sites now have the form $w, w+2C, w+6C$, with $\|C\|_{\mathbb{T}} = 3/7 > 2/5$. They give the six-site quadratic word

$$w + C(t^2 - 5t + 6), \quad 0 \le t \le 5,$$

whose values in order are $w + 6C, w + 2C, w, w, w + 2C, w + 6C$. Thus a fixed finite rational grid guarantees a monochromatic six-site quadratic shape with coefficient in a fixed finite set of admissible C's.

For the six-term test, apply the construction of (4.2.1) with the order-five factor and six-site progression groups, using the BHK formulas at $k = 5$. The doubling projection onto homogeneous degree two now also kills the eigenvalue 32. Choose the tangent vector with auxiliary y-coordinate one. In the Haar-array argument take the scaled grid as the row set, use six identical columns, and sum over the finitely many shapes and column progressions. Affine reindexing and left multiplication by the constant and linear factors preserve the Haar law exactly as above. For at least one admissible C, this yields a uniform positive lower bound on the six-term structured probability at the joint vector multiplied by C. The bound depends only on the finite grid and the van der Waerden number, and so is available before selecting the finite nilfactor approximation. At this same C all the five-term functions have their uniform lower bound from Lemma 4.2.3. The open-neighborhood argument proves the conclusion. □

Corollary 4.2.5. *The assertion of Corollary 4.2.4 holds with two prescribed binary six-term tests c, c' and any finite number of binary five-term tests.*

Proof. On $[0, 179] \cap \mathbb{Z}$ use the eight possible quadratic coefficients

$$\mathcal{C} = \{3, 4, 10, 11, 17, 18, 24, 25\}.$$

For any two binary colorings of the interval there is one $C \in \mathcal{C}$ for which both colorings have a monochromatic set $a + bt + Ct^2$, $0 \le t \le 5$, with their own a, b. The finite proof consists of the nine reverse-unit-propagation certificates described in Section B.1.2; their checker is printed in Section B.4. For any proposed family $J \subseteq \mathcal{C}$ that a single coloring avoids, put one Boolean variable at each site. For each six-site set using a coefficient in J, add the clause requiring at least one red site and the clause requiring at least one blue site. Fix the color at zero by symmetry. For each supplied certificate the checker regenerates these clauses, then verifies each added clause by assuming its negation and using forced assignments to derive a contradiction. The final clause is empty. This rules out the specified J for one coloring. The checker then tests all 2^8 assignments of the coefficients to the first or the second blocking coloring. In each assignment some one part contains a certified family. In total the files contain 2,008 verified clauses.

For the transfer, scale the row coordinates by $1/7$ so the possible leading coefficients are $C/7$, all of norm $3/7$ modulo one. Draw the two Haar integration variables independently and choose six identical columns in each table as before. The resulting two binary columns have a common coefficient by the finite result. Sum over the finite set of joint templates, allowing distinct row and column progressions for each test. By independence and the affine Haar substitution, the expectation for a fixed template is the product of the two structured monochromatic probabilities at their respective representatives of the same joint parameter multiplied by $C/7$. Some product has a system-independent positive lower bound, hence so does each factor since every probability is at most one. The five-term functions are bounded below at all these parameters. The rest follows from (4.2.1) and equidistribution. □

4.3 The exact palette threshold for quadratic triple differences

Write $S_{\alpha,\delta} = \{d \ge 1 : \|\alpha d^2\|_{\mathbb{T}} > \delta\}$ for $0 < \delta < 1/2$ and irrational α.

Theorem 4.3.1. *For $r \ge 4$ the set $S_{\alpha,\delta}$ forces a monochromatic triple in every r-coloring of $\mathbb{N}$ if and only if $\delta < 2/r$. More strongly, when $\delta < 2/r$, for any one r-coloring f, finitely many binary colorings c_i, and any ordinary Bohr neighborhood U of zero, the set*

$$S_{\alpha,\delta} \cap U \cap R_3(f) \cap \bigcap_i R_5(c_i)$$

has positive lower density. In particular the smallest number of colors that can block all $S_{\alpha,\delta}$-triples is $\lceil 2/\delta \rceil$.

Proof. For the upper obstruction color n by the rth of the circle containing $\alpha n^2/2$. A monochromatic triple has phase second difference of norm strictly less than $2/r$.

Here is the finite configuration used in the lower bound. Put $a = 2/r$. For every $1 \le j < r$,

$$\max\left(\|ja\|_{\mathbb{T}}, \|ja/2\|_{\mathbb{T}}\right) \ge 2/r. \tag{4.3.1}$$

Check $r = 4, 5$ directly. For $r \ge 6$, write $x = \|j/r\|_{\mathbb{T}} \in [1/r, 1/2]$. If $x \le 1/4$, then $\|2x\|_{\mathbb{T}} = 2x \ge 2/r$. For $x \ge 1/4$ one has $\max(x, \|2x\|_{\mathbb{T}}) \ge 1/3 \ge 2/r$.

Choose a rational mesh $e = 1/N < 2/r - \delta$ with $r \mid N$ and N large enough that $e < a$. Color the finite grid $[0, 2] \cap e\mathbb{Z}$ by r colors. There is a monochromatic pair u, v such that, writing $d = v - u > 0$, either $\|d\|_{\mathbb{T}} > \delta$ or $\|d/2\|_{\mathbb{T}} > \delta$. To see this, assume all such pairs have different colors. For $x = 0, e, \dots, a - e$, the r sites $x + ja$ for $0 \le j < r$ have all r colors by (4.3.1). The site

$x + e$ must differ from $x + ja$ for $1 \le j < r$, by (4.3.1) and the choice of mesh. Hence it has the color of x. Iteration gives the same color at 0 and a, a contradiction.

The pair gives a monochromatic quadratic triple of the form $A + Bt + Ct^2$, $0 \le t \le 2$, with $\|C\|_{\mathbb{T}} > \delta$. If $\|d\|_{\mathbb{T}} > \delta$, take the values v, u, v with coefficient $C = d$. Otherwise take u, u, v with $C = d/2$. The possible tuples belong to a fixed finite set, and all their coefficients lie in $(2N)^{-1}\mathbb{Z}$.

Apply the progression parameter construction of Section 4.2 to the one r-color test at length three and to all binary tests at length five. For the former use the sum of the r products in (4.2.2), the three-site Hall–Petresco groups, and the order-two characteristic factor. The indicator functions of the colors can be approximated simultaneously by conditional weights $F_1, \dots, F_r$ on one finite nilfactor, with $\sum F_j = 1$. Before choosing the approximation tolerance, take J large enough for a three-term van der Waerden progression with r^{2N+1} column colors. Let P bound the number of choices of the grid triple and the three columns. These constants depend only on r, δ.

As before take a pure degree-two tangent vector in the joint orbit component, this time with auxiliary y-coordinate *one* in its Lie algebra. Let Z represent its coefficient on the r-color test. For Haar h in the three-site integration space, color the array $\exp(iZ)h(j)\Lambda$ for grid points $i \in [0, 2] \cap e\mathbb{Z}$ and integer column indices $0 \le j < J$ by the largest weight F_l. Three equally spaced columns have identical color vectors. The grid result supplies a triple of rows $A + Bt + Ct^2$ on which one color is repeated. Exactly the Haar substitution in Lemma 4.2.3 applies: $t \mapsto \exp((A + Bt)Z)$ belongs to B and affine column substitution preserves Haar. After summing over all tuples, at least one of the finitely many C with $\|C\|_{\mathbb{T}} > \delta$ satisfies $\Phi_f(p_{CZ}) \ge r^{-3}/P$. This is a uniform bound, so all approximation tolerances may be chosen in advance.

Multiply the joint tangent vector by this C before exponentiating. Its auxiliary point is $(0, C)$, its ordinary Bohr rotation is zero, and the binary functions are bounded below by Lemma 4.2.3 at *every* such scalar. The same open-neighborhood and uniform-null argument as in Theorem 4.2.1 proves the assertion. For $r < 4$ use the lower assertion for four colors, whose hypothesis is $\delta < 1/2$. □

Higher powers

The same finite-grid argument gives an exact threshold at the first order above the classical recurrence order when the palette is sufficiently large.

Corollary 4.3.2. *Fix $k \ge 2$, put $M = 2^{k-1}$, and suppose that $r \ge M^2$. For irrational α, $0 < \delta < 1/2$, and $S = \{d : \|\alpha d^k\|_{\mathbb{T}} > \delta\}$, every r-coloring of $\mathbb{N}$ has a monochromatic $(k+1)$-term progression of difference in S if and only if $\delta < M/r$.*

Proof. The avoiding coloring for $\delta \ge M/r$ uses equal $r$ths for the phase $\alpha n^k/k!$. The positive coefficients in a kth forward difference have sum M and so do the absolute values of the negative coefficients. Thus the alternating binomial sum of representatives in an interval of length $1/r$ has absolute value strictly less than M/r.

For the converse, the two-value interpolation step gives a polynomial of degree at most k taking prescribed endpoint labels u and v at the sites $0, \dots, k$. Its leading coefficient can be either $(v-u)/k!$ or $M(v-u)/k!$. To obtain the first value put v only at the last site. To obtain the second put v exactly at the sites with positive binomial coefficient in the forward difference. For every $1 \le j < r$,

$$\max(\|j/r\|_{\mathbb{T}}, \|Mj/r\|_{\mathbb{T}}) \ge M/r.$$

By symmetry it suffices to take $j \le r/2$. If $j < M$, then $M \le Mj \le M(M-1) \le r - M$. If $j \ge M$, use j/r itself. Now take the r equally spaced sites with spacing $a = k!/r$ and a rational mesh e

dividing a, small enough to preserve the strict inequalities relative to δ after perturbation by e. Exactly the argument of Theorem 4.3.1 finds a monochromatic pair whose interpolating polynomial has leading coefficient C satisfying $\|C\|_{\mathbb{T}} > \delta$. All possibilities come from one finite rational grid.

For completeness a graded auxiliary nilsystem tracking αn^k can be written explicitly. Let V consist of formal polynomials of degree at most $k-1$ with coefficients in $\mathbb{T}$, and let $S_n v(t) = v(t+n)$ for $n \in \mathbb{Z}$. Take the semidirect product with law $(n, v)(m, w) = (n + m, v + S_n w)$ and lattice $\mathbb{Z} \times \{0\}$. This is a nilpotent Lie group, since commutating with translation lowers polynomial degree. For $s = (1, \alpha((t+1)^k - t^k))$, the quotient coordinate of s^n is $\alpha((t+n)^k - t^k)$, whose constant coefficient is αn^k. Its coefficient vector is equidistributed in V by Weyl's criterion. The map $(n, v(t)) \mapsto (2n, 2^k v(t/2))$ is well-defined on coefficients, since the multiplier on t^j is the integer 2^{k-j}. It respects the group law and lattice and sends s to s^2.

Use the order-k version of (4.2.1). The same tangent projection selects only degree k and normalizes the auxiliary constant coefficient to one. In the grid-and-column argument, a polynomial of degree less than k with coefficients on $Z \in \mathrm{Lie}(G_k)$ gives a sequence in B after exponentiation along this one-parameter subgroup. It therefore disappears under the Haar average, just as the affine terms did for $k = 2$. Summing over the finitely many interpolants gives a uniform positive lower bound at some joint parameter multiplied by C with $\|C\|_{\mathbb{T}} > \delta$. Equidistribution and (4.2.1) conclude the proof. □

4.4 A divisibility consequence for difference sets

The following observation concerns actual integer differences and requires no distribution hypothesis on E. For infinite $E \subset \mathbb{N}$ write $\Delta(E) = \{v - u : u, v \in E,\ u < v\}$.

Theorem 4.4.1. *Let $c_1, \dots, c_m$ be binary colorings of $\mathbb{N}$ and let $s_1, \dots, s_{m+1}$ be positive integers satisfying $4s_l \mid s_{l+1}$. For some l and $d \in \Delta(E)$, the difference $s_l d$ belongs to every $R_4(c_i)$. In particular $B\Delta(E)$ has this property for arbitrary finite lists if B meets $q\mathbb{N}$ for every q.*

Proof. Suppose there is no common difference. Color each unordered pair from E by the $(m+1)$-tuple giving a failed test at each of its scaled differences. Infinite Ramsey gives an increasing subsequence (a_j) homogeneous for this finite edge coloring. By the pigeonhole principle, one test fails at two multipliers s, t with $4s \mid t$ on every pair of this subsequence. Take a stationary sign process f for that test as above, write U for its Koopman operator, and put $r(n) = \langle U^n f, f \rangle$. Use a free ultrafilter p on the indices and define

$$r_*(k) = \lim_{i \to p} \lim_{j \to p} r\big(k(a_j - a_i)\big) = \Big\| \operatorname*{w\text{-}lim}_{j \to p} U^{k a_j} f \Big\|_2^2 \geq 0.$$

Terms with $j > i$ suffice in the inner limit. For four binary sites the overlapping-parity inequality used in the proof of Theorem 3.5.1, following Wolf's observation [18, Section 2], is

$$8M_4(v) \geq \tfrac{1}{2}(1 - r(4v)) + 3r(v) + 2r(2v) + r(3v).$$

Here $M_4(v)$ is the monochromatic four-site probability. Indeed the two overlapping products of four consecutive signs have common mean γ and product of mean $r(4v)$. Nonnegativity of the product of one plus each of those signs gives $1 + 2\gamma + r(4v) \geq 0$. Expanding the four-site monochromatic indicator yields the displayed inequality.

At $v = s(a_j - a_i)$ the left side is zero. Taking the two limits makes all terms on the right nonnegative, hence $r_*(4s) = 1$. Thus $\|U^{4s(a_j - a_i)} f - f\|_2$ tends to zero in the iterated limit.

Telescoping powers shows the same for $U^{ut(a_j-a_i)}f$ for $u = 1, 2, 3$. The union bound now makes $M_4(t(a_j - a_i))$ tend to one, contradicting its vanishing on every pair. □

A transfer for steps of bounded support

Let $G = \bigoplus_{i\geq 1} \mathbb{Z}e_i$. Suppose fixed q, h, b have the following property. For every m, L there is a finite nonzero direction set $F \subset G$ supported on at most b coordinates per direction. Every list of m binary colorings of G has a common good L-term direction in F, but some q-coloring of G blocks F at length h. Then the original finite-partition question on $\mathbb{N}$ has a negative answer.

To see this, restrict to a finite coordinate subgroup $\mathbb{Z}^n$ containing F. Extend arbitrary test colorings from that subgroup to its cosets by copying them whenever this restriction is needed. Choose a base P much larger than $2^b h \max_{v\in F,j} |v_j|$ and put $\ell(v) = \sum_{j=0}^{n-1} v_j P^j$. For P this large, $\ell(v) \neq 0$ for all $v \in F$. The packet $|\ell(F)| = \{|\ell(v)| : v \in F\}$ has the positive testing property on $\mathbb{N}$ by pullback. Indeed compactness gives a finite set of sites witnessing the positive property for every list on $\mathbb{Z}^n$. Translate their images by a single large constant before pulling back the bits, and reverse a progression when the projected difference is negative.

For the upper bound color an integer by applying the original q-coloring to its first n base-P digits. Along a progression of length $2^b h$ with signed step $\ell(v)$, normalize the uncarried digits $z_j + tv_j$ from the bottom upward. On a consecutive interval where the incoming carry is fixed, an inactive coordinate sends a constant carry upward. At an active coordinate the affine digit varies by less than P over the entire run and crosses at most one boundary on each such interval. Thus normalization partitions the time indices into at most 2^b consecutive intervals on each of which all carries are fixed. On some interval of length at least h the digit vectors are a true progression of direction v. This proves a uniform upper length $2^b h$ independent of n. Finally glue the integer packets at disjoint mixed-radix places as in Section 4.1, with the local q-colorings, to obtain the claimed counterexample.

4.5 The unrestricted quantifier

Theorem 4.1.1 rules out every bound on the number of partition cells that is fixed independently of the multicolor target. It does not provide one counterexample for all m, since its upper obstruction needs more colors as m increases. Theorem 4.2.1 and Theorem 4.3.1 give an example with a fixed five-color upper obstruction at length three, and it passes arbitrary finite binary lists at length five as well as two six-term tests together with five-term tests. For an unrestricted number of six-term tests the common-difference question remains unresolved here, and a finite list already fails at length seven. A counterexample for the original question requires finite packets for which the binary lengths and the numbers of tests both tend to infinity while the multicolor obstruction has one fixed palette and one fixed forbidden length.

4.6 A density bound for triples in one prescribed cell

The positive assertion for one color class can be strengthened using a finite inhomogeneous equation. Lev, Matolcsi, Pach and Varga proved a $2/7$ density bound for sets missing a value of $A + A - 2 \cdot A$ in a prime-order group. They proposed the following exact finite computation in [12, Section 3]. Here $2 \cdot A = \{2a : a \in A\}$.

Theorem 4.6.1 (Computer-assisted finite extremal bound)**.** *The largest possible size of an* $X \subseteq \{0, \dots, 461\}$ *with* $42 \notin X + X - 2 \cdot X$ *is* 126*. Consequently for every prime* $p > 7$ *and every* $A \subseteq \mathbb{F}_p$*,*

$$A + A - 2 \cdot A \neq \mathbb{F}_p \quad \Longrightarrow \quad |A|/p \leq 3/11.$$

Proof. The residues $0, 4, 8$ modulo eleven give 126 sites in the interval. They miss 42 in the stated sum of dilates, by checking modulo eleven.

For the upper bound we give a finite exhaustive proof; the complete program and an independent check of its pair-exclusion masks are given in Appendix B. The C++17 program `verify_kravitz_462.cpp` performs the enumeration with integer operations. Write a site as $r + 7i$ with $0 \leq r < 7$, $0 \leq i < 66$. Within each residue the forbidden equation on the indices is $x + z - 2y = 6$. Represent a choice of indices by a 66-bit mask. List all the inclusion constraints $\{x, y, z\}$, retaining repeated-variable solutions as two-element sets. To enumerate choices of a given size, add indices in increasing order. Reject a completed forbidden set and remove its last available vertex as soon as all other vertices have been chosen. This search gives no valid choice of size 21. The numbers of choices of sizes $20, 19, 18$ are respectively $1, 38, 838$. The program also enumerates all choices of sizes 16 and 17, generating smaller sizes as needed.

Search for 127 sites, processing the seven residues in order of decreasing final size, breaking ties by increasing residue. If the next residue is to be one of b remaining ones and n further sites are needed, its selected size must be at least $\lceil n/b \rceil$. For each unassigned residue maintain its available indices. For any two already selected sites x, y in distinct residues, exclude from any third residue the valid sites

$$42 + 2x - y, \qquad 42 + 2y - x, \qquad (x + y - 42)/2$$

when integral and inside the interval. Three sites of a solution have either all residues equal or all residues distinct modulo seven. Consequently the local mask checks and these pair exclusions suffice.

An upper bound on the size remaining in each residue comes from its available indices. Indices at distance three or six cannot both be selected, by repeated-variable solutions. Independently in each index class modulo three, greedily taking the least available index and discarding its next two neighbors maximizes the possible size under these two distance constraints. This is an upper bound when the other constraints are restored. Cap it also by the preceding residue size, strictly for a tie in the wrong order. If necessary reduce the cap whenever no precomputed mask of that size fits inside the available indices. If the sum of the caps is below the remaining target, prune the branch. Otherwise try every admissible choice for the next residue. The ordering chooses each hypothetical 127-point set along a unique branch, and all pruning steps give necessary conditions.

Across the seven possible first residues the search reaches 1,404,585 nodes with two residues assigned, 227,496,552 with three, and 3,958 with four. No node with five assigned residues survives. The bit-mask encoding of the cross-residue exclusions can be checked on singleton masks, where it is exactly the displayed three-site list, and then extended by unions. The verifier checks the lower example as well. The independent mask check verifies all $\binom{7}{2}66^2 = 91{,}476$ pairs of singleton masks against the three displayed formulas; see Section B.1.3.

If A misses a value in $\mathbb{F}_p$, it misses a nonzero value unless it is empty. Dilate A by a unit to make 42 the missing value. For any translate of the 462 integer indices, the indices whose residues land in A form a solution-free integer set, so at most 126 are selected. Averaging over the translation gives $462|A|/p \leq 126$, as claimed. This counts indices separately if $p < 462$. □

Corollary 4.6.2. *For every prime* $p > 11$*, the density bound of Theorem 4.6.1 can be sharpened to*

$$|A|/p \leq 3/11 - 1/(462 \cdot 20{,}000).$$

Proof. After dilation let 462 be missing and write $a_n = 1_A(n \bmod p)$. Every $w_n = \sum_{i=0}^{461} a_{n+11i}$ is at most 126 by the integer part of Theorem 4.6.1. If equality holds at 20,000 consecutive starts, translate these to start at zero. Comparing w_n with w_{n+11} gives $a_n = a_{n+5082}$ for $0 \le n \le 19988$. The 5,082-site word then gives a subset of $\mathbb{Z}/5082\mathbb{Z}$ missing 462 there. Indeed, for $0 \le x, y < 5082$ the integer $z = 462 + 2y - x$ lies between -4619 and 10624. Translating x, y, z by 5082 puts all three in the region governed by the displayed periodicity. The cyclic word has $11 \cdot 126 = 1386$ occupied sites by summing $w_0, \dots, w_{10}$.

The high digits in each of the 462 fibers modulo 462 form a subset of $\mathbb{Z}/11\mathbb{Z}$ missing 1 in its sum of dilates. Every such set has at most three points, and the three-point possibilities are precisely the translates of $\{0, 2, 4\}$. To check this, multiply the set by two so that 2 is missing. Repeated-variable solutions forbid distances $1, 2$, so four sites do not fit and the cyclic gaps of a three-set must be $3, 3, 5$ or $3, 4, 4$ up to cyclic order. In the first case a translate of $\{0, 3, 6\}$ represents 2 as $0 + 3 - 2 \cdot 6$ modulo eleven. The second case is a translate of $\{0, 4, 8\}$ and avoids 2. Undo the dilation. The total count $1386 = 462 \cdot 3$ gives equality in every fiber.

In the eleven fibers $u_r = 42r \bmod 462$, $r \in \mathbb{Z}/11\mathbb{Z}$, write the three selected sites as $F_r + 462\{0, 2, 4\}$. The class $R_r = (F_{r-1} + F_{r+1} - 2F_r)/462$ is well defined modulo eleven. Using the three indicated high digits shifts R_r by $2\{-4, -3, \dots, 4\}$, which misses only $1, -1$ modulo eleven. Hence the exclusion of 462 forces $R_r \in \{0, 2\}$. Their sum is zero modulo eleven by telescoping, so all eleven R_r are equal. The cyclic second difference of F_r is therefore constant, say K modulo 5082, where $462 \mid K$. Summing first differences gives $11(F_1 - F_0) + 55K = 0 \pmod{5082}$, which forces $462 \mid F_1 - F_0$. This contradicts $F_1 - F_0 \equiv 42 \pmod{462}$. Thus every 20,000 consecutive starts contain a deficit of at least one. Averaging gives the claim. □

Corollary 4.6.3. *Let T be an invertible ergodic probability-preserving transformation and let A have measure greater than $3/11$. Then the set of $d \in \mathbb{N}$ with $\mu(A \cap T^{-d}A \cap T^{-2d}A) > 0$ contains an ordinary Bohr neighborhood of zero in $\mathbb{N}$ after removing a set of zero upper Banach density.*

Proof. Put $\delta = \mu(A)$, $a_0 = 3/11$, and $\eta = (\delta - a_0)/4$. We first obtain a bound depending only on δ, before making any finite-factor approximation. Let F be any $[0, 1]$-valued function of mean δ on a finite two-step nilsystem, and put $\theta = \eta$. Since $F \le \theta + \mathbf{1}_{\{F \ge \theta\}}$, the marked set $\{F \ge \theta\}$ has Haar measure at least $a_0 + 3\eta$.

Fix a pure quadratic parameter Z and let h have Haar law on the three-site integration space B/Λ_B of (4.2.2). Form the array

$$\exp(iZ/21)h(j)\Lambda, \qquad 0 \le i < 462, \quad 0 \le j < J,$$

and mark an entry when its F-value is at least θ. Evaluation at each integer j maps B/Λ_B onto the original nilmanifold and preserves Haar probability, so every entry has the same marginal distribution. Let $g(h)$ be the proportion of columns with at least 127 marked entries. The overall proportion of marked entries is at most $a_0 + g(h)$, whence $\int g \, d\bar{\mu} \ge 3\eta$. It follows that

$$\bar{\mu}\{h : g(h) \ge \eta\} \ge \frac{2\eta}{1 - \eta} \ge \eta.$$

On this event, some one marked-row pattern occurs on a proportion at least $\eta/2^{462}$ of all columns and has at least 127 marked rows. Choose J, depending only on η, large enough that every subset of $[0, J-1]$ of this density contains a nonconstant three-term arithmetic progression. Three equally spaced columns therefore have the same pattern. Theorem 4.6.1 supplies marked rows i_0, i_1, i_2 with $i_0 - 2i_1 + i_2 = 42$. Their interpolating quadratic has leading coefficient 21. There are only finitely many possible row-and-column templates, say at most $P = P(\eta)$. For at least one template on

the event just described, the product of the three F-values is at least θ^3. For each fixed template, affine column substitution and left multiplication by its constant and linear row factors preserve Haar law, exactly as in Lemma 4.2.3. Its expected product is the structured single-cell correlation $\Phi(p_Z)$. Consequently

$$P\Phi(p_Z) \geq \eta\theta^3, \qquad \text{so} \qquad \Phi(p_Z) \geq c(\delta) := \eta^4/P > 0.$$

This lower bound is independent of the nilsystem, F, and Z.

Apply the three-site BHK decomposition to $\mathbf{1}_A$, using a conditional expectation on a finite two-step nilfactor of mean δ. The finite-factor approximation can now be chosen so that its uniform correlation error is less than $c(\delta)/8$. In the two-step progression parameter space, every point above the identity of the horizontal rotation has a pure-quadratic representative. Indeed, a representative $g_0 g_1^t g_2^{\binom{t}{2}}$ of such a point has $g_1 \in G_2\Lambda$. Remove the lattice part by right multiplication by a lattice polynomial, and remove the constant and central linear parts through the subgroup B. The remaining central coefficient belongs to the connected abelian group G_2; surjectivity of its exponential map gives a representative $t \mapsto \exp(t^2 Z)$. Thus $\Phi \geq c(\delta)$ on the entire identity fiber. Compactness of that fiber and continuity of Φ give an ordinary neighborhood of the horizontal identity above which $\Phi > c(\delta)/2$. The times when the horizontal rotation visits this neighborhood form an ordinary Bohr neighborhood of zero. Finally remove the set on which the uniform-null error in (4.2.1) has absolute value at least $c(\delta)/8$. This exceptional set has zero upper Banach density, and at every remaining Bohr time the original triple correlation is positive. □

Chapter 5

Winding squares and polynomial obstructions

5.1 A criterion for ternary obstructions

Write $R_h(c)$ for the positive steps of monochromatic h-term progressions under c. A set is r-large if it meets every $R_h(c)$ for colorings with at most r colors. Write $D \in \mathcal{R}_2$ if every finite partition of D has a 2-large cell. Equivalently, for every finite list of binary colorings c_i and every L, some $d \in D$ belongs to every $R_L(c_i)$. Indeed an empty intersection assigns each d to a failed test. Conversely, one test for each failed cell suffices, upon taking the maximum of the forbidden lengths. The starts for the tests may differ.

Regard the three colors as $\mathbb{Z}/3\mathbb{Z}$. A square with positive sides d, e reads, in its boundary order,

$$f(n), \qquad f(n+d), \qquad f(n+d+e), \qquad f(n+e), \qquad f(n).$$

It is a *winding square* if one edge keeps its color and the other three steps go once around the three colors, all in the same orientation. A square with equal opposite colors cannot wind.

Theorem 5.1.1 (Flat-square obstruction)**.** *Let $E \subseteq \mathbb{N}$ and $h \geq 2$. Suppose that a ternary coloring $f : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$ has no monochromatic h-term progression of difference in E, and that no square with both sides in E winds. There exist binary colorings a, b of $\mathbb{Z}$ such that*

$$E \cap R_3^{\mathbb{Z}}(a) \cap R_h^{\mathbb{Z}}(b) = \varnothing. \tag{5.1.1}$$

Thus two tests at the common length $\max(h, 3)$ suffice. For $h = 3$, a single binary coloring blocks all five-term progressions with difference in E. For $h = 2$, a single binary coloring blocks all three-term progressions with difference in E.

A one-sided coloring with these avoidance properties gives a two-sided one by taking a limit of forward translates. Every forbidden configuration uses finitely many entries. Binary obstructions on $\mathbb{Z}$ can simply be restricted to $\mathbb{N}$.

5.2 Integrating the color increments

Assume E is nonempty and put $g = \gcd(E)$. Restrict to $g\mathbb{Z}$, divide all coordinates and steps by g, and henceforth suppose $\gcd(E) = 1$. At the end each binary coloring c on the normalized group gives $n \mapsto c(\lfloor n/g \rfloor)$ on the original group. This works because every tested step is a multiple of g.

For $d \in E$ let $u_d(n) \in \{-1, 0, 1\}$ represent $f(n+d) - f(n)$ modulo three. The oriented sum around a square is a multiple of three in $[-4, 4]$. It equals 3 or -3 precisely on a winding square. In their absence we have

$$u_d(n) + u_e(n+d) = u_e(n) + u_d(n+e). \tag{5.2.1}$$

Use $u_{-d}(n) = -u_d(n-d)$ for reversed steps. Translating (5.2.1) gives the same identity for signed steps.

Choose a finite word in $\pm E$ whose sum is one. Let $V(n)$ be the sum of its increments when the word starts at n. Commuting a step d past the successive letters by (5.2.1) gives

$$V(n) + u_d(n+1) = u_d(n) + V(n+d).$$

Choose an integer $H(0)$ representing $f(0)$ and define $H(n+1) - H(n) = V(n)$ for all $n \in \mathbb{Z}$. Then $H(n) \equiv f(n) \pmod 3$. The last identity says that $H(n+d) - H(n) - u_d(n)$ is independent of n. Thus for each $d \in E$ there is an integer k_d with

$$H(n+d) - H(n) = u_d(n) + 3k_d. \tag{5.2.2}$$

The increments of H are bounded by the length of the fixed word. Choose a sequence $N \to \infty$ along which $H(N)/N \to s$. For each fixed d, summing $H(n+d) - H(n)$ over $0 \le n < N$ leaves d terms at each boundary. Bounded increments show that its average tends to ds along the same sequence. It follows from (5.2.2) that

$$m_d = ds - 3k_d \in [-1, 1].$$

Proof of Theorem 5.1.1. The assertion is immediate if $E = \varnothing$. Use the height H, slope s, and integers k_d constructed above, and define

$$a(n) = \left\lfloor \frac{ns}{3} \right\rfloor \pmod 2, \qquad b(n) = H(n) \pmod 2. \tag{5.2.3}$$

Suppose a is constant at $x, x+d, x+2d$, and put $z = xs/3$ and $\theta = m_d/3$. The two successive integer increments of $\lfloor ns/3 \rfloor$ are

$$k_d + \lfloor z + (j+1)\theta \rfloor - \lfloor z + j\theta \rfloor, \qquad j = 0, 1.$$

Each floor increment is either zero or $\operatorname{sgn}(\theta)$. At least one is zero, since $|2\theta| \le 2/3 < 1$. Both displayed increments are even, so k_d is even. If b is constant on an h-term progression of this same difference, with any starting point y, equation (5.2.2) now forces every $u_d(y + jd)$ on that progression to be even. Since u_d takes values in $\{-1, 0, 1\}$, all these increments vanish. The original f-colors on the h sites coincide, a contradiction. The two starting points x and y are independent.

For the single-test conclusions, instead use

$$c(n) = H(n) - \lfloor ns/3 \rfloor \pmod 2.$$

On a c-monochromatic progression starting at x, set $J_j = \lfloor xs/3 + jm_d/3 \rfloor$. Equation (5.2.2) and the evenness of $3k_d - k_d$ imply

$$u_d(x + jd) \equiv J_{j+1} - J_j \pmod 2. \tag{5.2.4}$$

Because $|m_d|/3 \le 1/3$, successive nonzero increments of J are at least three indices apart. Every four consecutive increments therefore contain two consecutive zeros. Five c-monochromatic sites would give three consecutive f-monochromatic sites, proving the assertion for $h = 3$. Similarly, every two consecutive increments contain a zero, which proves the assertion for $h = 2$. The normalization by $\gcd(E)$ and the one-sided-to-two-sided passage described above preserve each conclusion. □

5.3 What a counterexample would have to exhibit

For a ternary h-blocker f on D, form a graph on D by joining d to e if some square with those sides winds. If $D \in \mathcal{R}_2$, this graph has infinite chromatic number. Otherwise its independent color classes, each covered by the two failures of Theorem 5.1.1, would give a finite partition of D with no 2-large cell.

For $h \geq 3$, a finite coloring of this graph would in fact give a finite binary obstruction at the original length h.

There is a uniform version in the starting point. A set of starts is syndetic if some fixed length of interval always contains a start.

Corollary 5.3.1. *If $D \in \mathcal{R}_2$, then for each $k \geq 1$ and $h \geq 2$ there exist a finite $F \subseteq D$ and $B \geq 1$ with the following property. For every coloring of F with k colors, every ternary h-blocker f on D, and every interval of B consecutive starts, some start in the interval gives a winding square with two same-colored sides $d < e$ from F. Either prescribed orientation of winding can be required.*

Proof. Put $L = \max(h, 3)$. Compactness of the space of $2k$ binary colorings gives a finite $F \subseteq D$ meeting every intersection of their length-L tests. Indeed, having prescribed finite monochromatic progressions is an open condition. If the conclusion failed for this F, take a sequence of arbitrarily long intervals avoiding one prescribed orientation. Pass to a subsequence with the same coloring of F and the same prescribed orientation. Translating the intervals to their midpoints and taking a limit gives a two-sided ternary h-blocker with no prescribed-orientation square at any start for any same-cell $d < e$ from F.

Take a shift-invariant probability measure on the orbit closure of this blocker, for instance a subsequential limit of orbit averages. For each such d, e, the expected signed boundary sum in (5.2.1), with the right side subtracted, is zero by shift invariance. This sum takes only the values $0, 3, -3$. Since one sign is excluded throughout the orbit closure, the other has measure zero. Some point thus has no winding square at any integer translate for any of these finitely many pairs. It still blocks h-term progressions on F. Apply Theorem 5.1.1 to each cell. The resulting $2k$ binary tests altogether exclude F, contrary to the choice of F. □

The winding condition alone can hold even for an arithmetically simple forbidden set. Take $E = \{d \geq 1 : d \equiv 1 \pmod 3\}$ and independently assign to each n one of the two colors different from n mod 3, each with probability $1/2$. No E-progression of length three can be monochromatic, since its three residue classes exclude all three possible common colors. For any fixed distinct $d, e \in E$ and $n \equiv 0 \pmod 3$, the assignments

$$(f(n), f(n+d), f(n+d+e), f(n+e)) = (2,0,1,2) \quad \text{or} \quad (2,2,1,0)$$

give opposite windings, each with probability $1/16$. Infinitely many starts have disjoint sets of four sites, so almost surely both orientations occur for every one of the countably many pairs. Nevertheless the periodic bit $\mathbf{1}_{3\mathbb{Z}}$ blocks all triples on E.

5.4 Four colors when pairs are blocked

Proposition 5.4.1. *Let $f : \mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ satisfy $f(n+d) \neq f(n)$ for all n and $d \in E$. Fix a cyclic order of the four colors. If no square with sides in E traverses these four distinct colors in the chosen order or its reverse, then some $\alpha \in \mathbb{T}$ satisfies $\|d\alpha\|_{\mathbb{T}} \geq 1/4$ for every $d \in E$. One binary coloring therefore blocks every triple of such a step.*

Proof. Normalize $\gcd(E) = 1$ as before and relabel the colors in the chosen cyclic order. For positive $d \in E$ use the representative $u_d(n) \in \{1, 2, 3\}$ of $f(n + d) - f(n) \pmod 4$. For reversed steps use the negative of the corresponding forward increment. The difference between the two positive path sums across a square belongs to $\{-4, 0, 4\}$. A nonzero difference requires two increments equal to 3 on one path and two equal to 1 on the other. These are exactly the two orientations of the excluded traversal. Hence (5.2.1) holds. The height construction now gives integers k_d and a real slope s with $1 \le ds - 4k_d \le 3$. Take $\alpha = s/4$ in the normalized group, or $s/(4g)$ in the original group. Three successive points of increment $d\alpha$ cannot lie in one half-open semicircle when $\|d\alpha\|_{\mathbb{T}} \ge 1/4$. □

In particular, in a proper four-color witness against $D \in \mathcal{R}_2$, each of the three undirected Hamiltonian cycles on the four colors separately defines a graph of infinite chromatic number on D. Here an edge means that some square realizes that specified cycle. For three colors blocking pairs, a stationary edge is impossible, so Theorem 5.1.1 also recovers a binary triple obstruction directly.

5.5 The full inverse image of the exponent-three directions

In $\bigoplus_j \mathbb{F}_3 e_j$, the coloring $x \mapsto \sum_j x_j^2 \pmod 3$ blocks monochromatic triples of step $e_i - e_j$. Lifting the direction set to $\bigoplus_j \mathbb{Z} e_j$ allows arbitrary multiples of three in every coordinate. The following calculation handles all these lifts simultaneously.

Proposition 5.5.1. *Let $G = \bigoplus_{j \ge 1} \mathbb{Z} e_j$ and define $S(x) = \sum_j x_j^2$. The binary coloring*

$$c(x) = \mathbf{1}_{\{0,1,2\}}(S(x) \bmod 9)$$

blocks every eight-term progression whose direction has a number of nonzero coordinates modulo 3 that is not divisible by 3. In particular it blocks every integral lift of $e_i - e_j$ modulo 3 for $i \ne j$, with no restriction on how many other coordinates of the lift are nonzero multiples of 3.

Proof. Fix a direction v as in the statement and a starting point x. The coefficient $A = S(v)$ of t^2 in $S(x + tv)$ is congruent modulo 3 to the number of coordinates of v that are nonzero modulo 3. Thus $2A$ is invertible modulo 9, so for suitable residues b, C we have

$$S(x + tv) \equiv C + A(t + b)^2 \pmod 9.$$

Each of the three unit squares $1, 4, 7$ modulo 9 has two square roots modulo 9. Any eight consecutive integers represent all but one residue modulo 9, so their squares attain all three of these values. The residues $C + A$, $C + 4A$, and $C + 7A$ form a full residue class modulo 3 inside $\mathbb{Z}/9\mathbb{Z}$. Exactly one belongs to $\{0, 1, 2\}$, so the progression has both colors. □

The length eight is sharp for this particular bit. For $v = e_1 - e_2$ and $x = 2e_3$, the seven values $S(x + tv) = 4 + 2t^2$ for $-3 \le t \le 3$ belong to $\{3, 4, 6\}$ modulo 9. All seven sites have color zero.

5.6 A binary test for polynomial colorings

For a positive integer $Q \ge 2$ and an integer $d \ge 1$, consider colorings obtained by reducing an integer polynomial of degree at most d modulo Q. The following bound is independent of the number of variables. We state it for a map P whose restrictions to integer lines are polynomials, which also permits sums such as $P(x) = \sum_j x_j^2$ on the direct sum of countably many copies of $\mathbb{Z}$.

Theorem 5.6.1. *Let G be an abelian group and suppose that $P : G \to \mathbb{Z}$ has the following property. For every $x, v \in G$, there is a polynomial $F_{x,v} \in \mathbb{Z}[t]$ of degree at most d satisfying $F_{x,v}(t) = P(x+tv)$ for all $t \in \mathbb{Z}$. There are a length $L = L(d, Q)$ and a binary coloring c of G such that, whenever c is constant on $x, x+v, \ldots, x+(L-1)v$, the function $t \mapsto P(x+tv) \bmod Q$ is constant on all of $\mathbb{Z}$.*

Proof. For each prime power p^a exactly dividing Q, define

$$E_p = a - 1 + \lfloor \log_p d \rfloor + v_p((d-1)!), \qquad K_p = 2E_p + 2, \qquad L = \prod_{p \mid Q} p^{K_p}.$$

For $y \in \mathbb{Z}$, let $b_p(y)$ be 1 if the highest digit of the standard representative of y modulo p^{K_p} in base p is zero, and 0 otherwise. Set $c(x) = \sum_{p|Q} b_p(P(x)) \bmod 2$.

Fix x, v and abbreviate $F_{x,v}$ to F. If F takes different values modulo Q, it takes different values modulo p^a for some p^a exactly dividing Q. At least one coefficient of t^j in F, with $1 \le j \le d$, then has valuation at most $a-1$. The corresponding coefficient of F' has valuation at most $a - 1 + \lfloor \log_p d \rfloor$.

We claim that $e = v_p(F'(t_0)) \le E_p$ for some $t_0 \in \{0, \ldots, d-1\}$. Otherwise every value of F' at those d sites would be divisible by p^{E_p+1}. In the Lagrange interpolation formula for F' at those sites, the denominator of the term indexed by i is $(-1)^{d-1-i} i!(d-1-i)!$, which divides $(d-1)!$ up to sign. Every coefficient of F' would therefore have valuation at least $E_p + 1 - v_p((d-1)!)$, contradicting the preceding bound.

Write $m = K_p - 1 - e$. Since $m \ge E_p + 1$, expansion in integer powers of jp^m gives

$$F(t_0 + jp^m) \equiv F(t_0) + jp^{K_p - 1} \frac{F'(t_0)}{p^e} \pmod{p^{K_p}}.$$

As j runs from 0 to $p-1$, these values have identical lower digits and run through all possibilities for their highest digit. For each j use the Chinese remainder theorem to choose $t_j \in \{0, \ldots, L-1\}$ congruent to $t_0 + jp^m$ modulo p^{K_p} and to zero modulo every ℓ^{K_ℓ} with $\ell \mid Q$ and $\ell \ne p$. At the times t_j the bit $b_p(F(t_j))$ takes both values, while every other bit in the sum defining c is fixed. Thus c takes both values on the asserted segment. □

Applied to integer direction packets of increasing rank, this theorem rules out a counterexample strategy with a scalar polynomial obstruction of uniformly bounded degree and fixed modulus. Its binary length is independent of rank, so even the one-part test eventually fails.

Seven sites for the lifted directions

For the particular family of lifts, three binary tests suffice already at length seven. For $a \in \{2, 5, 8\}$ use the indicator of $a^{-1}S(x) \bmod 9 \in \{0, 2, 4, 6\}$. A lifted direction has $S(v) \equiv a \pmod 9$ for one of these three units. In that test the completed square now has coefficient one. Seven consecutive residues have squares containing $0, 1, 4$, since omitting two adjacent residues cannot omit all the roots of any of those three squares. Every translate of $\{0, 1, 4\}$ receives both colors. Indeed, $7, 8$ are the only consecutive residues of one color, and $7 + 4 \equiv 2 \pmod 9$ has the other color.

Appendix A

The localized hierarchy: supporting construction

A.1 The statement and the plan

We write $\mathbb{N} = \{1, 2, 3, \ldots\}$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. All arithmetic progressions in $\mathbb{N}$ have positive starting point and positive common difference. Their length is the number of terms, not the number of gaps.

Definition A.1.1. A set $D \subseteq \mathbb{N}$ is *k-large* if, for every coloring $c : \mathbb{N} \to \{0, \ldots, k-1\}$ and every $h \geq 2$, there are $b, d \in \mathbb{N}$ such that $d \in D$ and

$$c(b) = c(b+d) = \cdots = c(b+(h-1)d).$$

It is *large* if it is k-large for every finite k.

These notions were systematically studied by Brown, Graham and Landman [3]. We will separate every successive pair of color thresholds, with an additional restriction on where the allowed differences lie.

A subset of $\mathbb{Z}$ is *thick* if it contains intervals of arbitrarily large finite length. It is *piecewise syndetic* if finitely many of its translates have a thick union. When applying this definition to a subset of $\mathbb{N}$, we regard it as a subset of $\mathbb{Z}$. The *host* $\mathcal{H}$ below is the prescribed set in which we must construct our differences.

Theorem A.1.2 (Localized hierarchy)**.** *Let $\mathcal{H} \subseteq \mathbb{N}$ and suppose that $\mathbb{N} \setminus \mathcal{H}$ is not piecewise syndetic. For every integer $k \geq 1$, there are a set $D_{k,\mathcal{H}} \subseteq \mathcal{H}$ and a coloring*

$$c_{k,\mathcal{H}} : \mathbb{N} \longrightarrow \mathbb{Z}/(k+1)\mathbb{Z}$$

with the following properties.

1. *$D_{k,\mathcal{H}}$ is k-large. More strongly, it has the nilsystem-window property $\mathcal{P}_k$ in Definition A.2.1.*
2. *$c_{k,\mathcal{H}}$ has no monochromatic arithmetic progression of length*

$$H_k = 400 \cdot 2^k \tag{A.1.1}$$

   whose common difference belongs to $D_{k,\mathcal{H}}$.
3. *For each $m \geq 1$, the set $D_{k,\mathcal{H}} \setminus m\mathbb{N}$ is finite. In particular, $D_{k,\mathcal{H}}$ has zero upper Banach density.*

Thus $D_{k,\mathcal{H}}$ is not $(k+1)$-large. The forbidden length H_k is independent of the host $\mathcal{H}$.

The condition on $\mathcal{H}$ is equivalent to *thick syndeticity*: for each $\ell \geq 1$, the starting points of length-ℓ intervals contained in $\mathcal{H}$ have bounded gaps in $\mathbb{N}$. We prove this equivalence in Section A.2.2. Consequently the theorem can be summarized as follows:

Every thickly syndetic set contains the entire strict color hierarchy.

Here "the entire hierarchy" means that for each k a suitable subset can be chosen; no nesting of those subsets is asserted. Taking $\mathcal{H} = \mathbb{N}$ recovers the unrestricted separation.

Example A.1.3 (Prescribing excluded differences)**.** We may require all the constructed differences to avoid the squares, or any other fixed set of zero upper Banach density. Such an excluded set is not piecewise syndetic. For the squares, the number in any interval of length N is at most $\sqrt{N}+1$, so their upper Banach density is zero. The implication from zero upper Banach density to non-piecewise-syndeticity is recalled in Section A.9.4.

Remark A.1.4 (Order of the quantifiers). The theorem says that for every host $\mathcal{H}$ a suitable set can be constructed *inside that host*. It does not say that one fixed D_k retains k-largeness after every deletion of a non-piecewise-syndetic set. Indeed our $D_{k,\mathcal{H}}$ itself has zero upper Banach density, so deleting that set would leave nothing. The distinction is important in the induction.

A.1.1 Why the proof has several parts

The positive and negative conclusions ask for opposite kinds of control. To prove k-largeness we must defeat every coloring with k colors. To prove failure of $(k+1)$-largeness we must exhibit one coloring with $k+1$ colors that works simultaneously for all permitted differences.

We first solve these two problems at a finite-dimensional stage. The dimension at that stage depends on the recurrence problem being tested, but the forbidden progression length does not depend on the dimension or the host. We then extract finite sets of differences and place them at widely separated integer scales. The separation of scales permits a single obstructing coloring to combine the local ones.

There are two distinct indices throughout the argument. The integer s counts colors in the induction. The integer j counts finite packets within one induction step. The localized version requires the old s-color recurrence set to depend on j, although its forbidden length remains fixed.

Part of the proof	Its role
Uniform induction hypothesis	Permits a fresh lower-color construction in each rescaled host.
Torus directions	Supply new differences while meeting nilsystem return conditions.
Moving palette	Adds one obstructing color without losing the old obstruction.
Return-time theorem	Preserves usable differences after excluding a non-piecewise-syndetic set.
Finite extraction	Handles a compact family of recurrence tests with finitely many differences.
Mixed-radix assembly	Combines packets without increasing the number of colors.

A.1.2 External inputs and the gain over the unrestricted proof

The principal input is Theorem A of Glasscock, Koutsogiannis, Le, Moreira, Richter and Robertson [8]. We use its linear specialization, stated precisely in Section A.2.4. The nilsystem orbit-closure facts come from Leibman [14], and the countable parametrization uses rational-coordinate theory for nilpotent lattices [17, Chapter II]. The torus construction follows the idea in Alweiss's example [2]; its required finite-dimensional ingredients are proved here.

In the unrestricted proof, it suffices to lift one suitable recurrence configuration from a pronilfactor. For the localized theorem, a lifted difference must also avoid a prescribed excluded set. Theorem A gives exactly the additional control: among suitable factor return times, the failures of lifting form a set that is not piecewise syndetic. Such failures, together with the prescribed excluded set, cannot exhaust a piecewise syndetic supply of candidate differences.

The linear algebra, all coloring estimates, the set-theoretic facts about excluded sets, and the finite-packet and digit arguments are given below. The local coloring formulas and the final digit formula are explicit. The choices of lower-color sets and finite packets are existence choices; no computability claim or effective packet-size bound is made. Section A.11 explains what can instead be done with Qiu's Theorem B [16]. In particular, localization is not exclusive to Theorem A: Qiu together with the independent syndeticity lemma recorded as Lemma 3.6 of [8] also proves it. Theorem A gives the cleaner main argument and the sharper relative exceptional-set control described in Section A.6.

A.2 The induction hypothesis and the dynamical input

A.2.1 Progressions inside a prescribed nilsystem window

A finite-dimensional nilsystem is a translation T on a compact nilmanifold $Y = G/\Gamma$, where G is a nilpotent Lie group and Γ is a discrete cocompact subgroup. We allow G and Y to be disconnected. For $y \in Y$ and an open neighborhood V of y, write

$$N_T(y, V) = \{n \in \mathbb{N} : T^n y \in V\}.$$

Definition A.2.1. For $s \geq 1$, a set $D \subseteq \mathbb{N}$ has property $\mathcal{P}_s$ if, for every finite-dimensional nilsystem (Y, T), every $y \in Y$, every open neighborhood V of y, every coloring of $N_T(y, V)$ with at most s colors, and every $h \geq 2$, there is a monochromatic h-term progression in $N_T(y, V)$ with common difference in D.

Taking Y to be a one-point system shows that $\mathcal{P}_s$ implies s-largeness. The stronger formulation is needed because eliminating one color leaves the remaining colors available only at visits to an open nilfactor window. Ordinary s-largeness would not directly apply to this restricted set of times.

Lemma A.2.2 (Dilation)**.** *If B has $\mathcal{P}_s$ and $a \in \mathbb{N}$, then aB has $\mathcal{P}_s$.*

Proof. Given a coloring of $N_T(y, V)$, color $n \in N_{T^a}(y, V)$ by the original color at an. Apply $\mathcal{P}_s$ of B in the nilsystem (Y, T^a). Multiplying the resulting progression of integer times by a gives the required progression for T, with difference in aB. □

The induction assertion we actually prove is uniform over hosts:

$$\begin{aligned} (\mathrm{I}_s): \quad & \text{for every } \mathcal{H} \subseteq \mathbb{N} \text{ with non-piecewise-syndetic complement,} \\ & \text{there is } B \subseteq \mathcal{H} \text{ with } \mathcal{P}_s \text{ and an } (s+1)\text{-color} \\ & \text{obstruction of length } H_s, \text{ where } H_s \text{ depends only on } s. \end{aligned} \tag{A.2.1}$$

The additional divisibility and density conclusions will follow from the assembly. The initial stage uses an empty old set and a constant old coloring; no property involving zero colors is required.

A.2.2 Piecewise syndeticity and the calculus of excluded sets

A set $A \subseteq \mathbb{Z}$ is *syndetic* if some integer $r \geq 1$ has the property that every interval of length r meets A. For subsets of $\mathbb{N}$ we say *syndetic in* $\mathbb{N}$ when only intervals in $\mathbb{N}$ are considered. A set syndetic in $\mathbb{N}$ is piecewise syndetic as a subset of $\mathbb{Z}$: finitely many translates cover a positive half-line, which is thick. It need not be syndetic in all of $\mathbb{Z}$.

Lemma A.2.3 (Thickly syndetic complements)**.** *For $E \subseteq \mathbb{Z}$, the following are equivalent.*

1. *E is not piecewise syndetic.*
2. *For every $\ell \geq 1$, the set*

$$A_\ell = \{b \in \mathbb{Z} : \{b, b+1, \ldots, b+\ell-1\} \subseteq \mathbb{Z} \setminus E\}$$

is syndetic.

For $E \subseteq \mathbb{N}$, the corresponding one-sided condition says that $\mathbb{N} \setminus E$ is thickly syndetic in $\mathbb{N}$.

Proof. A set is not thick precisely when its complement is syndetic. Also E is piecewise syndetic precisely when, for some ℓ, the set

$$\bigcup_{i=0}^{\ell-1} (E - i)$$

is thick. Indeed every finite set of translation offsets is contained in a translate of a finite interval, and enlarging the set of offsets preserves thickness. The complement of the displayed union is exactly A_ℓ. This proves the equivalence. For subsets of $\mathbb{N}$, the finitely many boundary positions created by fixed translation offsets do not affect thickness or piecewise syndeticity. Equivalently, one can apply the same argument to positive intervals; the missing negative half-line creates no additional condition on their starting points. □

Lemma A.2.4 (Excluded-set calculus)**.** *The following facts hold.*

1. *A subset of a non-piecewise-syndetic set is not piecewise syndetic. Finite unions of non-piecewise-syndetic sets are not piecewise syndetic.*
2. *If $P \subseteq \mathbb{Z}$ is piecewise syndetic and $E \subseteq \mathbb{Z}$ is not piecewise syndetic, then $P \setminus E$ is piecewise syndetic.*
3. *If $E \subseteq \mathbb{N}$ is not piecewise syndetic and $\lambda \geq 1$, then*

$$E^{(\lambda)} = \{d \in \mathbb{N} : \lambda d \in E\}$$

is not piecewise syndetic.

Proof. The subset assertion follows because piecewise syndeticity is upward closed. For finite unions, it is enough to prove that the intersection of two thickly syndetic sets is thickly syndetic. Let $A, B \subseteq \mathbb{Z}$ be thickly syndetic and fix a desired block length ℓ. Choose r so that every interval of length r contains the start of a length-ℓ block in B. Starts of length-$(r+\ell)$ blocks in A are syndetic. Within each such A-block, the choice of r supplies a length-ℓ block in B, hence a length-ℓ block in $A \cap B$. Its start lies within r places of the start of the A-block. Therefore the starts of the common length-ℓ blocks are syndetic. Lemma A.2.3 now proves the finite-union assertion by taking complements.

For the second assertion, if $P \setminus E$ were not piecewise syndetic, then both sets on the right side of

$$P = (P \setminus E) \cup (P \cap E)$$

would be non-piecewise-syndetic. Their union would have the same property, contradicting the assumption on P.

For the final assertion, positive integer dilations preserve piecewise syndeticity. To see this directly, suppose $F + A$ is thick for a finite $F \subseteq \mathbb{Z}$. If it contains $\{b, \dots, b + N - 1\}$, then

$$\lambda A + \lambda F + \{0, 1, \dots, \lambda - 1\}$$

contains the interval $\{\lambda b, \dots, \lambda(b + N) - 1\}$. Thus λA is piecewise syndetic. If $E^{(\lambda)}$ were piecewise syndetic, so would be $\lambda E^{(\lambda)} \subseteq E$, which is impossible. □

In particular, if $\mathcal{H}$ is an allowed host, then so is every rescaled host

$$\mathcal{H}^{(\lambda)} = \{d \in \mathbb{N} : \lambda d \in \mathcal{H}\}. \tag{A.2.2}$$

Its complement is $(\mathbb{N} \setminus \mathcal{H})^{(\lambda)}$. This is the elementary fact that makes localization compatible with the integer scales used later.

A.2.3 Basic nilsystem facts

We use the following standard facts about niltranslations; see [14].

1. The orbit closure of a point under a niltranslation is a minimal nilsystem. Its forward orbit is dense in that orbit closure.
2. Such an orbit closure is a finite union of connected closed subnilmanifolds. For a translation, its connected components are cyclically permuted.
3. Products and powers of niltranslations are again niltranslations.

All topological systems below are compact metric systems whose transformations are homeomorphisms. For a minimal system (X, T), write

$$\pi : X \longrightarrow Z$$

for its maximal infinite-step pronilfactor. The space Z is an inverse limit of finite-dimensional minimal nilsystems, and every nilsystem factor of X factors through Z. Every nonempty open subset of Z contains the full inverse image of a nonempty open set in a finite nilsystem factor. For clarity, an inverse-limit cylinder may initially involve finitely many coordinates; passing to a later common coordinate combines those conditions into one open set. Surjectivity of the finite-coordinate projections guarantees that this open set can be chosen nonempty.

A.2.4 The return-time theorem and its exact use

For open sets $U_0, \dots, U_{h-1}$ in a system X, define

$$R_h^X(U_0, \dots, U_{h-1}) = \left\{ d \in \mathbb{Z} : \bigcap_{j=0}^{h-1} T^{-jd} U_j \neq \varnothing \right\}.$$

Write $R_h^X(U)$ when all the sets equal U. The integer d is a difference of orbit times, not a starting position.

Theorem A.2.5 (Glasscock–Koutsogiannis–Le–Moreira–Richter–Robertson, linear form)**.** *Let (X, T) be minimal and invertible and let $\pi : X \to Z$ be its maximal infinite-step pronilfactor. For nonempty open $U_0, \dots, U_{h-1} \subseteq X$, the set*

$$R_h^Z\big((\pi U_0)^\circ, \dots, (\pi U_{h-1})^\circ\big) \setminus R_h^X(U_0, \dots, U_{h-1}) \tag{A.2.3}$$

is not piecewise syndetic in $\mathbb{Z}$.

This is Theorem A of [8], applied to the essentially distinct polynomials $0, n, \dots, (h-1)n$. Their pairwise differences are nonconstant, as required. In particular, that theorem allows the zero polynomial in this tuple. We retain the interiors in the statement; we will not assume that images of arbitrary open sets under π are open.

Lemma A.2.6 (Lifting while excluding prescribed differences)**.** *Let $U \subseteq X$ be nonempty and open and suppose $\pi(U) = W$ is open in Z. Let $P \subseteq R_h^Z(W) \cap \mathbb{N}$ be piecewise syndetic and let $E \subseteq \mathbb{Z}$ be not piecewise syndetic. Then*

$$(P \setminus E) \cap R_h^X(U)$$

is piecewise syndetic, and in particular is nonempty.

Proof. Because $\pi(U) = W$ is open, Theorem A.2.5 says that

$$F = R_h^Z(W) \setminus R_h^X(U)$$

is not piecewise syndetic. The set $E \cup F$ has the same property by Lemma A.2.4. That lemma therefore makes $P \setminus (E \cup F)$ piecewise syndetic. Since $P \subseteq R_h^Z(W)$, this set equals the one in the assertion. □

The strength here is not merely the existence of a lifted difference. It is that an independently prescribed non-piecewise-syndetic excluded set can be removed as well. We do not replace "not piecewise syndetic" by "zero upper Banach density" for the exceptional set in Theorem A.2.5; that stronger statement is not being assumed.

A.3 The finite-stage construction

Fix a host $\mathcal{H} \subseteq \mathbb{N}$ whose complement $E = \mathbb{N} \setminus \mathcal{H}$ is not piecewise syndetic. Fix $s \geq 1$ and an old set $B \subseteq \mathcal{H}$ with $\mathcal{P}_s$. Suppose also that

$$\chi : \mathbb{N} \to \{0, \dots, s\}$$

has no monochromatic B-progression of some length $H \geq 2$. The letter H without calligraphic type denotes this old forbidden length; $\mathcal{H}$ denotes the host. We want to add one color to the positive recurrence statement and one color to the obstructing palette. Put

$$u = s + 1, \qquad q = u + 1 = s + 2, \qquad Q = 100q, \qquad \delta = Q^{-1}, \qquad \eta = (10^6 Q^2)^{-1}. \tag{A.3.1}$$

The torus dimension M will be chosen later. Fix real numbers $\alpha_1, \dots, \alpha_M$ such that $1, \alpha_1, \dots, \alpha_M$ are linearly independent over $\mathbb{Q}$. For example, one can use the square roots of distinct primes. Write $\alpha = (\alpha_1, \dots, \alpha_M)$.

For $v \in \mathbb{T}^M$, use centered representatives $\theta_j \in [-1/2, 1/2)$ for its coordinates. Let O_M consist of those v for which some index i satisfies

$$|\theta_i + \delta| < \eta, \qquad |\theta_j| < \eta \quad (j \neq i), \qquad \sum_{j \neq i} |\theta_j| < 2\delta. \tag{A.3.2}$$

All these coordinates are away from the endpoints of the centered interval, so O_M is open in $\mathbb{T}^M$. Define

$$G_M = \{d \in \mathbb{N} : d\alpha \in O_M\}, \tag{A.3.3}$$

$$B_M = \left\{ d \in B : \sum_{j=1}^{M} \|d\alpha_j\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{4\pi q H} \right\}, \tag{A.3.4}$$

$$S_M = S_M(\mathcal{H}, B) = (G_M \cap \mathcal{H}) \cup B_M. \tag{A.3.5}$$

Because $B \subseteq \mathcal{H}$, the entire set S_M lies in $\mathcal{H}$. Only the new torus directions need to be explicitly intersected with the host.

A new direction has one coordinate of size approximately $-1/Q$, while its other coordinates are individually much smaller and have small total absolute size. An old direction in B_M moves the entire torus only slightly. The same coloring will handle these two types of motion.

Proposition A.3.1 (Localized finite-stage step)**.** *Let $h \geq 2$ and let $C \geq 0$ bound the nilmanifold dimension. For all sufficiently large M, depending only on h, C, q, the following hold.*

1. *Every coloring with at most $s+1$ colors of a return-time set $N_T(y, V)$ in a nilsystem of dimension at most C contains a monochromatic h-term progression with difference in S_M.*
2. *There is a q-coloring κ_M of $\mathbb{N}$ with no monochromatic S_M-progression of length*

$$L = \max\{2Q, H\} = \max\{200q, H\}. \tag{A.3.6}$$

The bound on M is independent of the host and of B. The length L is independent of M, h, C and of the host.

For the initial step the same conclusions hold with $s = 0$, $B = \varnothing$, $H = 2$, $u = 1$, $q = 2$, and a constant input coloring χ. No assumption $\mathcal{P}_0$ is made.

Section A.4 produces a syndetic supply of candidate torus directions compatible with the nilsystem window. Section A.5 constructs the obstruction even for the larger set $G_M \cup B_M$. Section A.6 uses Theorem A.2.5 to retain a suitable difference in the host.

A.4 Why the torus has enough useful directions

A.4.1 A linear-algebra lemma

The following elementary fact explains why increasing M helps. A subspace of small codimension cannot exclude all vectors having one distinguished coordinate and many individually small correction coordinates.

Lemma A.4.1. *Fix $b \geq 0$ and the positive numbers δ, η from (A.3.1). For all sufficiently large M, every real linear subspace $V \subseteq \mathbb{R}^M$ of codimension at most b contains a vector θ such that, for some i,*

$$\theta_i = -\delta, \qquad \max_{j \neq i} |\theta_j| < \eta, \qquad \sum_{j \neq i} |\theta_j| < 2\delta.$$

Consequently every connected subtorus of $\mathbb{T}^M$ of codimension at most b meets O_M.

Proof. Let $E = \mathbb{R}^M/V$ and let v_j be the image of the jth coordinate vector in E. Write $r = \dim E \leq b$. If $r = 0$, take $\theta = -\delta e_i$ and there is nothing more to prove. We may therefore assume $1 \leq r \leq b$.

Normalize the columns. Among all bases of E formed from r of the vectors v_j, choose one with maximal absolute determinant, computed in any fixed initial coordinates. Use these selected vectors as a new basis. Every v_j now has all coordinates in $[-1,1]$. Indeed, Cramer's rule expresses each coordinate as a determinant obtained by replacing one selected basis column, divided by the selected maximal determinant. Its absolute value cannot exceed one. This normalization is what makes the eventual bound independent of the particular subspace V.

Find many close columns. Put

$$\beta = \eta/\delta, \qquad R = \lceil 8/\beta \rceil, \qquad \tau_0 = \frac{\beta}{8b}, \qquad K = \left\lceil \frac{16b}{\beta} \right\rceil.$$

Here $0 < \beta < 1$. Partition $[-1,1]^r$ into at most $K^r \le K^b$ boxes of side length at most τ_0. If $M > RK^b$, one box contains at least $R+1$ indexed columns. Repeated vectors cause no problem; the indices, rather than the vector values, are what we count. Call one column v_i, let I be the indices of R other columns in that box, and put

$$\bar{v} = \frac{1}{R} \sum_{j \in I} v_j.$$

Each coordinate of $v_i - \bar{v}$ in our selected basis has absolute value at most τ_0. Consequently

$$v_i - \bar{v} = \sum_j c_j v_j, \qquad \sum_j |c_j| \le r\tau_0 \le \beta/8, \qquad \max_j |c_j| \le \tau_0,$$

where the c_j are zero outside the selected basis indices.

Correct the average exactly. Move $c_i v_i$ to the left and divide by $1 - c_i \ge 1 - \beta/8 > 0$. This gives

$$v_i = \sum_{j \ne i} t_j v_j, \qquad t_j = \frac{R^{-1} 1_I(j) + c_j}{1 - c_i}.$$

Using $R^{-1} \le \beta/8$ and $\tau_0 \le \beta/8$, we obtain

$$\sum_{j \ne i} |t_j| \le \frac{1 + \beta/8}{1 - \beta/8} \le \frac{9}{7} < 2, \qquad \max_{j \ne i} |t_j| \le \frac{\beta/4}{1 - \beta/8} \le \frac{2\beta}{7} < \beta.$$

Set $\theta_i = -\delta$ and $\theta_j = \delta t_j$ for $j \ne i$. Its image in E is zero, so $\theta \in V$, and the required bounds follow.

For the last assertion, apply the subspace statement to the Lie algebra of a connected subtorus and reduce θ modulo $\mathbb{Z}^M$. The coordinates just constructed satisfy (A.3.2). □

A.4.2 A syndetic supply of return steps with the desired torus motion

Lemma A.4.2. *Fix $h \ge 2$ and $C \ge 0$. Choose M large enough for Lemma A.4.1 with $b = hC$. For every nilsystem (Y,T) of dimension at most C, every $y \in Y$, and every open neighborhood V of y, the set*

$$P_h(y,V) = \{d \in \mathbb{N} : d\alpha \in O_M,\ T^{jd} y \in V\ (1 \le j \le h)\} \tag{A.4.1}$$

is syndetic in $\mathbb{N}$. In particular it is piecewise syndetic as a subset of $\mathbb{Z}$.

Proof. Consider the orbit of $(y, \dots, y, 0)$ in $Y^h \times \mathbb{T}^M$ under

$$(T, T^2, \dots, T^h, R_\alpha).$$

Let K be its orbit closure and let K_0 be the connected component containing the starting point. By the nilsystem facts above, K is minimal and is a finite union of connected closed subnilmanifolds.

The projection of K onto $\mathbb{T}^M$ is the whole torus, because $1, \alpha_1, \dots, \alpha_M$ are rationally independent. The projections of the finitely many components of K are translates of the same connected subtorus. If that subtorus were proper, finitely many of its cosets could not cover $\mathbb{T}^M$. For example, each such coset has Haar measure zero. Thus K_0 projects onto all of $\mathbb{T}^M$.

Present Y with a discrete lattice, so that the dimension of its ambient Lie group is $\dim Y \le C$. Write

$$K_0 = L \cdot (y, \dots, y, 0)$$

for a connected closed subgroup L of the product ambient group, and let $\mathfrak{l} = \mathrm{Lie}(L)$. Its projection onto the M torus directions is surjective. In particular $\dim \mathfrak{l} \ge M$. The first h factors together have Lie algebra dimension at most hC, so the kernel of their projection from $\mathfrak{l}$ has dimension at least $M - hC$. Equivalently, the pure-torus subspace

$$E_0 = \{v \in \mathbb{R}^M : (0, \dots, 0, v) \in \mathfrak{l}\}$$

has dimension at least $M - hC$.

Exponentiate E_0 in the torus and take its closure H_0. This is a connected subtorus of codimension at most hC. The corresponding group elements fix every Y-coordinate of the starting point. Since K_0 is closed, we obtain

$$\{(y, \dots, y)\} \times H_0 \subseteq K_0.$$

Lemma A.4.1 gives $H_0 \cap O_M \neq \varnothing$. Consequently the relatively open set

$$K \cap (V^h \times O_M)$$

is nonempty. Let S denote the product transformation on K and let $\mathcal{O} = K \cap (V^h \times O_M)$. Minimality implies that every forward orbit enters the nonempty open set $\mathcal{O}$. Consequently the sets $S^{-t}\mathcal{O}$, $t \ge 0$, cover K. By compactness, finitely many of them cover K; choose R larger than all their indices. For every $n \ge 1$, the point $S^n(y, \dots, y, 0)$ therefore enters $\mathcal{O}$ after at most $R - 1$ further iterates. Thus every interval $\{n, \dots, n + R - 1\}$ meets $P_h(y, V)$, proving the syndeticity assertion. No uniform bound on R is claimed. □

The useful feature is the dependence of M only on h, C, q. It does not depend on the translation, the point, the size of the open window, or any coloring. The small window may make the required return time very large, but it does not require a larger torus dimension.

Remark A.4.3 (Why the set of candidates, not just one candidate, matters). The geometric argument has not changed when we prescribe a host. It first supplies the same syndetic set $P_h(y, V)$ independently of the host and the coloring. Only afterward do we exclude the host's complement and the exceptional lifting set. Lemma A.2.4 guarantees that a piecewise syndetic set of candidates remains. Selecting one candidate before the lifting argument would not provide this freedom.

A.5 The local coloring: one moving empty color

We now prove the obstructing-coloring part of Proposition A.3.1. This part works for every M and is entirely elementary. In fact the coloring blocks $G_M \cup B_M$, so it also blocks its subset $S_M = (G_M \cap \mathcal{H}) \cup B_M$.

A.5.1 The palette before the formula

There are u old input colors and $q = u + 1$ new output colors. At each integer position we will choose an injection of the old palette into the new palette. Thus exactly one new color is unused at that position. The unused color changes as the position changes.

We need two features. On a G_M-progression, the unused color must eventually run through all q possibilities, which prevents any fixed output color from appearing at every site. On a B_M-progression, the injection must change so little that a constant output still forces a constant input.

Here is the general family of injections. For $J \in \mathbb{Z}$ and $i \in \{0, \dots, u-1\}$, choose the unique integer

$$t \in \{J-u, J-u+1, \dots, J-1\} \quad \text{with} \quad t \equiv i \pmod{u},$$

and put

$$\iota_J(i) = t - 1 \pmod{q}. \tag{A.5.1}$$

Example A.5.1 (Two input colors and three output colors)**.** For $u = 2$ and $q = 3$, the injections repeat after six steps.

$J \pmod 6$	Output of input 0	Output of input 1	Missing output
0	0	1	2
1	2	1	0
2	2	0	1
3	1	0	2
4	1	2	0
5	0	2	1

In moving down one row, one input moves into the previously empty output color. An output that occurs in both rows retains the same input label.

Lemma A.5.2. *The map ι_J is injective and omits exactly the color $J - 1 \pmod q$. Moreover,*

$$|J - J'| \le 1, \qquad \iota_J(i) = \iota_{J'}(i') \quad \Longrightarrow \quad i = i'. \tag{A.5.2}$$

Proof. The integer outputs before reduction modulo q are

$$J-u-1, J-u, \dots, J-2 = J-q, J-q+1, \dots, J-2.$$

They are $q - 1$ consecutive integers, so they are distinct modulo q and omit $J - 1$. When J is increased to $J + 1$, the representative $J - u$ leaves the input interval and J enters it. These two representatives correspond to the same input because their difference is u. Every other input keeps its representative and therefore its output. The changing input moves from output $J \pmod q$ to output $J - 1 \pmod q$, the previously missing color. Its old output is missing in the new chart. Thus every output present in both consecutive charts belongs to the same input in both. Together with injectivity within a chart, this proves (A.5.2). □

A.5.2 Choosing the chart at an integer

Let $\mathrm{e}(t) = \exp(2\pi i t)$ and define

$$\Phi(n) = \sum_{j=1}^{M} \mathrm{Re}(\mathrm{e}(n\alpha_j) - 1), \qquad J(n) = \lfloor q\Phi(n) \rfloor. \tag{A.5.3}$$

The new coloring is

$$\kappa_M(n) = \iota_{J(n)}(\chi(n)) \in \mathbb{Z}/q\mathbb{Z}. \tag{A.5.4}$$

We use the elementary inequality

$$|\mathrm{e}(t) - 1| \le 2\pi \|t\|_{\mathbb{R}/\mathbb{Z}}. \tag{A.5.5}$$

A.5.3 Old differences: the old obstruction is preserved

Let $d \in B_M$ and consider the first H terms of a progression with starting point b. For any $0 \le r, t < H$, the triangle inequality and (A.5.5) give

$$|q\Phi(b + td) - q\Phi(b + rd)| \le 2\pi q|t - r| \sum_{j=1}^{M} \|d\alpha_j\|_{\mathbb{R}/\mathbb{Z}} \tag{A.5.6}$$

$$< \frac{H - 1}{2H} < \frac{1}{2}. \tag{A.5.7}$$

This is a pairwise estimate, so all the real numbers $q\Phi(b + td)$ lie in an interval of length less than $1/2$. Their floors therefore lie in at most two consecutive charts.

If the output colors $\kappa_M(b+td)$ were all equal, Lemma A.5.2 would force all input colors $\chi(b+td)$ to be equal. Because $d \in B$, this contradicts the old length-H obstruction. Thus no B_M-progression of length H is monochromatic for κ_M.

A.5.4 New differences: every output color becomes unavailable

Let $d \in G_M$ and let θ_j be the centered coordinates of $d\alpha$. Choose the distinguished coordinate i from (A.3.2). We will show that on every progression of length $2Q$, the integers $J(n)$ visit every residue class modulo q. Since chart J omits color $J - 1$, this will rule out monochromaticity independently of the input coloring χ.

First estimate: the chart moves at most one step at a time. For consecutive sites,

$$|q\Phi(b + (r + 1)d) - q\Phi(b + rd)| \le 2\pi q \sum_{j} |\theta_j| < 2\pi q(3/Q + \eta) < 1. \tag{A.5.8}$$

Consequently $|J(b + (r + 1)d) - J(b + rd)| \le 1$.

Second estimate: the real values nevertheless have a large range. Choose an integer $0 \le t < Q$ such that

$$\|b\alpha_i - t/Q\|_{\mathbb{R}/\mathbb{Z}} \le \frac{1}{2Q}.$$

Such a t exists because the Q points $0, 1/Q, \dots, (Q-1)/Q$ are evenly spaced around the circle. Write $\theta_i = -1/Q + \epsilon$, where $|\epsilon| < \eta$. Then

$$\|b\alpha_i + t\theta_i\|_{\mathbb{R}/\mathbb{Z}} \le \frac{1}{2Q} + Q\eta < \frac{1}{Q}.$$

Since $Q \ge 200$, the phase

$$z = \mathrm{e}(b\alpha_i + t\theta_i)$$

has real part greater than 0.99.

Set $m = Q/2$, which is an integer. The three indices $t, t+m, t+2m$ all lie in $\{0, \dots, 2Q-1\}$. Consider the second difference

$$D = \Phi(b + td) - 2\Phi(b + (t+m)d) + \Phi(b + (t+2m)d).$$

Its distinguished-coordinate contribution is

$$\mathrm{Re}\big(z(\mathrm{e}(m\theta_i) - 1)^2\big).$$

Since $m\theta_i = -1/2 + m\epsilon$, we have

$$|\mathrm{e}(m\theta_i) + 1| \le \pi Q\eta.$$

Write $\mathrm{e}(m\theta_i) = -1 + w$. Expanding $(-2+w)^2$ shows that the distinguished contribution exceeds

$$3.96 - 4\pi Q\eta - \pi^2 Q^2 \eta^2.$$

The remaining coordinates have total contribution of absolute value at most

$$\sum_{j \ne i} |\mathrm{e}(m\theta_j) - 1|^2 \le \pi^2 Q^2 \sum_{j \ne i} \theta_j^2 \tag{A.5.9}$$

$$\le \pi^2 Q^2 \left(\max_{j \ne i} |\theta_j|\right) \sum_{j \ne i} |\theta_j| \tag{A.5.10}$$

$$< 2\pi^2 Q\eta. \tag{A.5.11}$$

Our choice $\eta = (10^6 Q^2)^{-1}$ therefore gives $D > 3.9$. If R is the range of Φ on the $2Q$ sites, then $D \le 2R$. Hence

$$R > 1.95.$$

The range of $q\Phi$ exceeds $1.95q > q$. Together with (A.5.8), this forces the integer sequence J to visit at least q consecutive integers. Indeed, a sequence of integers whose adjacent changes have magnitude at most one must pass through every integer between its minimum and maximum. Thus every residue modulo q is visited, and every output color is absent at some site. No G_M-progression of length $2Q$ is monochromatic.

Combining the old and new differences proves the second assertion of Proposition A.3.1, with $L = \max\{2Q, H\}$.

Remark A.5.3 (Why the dimension disappears). The first-difference bound uses the total sum $\sum_j |\theta_j|$. The second-difference error uses the maximum small coordinate times the sum of the small coordinates. Neither bound contains a factor M. This is why the same forbidden length works even as the auxiliary torus dimension increases.

A.6 Forcing one additional color inside the host

We prove the positive assertion of Proposition A.3.1. Fix h, C, choose M by Lemma A.4.2, and let c be an $(s+1)$-coloring of $N_T(y, V)$ in a nilsystem of dimension at most C. Extend c to a two-sided sequence $\omega \in \{1, \dots, s+1\}^{\mathbb{Z}}$ in any way. The colors outside the prescribed positive return-time set will not enter the final progression.

A.6.1 The joined system and its factors

Let Y_0 be the orbit closure of y and put $V_0 = V \cap Y_0$. Let σ be the left shift, so $(\sigma\omega)(n) = \omega(n+1)$. Consider the point

$$(\omega, y) \in \{1, \dots, s+1\}^{\mathbb{Z}} \times Y_0$$

under the product transformation (σ, T). Its forward limit set is the nonempty compact invariant set

$$\bigcap_{N \geq 1} \overline{\{(\sigma^n \omega, T^n y) : n \geq N\}}.$$

Choose a nonempty minimal subsystem X of that limit set. Its projection onto Y_0 is a nonempty compact invariant subset of the minimal nilsystem Y_0, hence is all of Y_0. Writing T also for the transformation on X, factor that projection as

$$X \xrightarrow{\pi} Z \xrightarrow{\rho} Y_0,$$

where Z is the maximal infinite-step pronilfactor of X. This factorization is available because Y_0 is a nilsystem factor.

Let $C_1 \subseteq X$ be the clopen cylinder where the symbol at coordinate zero is the first color. Put

$$W = \rho^{-1}(V_0).$$

This is a nonempty open set in Z. We do not join the auxiliary torus to X at this point. The return-time theorem will control the difference directly, so there is no need to recover its torus displacement from two orbit coordinates.

There are two cases: either the first color projects over the entire factor window W, or a nonempty open part of that window has no points of the first color above it.

A.6.2 First case: the first color covers the factor window

Assume $W \subseteq \pi(C_1)$ and set

$$U = C_1 \cap \pi^{-1}(W).$$

Then U is nonempty and open, and

$$\pi(U) = W. \tag{A.6.1}$$

To check the reverse inclusion in this equality, take $z \in W$. The covering assumption supplies a point of C_1 above z, and that point automatically belongs to $\pi^{-1}(W)$. Thus the interiors in Theorem A.2.5 cause no loss here.

Surjectivity of ρ gives

$$R_h^Z(W) = R_h^{Y_0}(V_0). \tag{A.6.2}$$

One inclusion follows by projecting a configuration. For the other, lift its starting point arbitrarily through ρ; equivariance puts every lifted iterate in the full inverse image W.

Apply Lemma A.4.2 to the original pointed nilsystem (Y, T, y) and its window V. Its syndetic set $P = P_h(y, V)$ consists of positive differences in G_M. Since $y \in V$ and all its relevant iterates remain in Y_0, it also satisfies

$$P \subseteq R_h^{Y_0}(V_0) = R_h^Z(W).$$

There is no need to apply the geometric lemma to a larger factor or to a nilsystem presentation of Y_0.

Let $E = \mathbb{N} \setminus \mathcal{H}$. Lemma A.2.6, with this excluded set, shows that

$$P \cap \mathcal{H} \cap R_h^X(U) \tag{A.6.3}$$

is piecewise syndetic. Choose d in that set. Then $d \in G_M \cap \mathcal{H} \subseteq S_M$, and there is $x \in X$ such that

$$x, T^d x, \ldots, T^{(h-1)d} x \in U.$$

These points all have the first color and all project to V_0. Notice that we have obtained a piecewise syndetic set of usable differences in this case, not only one. There is also a stronger relative statement:

$$P \setminus \left(\mathcal{H} \cap R_h^X(U)\right) \quad \text{is not piecewise syndetic.} \tag{A.6.4}$$

Indeed this loss is contained in the union of E and the exceptional set in Theorem A.2.5. Consequently every piecewise syndetic subset of P still has a piecewise syndetic intersection with the usable differences. A conclusion giving merely some syndetic set of lifted differences does not give this control relative to every such subset of P.

A.6.3 Second case: remove the first color

Assume $W \not\subseteq \pi(C_1)$. The set $\pi(C_1)$ is compact, hence closed, because C_1 is compact. Therefore $W \setminus \pi(C_1)$ is nonempty and open. Choose a finite nilsystem factor $\zeta : Z \to Z'$ and a nonempty open set $V' \subseteq Z'$ such that

$$\zeta^{-1}(V') \subseteq W \setminus \pi(C_1). \tag{A.6.5}$$

Every point of X above this cylinder has one of the remaining s colors, and its Y_0-coordinate belongs to V_0.

Choose $x_0 \in X$ with $z' = \zeta\pi(x_0) \in V'$. Choose an open neighborhood O of 0 in $\mathbb{T}^M$ so small that

$$\sum_{j=1}^{M} \|v_j\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{4\pi q H} \quad \text{whenever } v \in O - O. \tag{A.6.6}$$

For example, a sufficiently small product of centered intervals has this property, by the triangle inequality. Consider visits of $(z', 0)$ to $V' \times O$ under the product niltranslation (T, R_α). Color a visiting time n by the symbol at coordinate zero of $T^n x_0$. By (A.6.5), this uses at most s colors.

Property $\mathcal{P}_s$ of B supplies positive times

$$b, b + d, \ldots, b + (h-1)d$$

which are such visits, all have the same color, and have $d \in B$. The first two torus visits show that $d\alpha \in O - O$. Consequently (A.6.6) puts d in B_M. Since $B \subseteq \mathcal{H}$ was chosen at the start of this finite stage, the difference also lies in the host. The corresponding configuration in X has one color and projects into V_0.

A.6.4 Returning to the original coloring

In either case there is a point $x \in X$ and a fixed positive difference $d \in S_M$ such that

$$x, T^d x, \dots, T^{(h-1)d} x$$

have one color and project into V_0. Specifying that color at the finitely many coordinates $0, d, \dots, (h-1)d$ is a clopen symbolic condition. Requiring the corresponding Y_0-coordinates to lie in V_0 is an open condition. Thus the configuration is witnessed by an open neighborhood of x in the ambient joined system.

Because X lies in the forward limit set of (ω, y), this neighborhood is visited at arbitrarily late positive times a along the original joined orbit. At each such time,

$$a, a + d, \dots, a + (h-1)d$$

are positive visits of y to V and have equal original colors. All their colors were part of the original coloring of $N_T(y, V)$, not of its arbitrary extension. This proves the positive assertion of Proposition A.3.1.

For $s = 0$, the symbolic sequence has only one color, so $C_1 = X$ and the second case cannot occur. The first case proves the initial assertion without any hypothesis $\mathcal{P}_0$.

Remark A.6.1 (Uniformity and its limits). The geometric dimension bound is applied only to the original nilmanifold of dimension at most C. The finite nilfactor Z' in the second case can have larger dimension; this is harmless because $\mathcal{P}_s$ already concerns every finite-dimensional nilsystem. Thus M depends only on h, C, q, not on the host, the coloring, or the symbolic extension. The piecewise-syndetic conclusion (A.6.3) is specific to the first case. We do not claim that the full set of successful differences is piecewise syndetic in the second case, or after the final packet assembly.

Remark A.6.2 (A simplification provided by the stronger theorem). Compared with the Qiu-only argument, the main proof no longer needs an initial joining with $\mathbb{T}^M$, open phase targets P_0, P_1, or a separate device for obtaining a positive lifted difference. The candidate set P already consists of positive integers with the required torus motion, and Theorem A.2.5 allows us to choose the lifted difference directly from it. The auxiliary torus is still used in the geometry, the coloring, and the second-case restriction to B_M.

A.7 Finite packets and the passage to one set of integers

Proposition A.3.1 provides a useful set $S_M(\mathcal{H}, B)$ for each fixed target length, dimension bound, host, and old set. Taking a union of these sets would not preserve the obstruction: the colorings κ_M change with M. Instead, we extract a finite packet from each stage and place those packets at carefully chosen integer scales.

There are two separate tasks. First, a finite packet must handle an entire compact family of recurrence tests. Second, the local obstructing colorings must be assembled without increasing the number of colors. We treat these tasks in that order.

A.7.1 A countable list of compact tests

The following parametrization lemma is proved in Appendix A.12. Its useful simplification is that every test starts at an identity coset. There is no variable starting-point parameter.

Lemma A.7.1 (Countable tests)**.** *There is a countable list of data*

$$\mathcal{T} = (N, \Lambda, A, m, K, V) \tag{A.7.1}$$

with the following properties. The group N is connected, simply connected and nilpotent, Λ is a lattice, and A is a lattice-preserving unipotent automorphism. The integer m is positive, $K \subseteq N$ is compact, and V is an open neighborhood of $o = e\Lambda$ in N/Λ. For $b \in K$, let

$$R_b(z\Lambda) = bA(z)\Lambda.$$

This is a niltranslation in a possibly disconnected presentation, and every power R_b^e is a niltranslation on a nilmanifold of the same dimension $\dim N$.

Every pointed nilsystem (Y, T, y) with an open neighborhood U of y is represented by one of these tests after restricting to multiples of its integer m: there are $b \in K$ and an identification $\phi : N/\Lambda \to Y_y$, where Y_y is the component of y, such that

$$\phi(o) = y, \qquad \phi \circ R_b = T^m \circ \phi, \qquad \phi(V) \subseteq U.$$

Fix an enumeration $\mathcal{T}_1, \mathcal{T}_2, \dots$ of these tests. At every stage we also include the one-point nilsystem with its whole space as window and $m = 1$. This ensures that the finite packets extracted below are nonempty.

A.7.2 What is fixed before a finite packet is chosen

Fix the desired global host $\mathcal{H}$ and one induction level s. For $s \geq 1$, assume the full statement (I_s) in (A.2.1), including its uniform forbidden length H_s. Set

$$H = H_s, \qquad q = s + 2, \qquad L = \max\{200q, H\}.$$

For $s = 0$, use $H = 2$, $q = 2$, and empty old sets at every stage. The numbers s, q, H, L are fixed throughout this assembly. The old recurrence sets need not be fixed.

Start with $\lambda_1 = 1$. At stage j, the scale λ_j has already been chosen. Define the local host

$$\mathcal{H}_j = \{d \in \mathbb{N} : \lambda_j d \in \mathcal{H}\}. \tag{A.7.2}$$

Its complement is not piecewise syndetic by Lemma A.2.4. Apply (I_s) to this host to choose

$$B_j \subseteq \mathcal{H}_j, \qquad B_j \text{ has } \mathcal{P}_s, \tag{A.7.3}$$

and a coloring $\chi_j : \mathbb{N} \to \mathbb{Z}/(s+1)\mathbb{Z}$ blocking all B_j-progressions of length H_s. Identify its colors with $\{0, \dots, s\}$ when applying the moving-palette formula. For $s = 0$, instead take $B_j = \varnothing$ and χ_j constant.

Set $h_j = j + 2$. From the first j compact tests, retain those whose integer m divides λ_j, and include the one-point test. Let C_j be the largest dimension of those tests. Choose M_j sufficiently large for Proposition A.3.1, and apply it with host $\mathcal{H}_j$, old set B_j, and input coloring χ_j. Write

$$S_j = S_{M_j}(\mathcal{H}_j, B_j) \subseteq \mathcal{H}_j \tag{A.7.4}$$

and let κ_j be the resulting local q-coloring. Every κ_j blocks its set S_j at the same length L.

For a retained test, use the transformation

$$R_b^{e_j}, \qquad e_j = \lambda_j / m.$$

Its exponent is fixed before compactness is applied. Although the exponent may be large, the nilmanifold dimension is unchanged.

Remark A.7.2 (Why the old set must be allowed to vary). Reusing a single old set B would require it to supply differences avoiding every excluded pullback $(\mathbb{N} \setminus \mathcal{H})^{(\lambda_j)}$. The non-piecewise-syndeticity of each individual pullback does not imply that their countable union is non-piecewise-syndetic. Nor does $\mathcal{P}_s$ justify deleting such sets from a fixed B. The correct induction is therefore uniform over hosts: choose B_j afresh inside $\mathcal{H}_j$. There is no circularity, because each such choice uses only the already established color level s, not the level $s+1$ under construction.

A.7.3 Why a finite packet works uniformly

Fix one retained test (N, Λ, A, m, K, V). Consider the compact parameter space

$$K \times \{1, \dots, s+1\}^{\mathbb{Z}}. \tag{A.7.5}$$

The first coordinate chooses the translation parameter b; the second chooses an arbitrary coloring ω of the integer times. For fixed positive integers a, d, with $d \in S_{M_j}$, impose the conditions

$$\omega(a) = \omega(a+d) = \cdots = \omega(a+(h_j-1)d), \qquad (R_b^{e_j})^{a+td} o \in V \quad (0 \le t < h_j). \tag{A.7.6}$$

They define an open subset of (A.7.5). The color equalities inspect only finitely many coordinates, so they are clopen conditions. Each fixed iterate depends continuously on b, so the window conditions are open.

Proposition A.3.1 says that every parameter in (A.7.5) satisfies (A.7.6) for some positive a and some $d \in S_{M_j}$. Thus these open sets cover the compact parameter space. Choose a finite subcover and retain the finitely many differences occurring in it. Repeat for the finitely many tests at this stage, and let F_j be the union of the retained differences. Then

$$\varnothing \neq F_j \subseteq S_j \subseteq \mathcal{H}_j, \qquad |F_j| < \infty. \tag{A.7.7}$$

The packet F_j works for every translation parameter and every coloring in every retained test. Because $F_j \subseteq S_j$, the same local coloring κ_j blocks F_j-progressions of length L.

The minimal symbolic extension and the finite nilfactor appearing in the proof of Proposition A.3.1 can depend on the coloring. They are not part of the parameter space (A.7.5). Compactness is applied only after a finite configuration has been obtained in the original nilsystem and coloring.

A.7.4 Choosing the next scale

After choosing F_j, choose an integer radix r_j such that

$$r_j > 2L \max F_j, \qquad (j+1)! \mid r_j, \tag{A.7.8}$$

and put

$$\lambda_{j+1} = r_j \lambda_j. \tag{A.7.9}$$

For example, $r_j = (j+1)!(2L \max F_j + 1)$ is an admissible choice. By induction,

$$j! \mid \lambda_j \qquad (j \ge 1). \tag{A.7.10}$$

Define the new set by

$$D = \bigcup_{j \ge 1} \lambda_j F_j. \tag{A.7.11}$$

The order of choices is important:

$$\lambda_j \longrightarrow \mathcal{H}_j \longrightarrow (B_j, \chi_j) \longrightarrow S_j \longrightarrow F_j \longrightarrow (r_j, \lambda_{j+1}).$$

The retained tests and the choice of M_j occur before S_j is formed. There is no need to choose a radix before knowing how large the finite packet is. The finite packet has already been selected inside $\mathcal{H}_j$, so choosing a larger later radix cannot spoil membership of $\lambda_j F_j$ in the global host.

A.7.5 The set D has the stronger recurrence property

Proposition A.7.3. *The set D in (A.7.11) is contained in $\mathcal{H}$ and has property $\mathcal{P}_{s+1}$. For the initial step $s = 0$, it has $\mathcal{P}_1$.*

Proof. For each j, the inclusions $F_j \subseteq \mathcal{H}_j$ and the definition (A.7.2) imply $\lambda_j F_j \subseteq \mathcal{H}$. Taking the union shows $D \subseteq \mathcal{H}$.

Fix a pointed nilsystem (Y, T, y), an open neighborhood U of y, a coloring of $N_T(y, U)$ with $s+1$ colors, and a desired length h. Use Lemma A.7.1 to select a test (N, Λ, A, m, K, V) and a parameter $b \in K$ representing the component return map and a window inside U. Choose j large enough that this test is among the first j tests, $j \geq m$, and $h_j \geq h$. By (A.7.10), $m \mid \lambda_j$, so the test is retained.

For $n \geq 1$, read the original coloring at time $\lambda_j n$. Whenever

$$(R_b^{\lambda_j/m})^n o \in V,$$

the original time $\lambda_j n$ is a visit to U, so its color is defined. Extend this coloring arbitrarily to all integer indices n. The defining property of F_j gives a monochromatic progression

$$a, a+d, \ldots, a+(h_j-1)d, \qquad a > 0, \quad d \in F_j,$$

all of whose iterates under $R_b^{\lambda_j/m}$ visit V. In the original system these are the positive times

$$\lambda_j a, \lambda_j a + \lambda_j d, \ldots, \lambda_j a + (h_j - 1)\lambda_j d.$$

They visit U, have the same original color, and have common difference $\lambda_j d \in D$. Taking the first h terms proves the assertion. The proof is identical for the initial one-color stage. □

The proof uses only the recurrence guarantee of each packet, not any compatibility between the old sets B_j. The obstruction also needs no such compatibility: it will use only the common output group $\mathbb{Z}/q\mathbb{Z}$ and the common forbidden length L.

A.8 Combining the local colorings by digits

We now construct a single q-coloring that blocks the whole union (A.7.11). This is a general device: it uses only the finite packets, their local colorings, and the radix inequality in (A.7.8).

A.8.1 Mixed-radix notation

Every nonnegative integer has a unique expansion

$$n = \sum_{j \geq 1} z_j(n)\lambda_j, \qquad 0 \leq z_j(n) < r_j. \tag{A.8.1}$$

Only finitely many digits are nonzero. This is the usual positional expansion, except that the radix at place j is r_j rather than a fixed radix such as ten. Equivalently,

$$z_j(n) = \left\lfloor \frac{n}{\lambda_j} \right\rfloor \pmod{r_j}, \qquad 0 \leq z_j(n) < r_j.$$

Use the additive group $\mathbb{Z}/q\mathbb{Z}$ as the output palette and define

$$c(n) = \sum_{j \geq 1} \bigl(\kappa_j(1 + z_j(n)) - \kappa_j(1)\bigr) \pmod{q}. \tag{A.8.2}$$

The subtraction makes a zero digit contribute zero. Therefore the sum is finite for each n, and it uses exactly the same q output labels as the local colorings. The shift by one makes all arguments of κ_j positive.

A.8.2 Why carries do not destroy the obstruction

Proposition A.8.1 (Digit assembly)**.** *The coloring* (A.8.2) *has no monochromatic D-progression of length $2L$.*

Proof. Suppose, to the contrary, that a progression of length $2L$ and common difference $d \in D$ is monochromatic. Choose j and $e \in F_j$ such that $d = \lambda_j e$. Write its sites as

$$n_t = n_0 + t\lambda_j e, \qquad 0 \leq t < 2L.$$

The lower digits are fixed. All n_t have the same residue modulo λ_j. Thus every digit in a place below j is constant on the progression.

The higher prefix changes at most once. The total span is

$$n_{2L-1} - n_0 = (2L-1)\lambda_j e < \lambda_j r_j = \lambda_{j+1}.$$

Consequently the integer

$$p_t = \left\lfloor \frac{n_t}{\lambda_{j+1}} \right\rfloor$$

has at most two values and changes at most once. The $2L$ sites split into at most two consecutive runs on which p_t is constant. One of those runs has at least L sites. On this run all digits above place j are constant. This argument includes a carry that propagates through several higher places, because p_t records all the higher digits at once.

The remaining digit sees the local obstruction. On the chosen run, both the lower residue and the higher prefix are fixed. Therefore the jth digit increases by exactly e at each step, with no wrap inside that run. All summands in (A.8.2) except the jth are constant. Monochromaticity of the global coloring forces

$$\kappa_j(1+z),\ \kappa_j(1+z+e),\ \ldots,\kappa_j(1+z+(L-1)e)$$

to be constant for some nonnegative digit z. This is a forbidden local progression of length L, since $e \in F_j \subseteq S_j$. □

Example A.8.2 (One carry is harmless)**.** For an illustration, take $L = 4$, $e = 3$, and a radix $r_j = 32 > 2Le$. Eight successive jth digits might be

$$26,\ 29,\ 0,\ 3,\ 6,\ 9,\ 12,\ 15.$$

The higher prefix changes between the second and third sites and stays fixed afterward. The final six sites contain a four-term run on which the local digit is an ordinary arithmetic progression of difference 3. The proof uses exactly this observation for arbitrary radices and packet differences.

Remark A.8.3 (Why finite extraction is essential)*.* The choice $r_j > 2L \max F_j$ requires a finite upper bound for the differences used at one digit place. For an entire infinite set S_j, no finite radix would provide this control. Finite extraction is therefore part of the mechanism that preserves the obstructing palette.

A.9 Completing the localized induction

We now collect the construction and verify the numerical bound. At each color level the countable test list is fixed as in Lemma A.7.1, and the packet procedure is run for the prescribed host.

A.9.1 The base case for every host

Let $\mathcal{H} \subseteq \mathbb{N}$ have non-piecewise-syndetic complement. Use

$$s = 0, \qquad H_0 = 2, \qquad q = 2, \qquad B_j = \varnothing, \quad \chi_j(n) = 0$$

at every packet stage. The local hosts are still $\mathcal{H}_j$ from (A.7.2). The initial case of Proposition A.3.1 forces one-color recurrence inside each bounded-dimensional compact test family with differences in $S_j \subseteq \mathcal{H}_j$. The same local sets have two-color obstructions of length

$$L_1 = \max\{200 \cdot 2, 2\} = 400.$$

Finite extraction and Proposition A.7.3 give $D_{1,\mathcal{H}} \subseteq \mathcal{H}$ with $\mathcal{P}_1$. Proposition A.8.1 gives a two-color obstruction of length

$$H_1 = 2L_1 = 800.$$

The argument works for every permitted host, with the same value H_1. This establishes (I_1), not just the case of the host $\mathbb{N}$.

A.9.2 The induction step and its uniformity

Assume (I_s) for some $s \geq 1$. Fix an arbitrary permitted global host $\mathcal{H}$. At stage j, Lemma A.2.4 makes $\mathcal{H}_j$ a permitted host, and (I_s) supplies $B_j \subseteq \mathcal{H}_j$ with $\mathcal{P}_s$ and an $(s+1)$-color obstruction of length H_s. Apply the finite-stage construction with $q = s+2$. All stages have the common local forbidden length

$$L = \max\{200(s+2), H_s\}.$$

The finite-packet argument gives a set $D_{s+1,\mathcal{H}} \subseteq \mathcal{H}$ with $\mathcal{P}_{s+1}$. The digit coloring uses $s+2$ colors and blocks length

$$H_{s+1} = 2\max\{200(s+2), H_s\}. \tag{A.9.1}$$

This bound is independent of the global host, all local hosts, and all choices of old sets. Hence (I_{s+1}) holds for every permitted host, completing the induction.

A.9.3 The advertised numerical bound

We have $H_1 = 800 = 400 \cdot 2$. For $s \geq 1$,

$$200(s+2) \leq 400 \cdot 2^s.$$

Thus if $H_s = 400 \cdot 2^s$, equation (A.9.1) gives

$$H_{s+1} = 2H_s = 400 \cdot 2^{s+1}.$$

The formula (A.1.1) follows for all $k \geq 1$. The auxiliary value $H_0 = 2$ is separate and is not obtained by substituting $k = 0$ into that formula.

A.9.4 Eventual divisibility and zero upper Banach density

For a set $A \subseteq \mathbb{N}$, define

$$d^*(A) = \limsup_{N\to\infty} \sup_{a\geq 0} \frac{|A \cap \{a+1, \ldots, a+N\}|}{N}.$$

We verify the additional sparsity conclusions directly from the assembled form $D = \bigcup_j \lambda_j F_j$.

Proposition A.9.1. *For every $m \geq 1$, the set $D \setminus m\mathbb{N}$ is finite. Moreover $d^*(D) = 0$.*

Proof. Choose $J \geq m$. For every $j \geq J$, we have $m \mid j! \mid \lambda_j$. Thus every packet $\lambda_j F_j$ with $j \geq J$ lies in $m\mathbb{N}$. The earlier packets have finite union, proving the first assertion.

For a uniform density estimate, fix J and put

$$E_J = \bigcup_{j<J} \lambda_j F_j.$$

Since each later scale is a multiple of λ_J, we have

$$D \subseteq E_J \cup \lambda_J\mathbb{N}.$$

Every interval of length N therefore contains at most

$$|E_J| + \frac{N}{\lambda_J} + 1$$

points of D. Taking the supremum over interval locations and then the limit superior as $N \to \infty$ gives $d^*(D) \leq 1/\lambda_J$. Since $\lambda_J \geq J!$ tends to infinity, $d^*(D) = 0$. □

For reference, every piecewise syndetic subset of $\mathbb{N}$ has positive upper Banach density. If $F + A$ contains arbitrarily long intervals and F is finite, then on each such interval the sum of the counts in its finitely many translated copies of A is at least the interval length. One translate therefore has density at least $1/|F|$ on that interval. The associated intervals for A have the same length; bounded shifts at the origin do not affect the limiting bound. Consequently $d^*(A) \geq 1/|F| > 0$. This justifies the examples in Section A.1 and shows that our final sets are not piecewise syndetic.

Property $\mathcal{P}_k$, the obstruction, and Proposition A.9.1 prove all parts of Theorem A.1.2.

A.9.5 The construction as a recipe

Here are the choices at one color level, with the two indices kept separate.

1. Fix the desired host $\mathcal{H}$ and the old color level s. Assume (I_s) when $s \geq 1$, or use the empty initial data when $s = 0$.
2. Fix $q = s + 2$, the old forbidden length H, and $L = \max\{200q, H\}$. Set $\lambda_1 = 1$.
3. At packet stage j, form $\mathcal{H}_j = \{d : \lambda_j d \in \mathcal{H}\}$. Choose the old set B_j and its input coloring inside this local host by (I_s).
4. Retain the compatible compact tests, with $h_j = j + 2$ and maximum dimension C_j. Choose M_j and form $S_j = (G_{M_j} \cap \mathcal{H}_j) \cup (B_j)_{M_j}$. Here $(B_j)_{M_j}$ denotes the small-torus-motion restriction in (A.3.4).
5. Extract a nonempty finite packet $F_j \subseteq S_j$ by a finite subcover. The moving-palette formula supplies its local obstructing coloring κ_j.
6. Choose $r_j > 2L \max F_j$ divisible by $(j + 1)!$, and put $\lambda_{j+1} = r_j \lambda_j$.
7. Set $D_{s+1,\mathcal{H}} = \bigcup_j \lambda_j F_j$ and use the digit coloring (A.8.2).

The host is enforced before each packet is chosen. Positive recurrence is carried by the packet tests. The obstruction is carried by the digit formula. Uniformity of the old forbidden length over all hosts is what lets these three requirements coexist.

A.10 A piecewise-syndetic criterion and the first separation

The return-time theorem also gives a useful general observation about the stronger induction property. It explains both the short first separation and why the full localized construction naturally produces non-piecewise-syndetic sets.

Proposition A.10.1 (One additional color from piecewise syndeticity)**.** *Let $s \geq 1$. If $D \subseteq \mathbb{N}$ is piecewise syndetic and has $\mathcal{P}_s$, then D is $(s + 1)$-large.*

Proof. Fix a coloring of $\mathbb{N}$ with $s + 1$ colors and a desired length h. Extend it to a two-sided sequence and choose a minimal symbolic subsystem in its forward limit set, as in Section A.6. Let C_1 be the first color cylinder and let $\pi : X \to Z$ be the maximal infinite-step pronilfactor.

If $\pi(C_1) = Z$, Theorem A.2.5 gives

$$\mathbb{Z} \setminus R_h^X(C_1) \quad \text{not piecewise syndetic.}$$

It cannot contain the piecewise syndetic set D. Choose $d \in D \cap R_h^X(C_1)$. This gives a monochromatic length-h configuration with a positive difference in D.

If $\pi(C_1) \neq Z$, the nonempty open set $Z \setminus \pi(C_1)$ contains a finite nilfactor cylinder $\zeta^{-1}(V')$. Choose $x_0 \in X$ above a point $z' \in V'$. The times at which $T^n z'$ visits V' carry at most s colors from the orbit of x_0. Apply $\mathcal{P}_s$ of D to this nilsystem window to obtain a monochromatic length-h progression with difference in D.

In either case the finite configuration is specified by open conditions in the symbolic system. Because the subsystem lies in the forward limit set of the original sequence, the same configuration appears at a positive starting position in the original coloring. □

Remark A.10.2. A set with $\mathcal{P}_s$ and an $(s+1)$-color obstruction cannot be piecewise syndetic, by Proposition A.10.1. Thus the sparsity of the constructed $D_{s,\mathcal{H}}$ is compatible with a genuine structural restriction, not just an artifact of the scales. This does not prevent a different, merely s-large set from being syndetic.

Once a set E with $\mathcal{P}_1$ and a binary obstruction is available, the first strict separation has a short modular proof. One may take $E = D_{1,\mathbb{N}}$, without importing a separate construction.

Proposition A.10.3. *Suppose E has $\mathcal{P}_1$ and a coloring $f : \mathbb{N} \to \{0,1\}$ has no monochromatic E-progression of length H. Then*

$$S = (\mathbb{N} \setminus 3\mathbb{N}) \cup 3E$$

is syndetic and 2-large, and a three-coloring blocks all its progressions of length $\max\{3, H\}$.

Proof. The set S is syndetic because it contains every positive integer not divisible by three. Lemma A.2.2 shows that $3E$ has $\mathcal{P}_1$. Since the window property is upward closed, S also has $\mathcal{P}_1$. Proposition A.10.1, with $s = 1$, proves that S is 2-large.

For the obstruction, write each positive integer uniquely as $3n + r$, with $r \in \{0,1,2\}$ and $n \geq 0$. On the three residue classes use, respectively, the ordered output palettes

$$(0,1), \qquad (1,2), \qquad (2,0),$$

choosing the first or second entry according as $f(n+1) = 0$ or 1. A difference not divisible by three visits all three residue classes in its first three terms. The sets of available colors $\{0,1\}, \{1,2\}, \{2,0\}$ have empty intersection, so such a progression cannot be monochromatic.

A difference in $S \cap 3\mathbb{N}$ has the form $3e$ with $e \in E$. It stays in one residue class, and dividing the changes in the indices by three gives a progression of quotient indices with difference e. Within a fixed residue class the palette map is injective, so constant output color would force constant input values of f. This is impossible on H terms. The two cases prove the stated obstruction length. □

The modular construction illustrates the missing-color idea without the dimension and compactness bookkeeping. It does not supply the stronger induction property $\mathcal{P}_2$ for S; in fact Proposition A.10.1 and the three-color obstruction show that S cannot have $\mathcal{P}_2$. The full hierarchy therefore still needs the moving torus palette and the window-sensitive induction.

A.11 Comparison with Qiu's theorem

This section records the alternative dynamical input in detail. It is not needed for the proof of the localized theorem above. Its purpose is to explain why the almost one-to-one modification is

harmless, how Qiu gives the unrestricted hierarchy, and how an additional syndeticity fact recovers the localized hierarchy as well. This comparison distinguishes the amount of recurrence needed for the construction from the sharper relative exceptional-set information supplied by Theorem A.

A.11.1 The modification and the positive lifting lemma

Qiu's Theorem B [16] supplies minimal systems X^*, Z^* and factor maps

$$\tau : X^* \to X, \qquad \sigma : Z^* \to Z, \qquad \pi^* : X^* \to Z^*, \qquad \sigma\pi^* = \pi\tau, \tag{A.11.1}$$

where τ and σ are almost one-to-one. For open $V_0, \dots, V_{h-1} \subseteq X^*$ whose images under π^* have a common point, the theorem gives simultaneous visits along distinct nonconstant integer polynomials p_j, $1 \le j < h$, with $p_j(0) = 0$. The zeroth visit is to V_0 at time zero. This statement requires minimality, not total minimality.

An almost one-to-one map has singleton fibers over a dense set of base points. In the following proof, the singleton-fiber property is used for σ; only surjectivity and equivariance of τ are needed.

Lemma A.11.1 (Positive recurrence lifting from Qiu)**.** *Let $U_0, \dots, U_{h-1}$ be open subsets of a minimal invertible system X. Suppose that for some $m \in \mathbb{N}$,*

$$I = \bigcap_{j=0}^{h-1} T^{-jm}(\pi(U_j))^\circ \neq \varnothing. \tag{A.11.2}$$

For each prescribed integer $a \ge 1$, there are $x \in X$ and $n \in \mathbb{Z}$ such that, with

$$d = m + an^2 > 0,$$

we have $T^{jd}x \in U_j$ for $0 \le j < h$.

Proof. The set I is nonempty and open. Choose $z \in I$ such that $\sigma^{-1}(z) = \{z^*\}$ is a singleton. This is possible because the singleton fibers occur over a dense subset of Z. For each j, set

$$V_j = \tau^{-1}(T^{-jm}U_j).$$

These are open subsets of X^*.

We claim that all their π^*-images contain z^*. Since $z \in T^{-jm}\pi(U_j)$, equivariance of π supplies $x_j \in T^{-jm}U_j$ with $\pi(x_j) = z$. Choose any lift $x_j^* \in \tau^{-1}(x_j)$. Then $x_j^* \in V_j$ and commutativity gives

$$\sigma(\pi^*(x_j^*)) = \pi(\tau(x_j^*)) = \pi(x_j) = z.$$

The singleton-fiber condition forces $\pi^*(x_j^*) = z^*$ for every j. Thus

$$z^* \in \bigcap_{j=0}^{h-1} \pi^*(V_j).$$

The lifts x_j^* need not coincide; the point is that their images coincide. This is exactly the intersection hypothesis in Qiu's theorem.

Apply that theorem with $p_j(n) = ajn^2$ for $1 \le j < h$. These polynomials are distinct, nonconstant, integer-valued, and vanish at zero. It gives $x^* \in X^*$ and $n \in \mathbb{Z}$ such that

$$x^* \in V_0, \qquad T^{ajn^2}x^* \in V_j \quad (1 \le j < h).$$

Let $x = \tau(x^*)$. The definition of V_j and equivariance of τ give

$$T^{j(m+an^2)}x \in U_j \qquad (0 \le j < h).$$

Since $m > 0$, the difference $m + an^2$ is positive even if Qiu's parameter n is zero or negative. It also satisfies $d \equiv m \pmod{a}$. □

The openness of I is the reason a singleton-fiber point can be chosen. An intersection consisting only of exceptional base points would not justify the same argument. We also make no claim that Z^* itself is a pronilsystem. All nilsystem geometry stays on Z and its finite factors; the modified spaces are used only to apply Qiu and project the resulting finite configuration back to X.

Remark A.11.2 (An intrinsic formulation avoids the diagram). The six-author paper formulates a result equivalent to Qiu's Theorem B as its Theorem 1.1 and proves the equivalence in Theorem 3.9 [8]. That formulation says that nonemptiness of the factor return-time set for essentially distinct integer polynomials implies nonemptiness of the original return-time set. It is an existence statement, not Theorem A's exceptional-set statement.

From this formulation, Lemma A.11.1 has a short proof. Use the polynomials

$$p_j(n) = j(m + an^2), \qquad 0 \le j < h.$$

Condition (A.11.2) witnesses a factor configuration at the parameter $n = 0$. The existence statement supplies an original-system configuration at some integer parameter, and its difference $m + an^2$ is positive. Thus a shorter exposition of the unrestricted proof can omit the modification diagram altogether by citing this equivalent formulation. The direct proof above is retained to make the role of almost one-to-one extensions explicit.

A.11.2 Recovering the unrestricted finite-stage argument

For this subsection only, take the host $\mathcal{H} = \mathbb{N}$. To replace Theorem A.2.5 by Lemma A.11.1 in the first case of Section A.6, join the original colored sequence to the auxiliary torus as well as the nilsystem:

$$(\omega, y, 0) \in \{1, \dots, s+1\}^{\mathbb{Z}} \times Y_0 \times \mathbb{T}^M.$$

Choose a minimal subsystem of its forward limit set. The joint orbit closure of $(y, 0)$ under (T, R_α) is a minimal nilsystem, so the subsystem projects onto it and this projection factors through its maximal pronilfactor Z. There are therefore maps

$$\pi : X \to Z, \qquad \rho : Z \to Y_0, \qquad \gamma : Z \to \mathbb{T}^M, \qquad \gamma(Tz) = \gamma(z) + \alpha.$$

As before let $W = \rho^{-1}(V_0)$ and suppose $W \subseteq \pi(C_1)$.

Choose $z_0 \in Z$ with $\rho(z_0) = y$, and put $t_0 = \gamma(z_0)$. Lemma A.4.2 supplies a positive $m \in G_M$ such that $T^{jm}y \in V_0$ for $0 \le j < h$. Only one such m is needed in this argument. Since $m\alpha \in O_M$ and subtraction on the torus is continuous, choose open neighborhoods

$$P_0 \ni t_0, \qquad P_1 \ni t_0 + m\alpha, \qquad P_1 - P_0 \subseteq O_M.$$

Put $P_j = \mathbb{T}^M$ for $j \ge 2$ and set

$$A_j = W \cap \gamma^{-1}(P_j), \qquad U_j = C_1 \cap \pi^{-1}(A_j).$$

The covering assumption gives $\pi(U_j) = A_j$, and these images are open. Moreover $T^{jm} z_0 \in A_j$ for every j, so (A.11.2) holds.

Apply Lemma A.11.1 with $a = 1$. It supplies a monochromatic configuration of positive difference $d = m + n^2$ whose Y_0-coordinates lie in V_0. The first two torus conditions force

$$d\alpha = \gamma(\pi(T^d x)) - \gamma(\pi(x)) \in P_1 - P_0 \subseteq O_M.$$

Thus $d \in G_M$. The second case of Section A.6 and the transfer back to the original coloring are unchanged. The geometric lemma is still applied only to the original nilsystem, so its dimension bound is not enlarged by joining the torus.

A.11.3 What simplifies in the unrestricted proof

Remark A.11.3 (Unrestricted hierarchy versus localization). For the unrestricted hierarchy, Qiu removes the need to discuss the ideal of non-piecewise-syndetic sets in the recurrence argument. Every rescaled host is just $\mathbb{N}$, so a single previously constructed set $B = D_s$ can be reused at all packet stages instead of choosing fresh sets B_j. The geometric lemma needs only one candidate return time, not its syndeticity. These are genuine simplifications when only the strict color hierarchy is wanted.

For an arbitrary prescribed host, however, the conclusion $d = m + n^2$ does not ensure $d \in \mathcal{H}$. For example, the set $\{m + n^2 : n \in \mathbb{Z}\}$ itself is not piecewise syndetic and could be excluded by the host. The phase condition $d\alpha \in O_M$ does not correct that problem. Theorem A.2.5 supplies the missing control in our main proof by allowing the prescribed excluded set to be removed from a piecewise syndetic supply of factor return times. There is also another way to obtain enough control from Qiu: retain the full set of linear progression differences and use its syndeticity, as we now explain.

A.11.4 Qiu plus syndeticity also gives localization

The following separate input is useful for an accurate comparison. It is the linear specialization of Lemma 3.6 in [8]; its proof there is independent of Theorem A.

Lemma A.11.4 (Nonempty progression-return sets are syndetic)**.** *Let (X, T) be minimal and invertible and let $U_0, \dots, U_{h-1} \subseteq X$ be nonempty and open. If*

$$R_h^X(U_0, \dots, U_{h-1}) \neq \varnothing,$$

then this return-time set is syndetic in $\mathbb{Z}$.

The cited lemma treats essentially distinct polynomial tuples, but only the linear case is needed here. To see the reduction behind it, choose one existing difference d_0 and put

$$V = \bigcap_{j=0}^{h-1} T^{-jd_0} U_j.$$

This is a nonempty open set. If $n \in R_h^X(V)$, then a point x with $T^{jn} x \in V$ satisfies $T^{j(n+d_0)} x \in U_j$ for every j. Thus

$$d_0 + R_h^X(V) \subseteq R_h^X(U_0, \dots, U_{h-1}).$$

The syndeticity input for the single-target set $R_h^X(V)$ gives the conclusion. The proof in [8, Lemma 3.6] obtains the required polynomial single-target input from the IP polynomial Szemerédi

theorem. We cite this input rather than treating syndeticity as a consequence of nonemptiness alone without justification.

Now use the torus-joined system and open targets U_j from the preceding Qiu argument, but allow an arbitrary host $\mathcal{H}$ again. The application of Qiu has already proved that

$$R := R_h^X(U_0, \dots, U_{h-1})$$

is nonempty. Lemma A.11.4 makes R syndetic in $\mathbb{Z}$. For *every* $d \in R$, the defining first two targets force

$$d\alpha \in P_1 - P_0 \subseteq O_M.$$

Also every configuration witnessing membership in R is of the first color and lies above the desired nilsystem window. Consequently

$$R \cap \mathbb{N} \subseteq G_M$$

is syndetic in $\mathbb{N}$. Because $\mathbb{N} \setminus \mathcal{H}$ is not piecewise syndetic, Lemma A.2.4 shows that

$$(R \cap \mathbb{N}) \cap \mathcal{H}$$

is piecewise syndetic and in particular nonempty. Choosing a difference there proves the localized first case. The second case, finite extraction, and the uniform induction with fresh sets B_j then proceed exactly as in the main proof.

Remark A.11.5 (What this alternative does and does not establish). The full localized main theorem can therefore be proved using Qiu together with Lemma A.11.4, without using Theorem A. The crucial set to which syndeticity is applied is the full linear-difference set R, not the particular differences of the form $m + n^2$ produced by the positivity device. There is no claim that those quadratic values form a syndetic set.

Theorem A still has two advantages for this exposition. It removes the torus joining and phase-target bookkeeping from the first case, and it gives the relative loss statement (A.6.4) for the original geometric candidate set P. The syndeticity shortcut only supplies a usable syndetic family somewhere in G_M; it does not compare all lifted return times to all factor return times. In particular, two arbitrary syndetic sets may be disjoint, so the shortcut by itself does not allow a prescribed piecewise syndetic subset of P to be retained. The general piecewise-syndetic boosting criterion in Proposition A.10.1 is another place where the exceptional-set conclusion is directly useful.

A.11.5 The short modular separation

For the short modular separation in Section A.10, Qiu also suffices without the general piecewise-syndetic criterion. When the first color projects onto all of Z, apply Lemma A.11.1 with every $U_j = C_1$, with $m = 1$ and $a = 3$. The resulting difference $1 + 3n^2$ is positive and not divisible by three, hence belongs to S. In the other case, $\mathcal{P}_1$ of $3E$ supplies the monochromatic configuration inside the pure second-color cylinder exactly as before.

A.12 Why the compact test list is sufficient

This appendix proves Lemma A.7.1. Its purpose is to justify a countable list of compact parameter families, despite the uncountably many possible niltranslations. The essential observation is that the group and lattice models can be listed countably, while the translation parameter is handled by compactness within each model.

A.12.1 Move the starting point to the identity

Let $Y = G/\Gamma$, let T be left translation by $g_0 \in G$, and write the given starting point as $y = g\Gamma$. Left translation by g^{-1} sends y to the identity coset Γ and conjugates T to translation by

$$a = g^{-1} g_0 g.$$

Transport the open window by the same homeomorphism. Thus we can work with the identity coset as starting point without changing the form of the recurrence problem.

The identity component of G/Γ is

$$G^\circ/(G^\circ \cap \Gamma).$$

It has some finite period $m \geq 1$ under the translation by a, since the compact manifold has only finitely many components. Write

$$a^m = b\gamma, \qquad b \in G^\circ, \quad \gamma \in \Gamma.$$

On the identity component, the return map is

$$z\Gamma \longmapsto b\gamma z \gamma^{-1}\Gamma.$$

Pass to the connected simply connected covering group N of G° and let Λ be the inverse image of $G^\circ \cap \Gamma$. The resulting quotient N/Λ is the same connected nilmanifold. Conjugation by γ lifts to an automorphism A preserving Λ, and a lift of b gives the affine map

$$R_b(z\Lambda) = bA(z)\Lambda. \tag{A.12.1}$$

The starting point is still $o = e\Lambda$.

A.12.2 Why the automorphism is unipotent

Let

$$F(x) = \gamma x \gamma^{-1} x^{-1}$$

on the identity component. Repeated application of F produces iterated commutators with γ. Since G is nilpotent, some iterate is identically the identity element. The derivative at the identity satisfies

$$dF_e = \mathrm{Ad}(\gamma) - I.$$

Differentiating the constant iterate and using the chain rule gives

$$(\mathrm{Ad}(\gamma) - I)^c = 0$$

for some c. Thus the induced automorphism of the Lie algebra of N is unipotent. This is the meaning of unipotence for A in Lemma A.7.1.

A.12.3 Why the affine map is still a niltranslation

Form the discrete semidirect product

$$G_A = N \rtimes_A \mathbb{Z}, \qquad \Gamma_A = \Lambda \rtimes_A \mathbb{Z}.$$

Its multiplication is

$$(x, r)(y, t) = (xA^r(y), r + t).$$

The quotient G_A/Γ_A identifies with N/Λ by choosing the representative $(x, 0)$. Under this identification, translation by $(b, 1)$ induces (A.12.1). The subgroup Γ_A is discrete and cocompact.

For completeness, we verify that G_A is nilpotent. Let $\mathfrak{n}$ be the Lie algebra of N, with lower central series $\mathfrak{n}_r$. The automorphism A preserves this series and is unipotent on each quotient $\mathfrak{n}_r/\mathfrak{n}_{r+1}$. On each such quotient choose a full flag lowered by $A - I$, as one can do from a Jordan basis for a unipotent linear map. Joining these flags, beginning with the deepest central quotient, gives

$$0 = F_0 \subset F_1 \subset \cdots \subset F_d = \mathfrak{n}$$

such that

$$[\mathfrak{n}, F_j] \subseteq F_{j-1}, \qquad (A - I)F_j \subseteq F_{j-1}.$$

The subspaces are A-invariant ideals. Their exponential subgroups satisfy

$$[G_A, \exp(F_j)] \subseteq \exp(F_{j-1}).$$

Indeed, conjugation by N is trivial on $\exp(F_j)/\exp(F_{j-1})$ by the bracket condition, and conjugation by the generator of $\mathbb{Z}$ is trivial there by the condition on $A - I$. Together with $G_A/N \cong \mathbb{Z}$, these subgroups give a central series for G_A.

The group $\mathbb{Z}$ is discrete, so $\dim G_A = \dim N$. Thus this presentation does not add a dimension to the nilmanifold. Every positive power R_b^e is translation by $(b, 1)^e$ on the same quotient.

A.12.4 Countably many models

Mal'cev's rational-coordinate theorem implies that a pair (N, Λ) can be described by a finite list of rational Lie structure constants and rational coordinates for a finite generating set of the lattice [17, Chapter II]. There are only countably many such finite rational descriptions. We may therefore choose a countable set of representatives for the isomorphism classes of pairs (N, Λ).

For a fixed pair, the lattice-preserving automorphisms form a countable set. They are determined by the images of finitely many lattice generators, each of which must lie in the countable group Λ. Uniqueness of the corresponding Lie-group automorphism follows because the logarithms of the lattice span the Lie algebra over $\mathbb{R}$ and N is simply connected. Retain the unipotent automorphisms. This gives countably many triples (N, Λ, A).

For each triple choose a compact exhaustion

$$K_1 \subseteq K_2 \subseteq \cdots \subseteq N, \qquad \bigcup_{R \geq 1} K_R = N,$$

and a countable neighborhood basis $V_1, V_2, \ldots$ at $o = e\Lambda$. For instance, exponential images of larger and larger coordinate cubes give a compact exhaustion of N. Now list all tuples

$$(N, \Lambda, A, m, K_R, V_\ell), \qquad m, R, \ell \in \mathbb{N}.$$

This is the promised countable test list.

Given an original pointed nilsystem and window, perform the normalization and component reduction above. Choose R containing the resulting translation parameter b, and choose V_ℓ inside the transported window. The corresponding test represents the required recurrence problem at multiples of m. This proves Lemma A.7.1.

A.12.5 Continuity used in finite extraction

For a fixed positive integer r, iterating (A.12.1) gives

$$R_b^r(z\Lambda) = bA(b)\cdots A^{r-1}(b)A^r(z)\Lambda.$$

The right side is continuous in b and z. Thus every fixed finite orbit-window condition is open in the parameter b. This supplies exactly the continuity required in Section A.7; no uniform continuity in an unbounded iterate is asserted or needed.

Appendix B

Finite computations and reproducible certificates

B.1 The three finite inputs

The computer-assisted arguments use exact integer and Boolean calculations. This appendix specifies the finite statements, explains why the algorithms prove them, and gives the complete verification programs. The nine clause-certificate files are included in the PDF as part of the embedded archive `partition_regular_verification.zip`. The archive also contains the programs, an independent modular-example checker, and the outputs displayed below. No part of these checks uses floating-point arithmetic, numerical optimization, or a probabilistic acceptance criterion.

B.1.1 Five sites with a prescribed quadratic coefficient

The finite assertion in Lemma 4.2.2 is that every binary coloring of $\{0, \dots, 122\}$ is constant on

$$\{a + bt + 10t^2 : 0 \le t \le 4\}$$

for some integers a, b, with every displayed value in the interval. Repeated values are allowed. The program in Section B.3 enumerates all these sets, represents a set by its bit mask, and retains only inclusion-minimal sets. This reduction preserves the existence of a coloring that splits every set: a coloring splitting a subset necessarily splits each of its supersets.

A node of the search tree specifies disjoint red and blue masks. If an edge is already monochromatic, the node is rejected. If exactly one vertex of an edge remains uncolored and all the colored vertices have one color, the remaining vertex is forced to have the other color. Conflicting forced assignments reject the node. When no further assignment is forced, branching on an uncolored vertex explores both possible colors. Thus each rejection is logically necessary, and the surviving branches exhaust all possible extensions. Global color reversal allows the color at vertex zero to be fixed. The completed search gives

```
1397 inclusion-minimal edges
6093 search calls
An avoiding binary coloring exists:  False
```

This proves the finite assertion. The ergodic-theoretic transfer is proved separately in Section 4.2.

B.1.2 Two six-site tests with one common coefficient

Let

$$\mathcal{C} = \{3, 4, 10, 11, 17, 18, 24, 25\}.$$

The finite assertion used in Corollary 4.2.5 is the following: for any two binary colorings of $\{0, \dots, 179\}$ there exists one $C \in \mathcal{C}$ such that both colorings are monochromatic on a set

$$\{a + bt + Ct^2 : 0 \le t \le 5\} \subseteq \{0, \dots, 179\},$$

with a, b allowed to depend on the coloring.

For a family $J \subseteq \mathcal{C}$, introduce one Boolean variable for each site. For each six-site set having coefficient in J, insert two clauses: at least one site is red, and at least one site is blue. A satisfying assignment would be one coloring avoiding every such monochromatic set. Fixing the color of vertex zero is harmless by color reversal. The checker in Section B.4 regenerates these clauses directly from J.

Each certificate is a sequence of clauses ending with the empty clause. To verify an added clause, the checker assumes that all its literals are false and repeatedly applies unit propagation to the preceding clauses. A conflict proves that the added clause follows from the preceding ones. Induction on the certificate lines proves the validity of all derived clauses; deriving the empty clause proves unsatisfiability. The nine certified families are

$$\{3,10,17\},\ \{3,10,18\},\ \{3,10,25\},\ \{3,4,11\},\ \{3,4,18\},$$
$$\{3,4,24\},\ \{4,10\},\ \{4,17,18,25\},\ \{10,11,18,24\}.$$

The checker then enumerates all 2^8 partitions of $\mathcal{C}$ into two labeled parts and verifies that at least one part contains a certified family. If two colorings had no common coefficient, assign each coefficient to a coloring that avoids it. One coloring would avoid a certified family, contradicting its unsatisfiability. This proves the finite assertion. The completed verification gives

```
Verified 9 families, 2008 clauses, and all 256 partitions.
```

After division by seven, every coefficient in $\mathcal{C}$ has distance $3/7$ from an integer, as required by the recurrence application. This argument concerns two six-term tests, not an arbitrary number of them.

B.1.3 The exact 462-point bound

The finite assertion in Theorem 4.6.1 is

$$\max\{|X| : X \subseteq [0,461] \cap \mathbb{Z},\ 42 \notin X + X - 2 \cdot X\} = 126.$$

The lower example is the set of sites congruent to $0, 4, 8$ modulo eleven. The proof program verifies its cardinality and checks every potentially forbidden triple.

For the upper bound, the program first treats the seven residue classes modulo seven separately. Within one class, the forbidden equation on the 66 indices is $x + z - 2y = 6$. The increasing-index search enumerates all independent masks of a prescribed size; whenever all but one vertex of a forbidden set have been selected, its last vertex is removed from the available mask. Every such removal is necessary, and increasing order visits each selected mask exactly once. The local enumeration has no mask of size 21. The counts at sizes $18, 19, 20$ are $838, 38, 1$.

The global search assigns residue classes in decreasing order of their selected cardinalities, resolving ties by increasing residue. Its current state stores the selected masks, all cross-residue exclusions, the remaining cardinality target, and a size cap for each unassigned residue. The caps are valid upper bounds obtained from the forbidden spacings three and six, the prescribed assignment order, and the absence of a fitting precomputed local mask. A branch is pruned only when these necessary bounds exclude its target. The three cross-residue exclusions are exactly

$$42 + 2x - y, \qquad 42 + 2y - x, \qquad (x + y - 42)/2,$$

when the value is integral and belongs to $[0, 461]$. Modulo seven, a solution has either all three residues equal or all three distinct. Hence the local checks and these pair exclusions detect every

forbidden triple. The detailed branch bounds are proved in Theorem 4.6.1 and implemented in Section B.5.

The exhaustive search with target 127 terminates with

```
NO 127.  Nodes by depth:
1 273 1404585 227496552 3958 0 0 0
```

No branch with five assigned residues survives. Every larger avoiding set would have a 127-point subset, so this proves the upper bound.

The independent test in Section B.6 compares the optimized cross-residue mask function to the three displayed integer formulas on every pair of singleton masks. The mask construction distributes over unions in each selected mask, so agreement on these pairs checks it for arbitrary masks. It also checks every enumerated local mask of sizes 18 through 21 directly against the integer equation. Compiled with undefined-behavior checking enabled, it gives

```
Cross-lane singleton checks:  91476 passed.
Local masks of size 18:  838
Local masks of size 19:  38
Local masks of size 20:  1
Local masks of size 21:  0
```

The 91,476 cases are all $\binom{7}{2} \cdot 66^2$ possible pairs in distinct residues.

B.2 Reproduction and file integrity

The verification archive is embedded in this PDF under the name

`partition_regular_verification.zip`

A PDF reader's attachments panel can extract it; some browser viewers do not expose attachments. The archive contains no executable binaries and requires no network access. After extraction, the following commands run all the checks:

```
python3 verify_five_point_pattern.py
python3 check_modular_examples.py

g++ -std=c++17 -O2 two_six/check.cpp -o check_two_six
./check_two_six two_six

g++ -std=c++17 -O3 verify_kravitz_462.cpp -o check_462
./check_462

g++ -std=c++17 -O2 -fsanitize=undefined \
    selftest_462.cpp -o selftest_462
./selftest_462
```

Python 3.10 or later suffices for the Python checks; run it without the `-O` option, which disables assertions. The C++ programs use C++17, and the 462-point programs additionally use the GCC/Clang type `unsigned __int128`. The full search can alternatively be split into the seven commands `./check_462 r`, $0 \le r \le 6$, where r specifies the first assigned residue. All seven cases are required.

The supplementary program `check_modular_examples.py` checks the period-five and period-eight quadratic obstructions, the modulo-nine eight-site and seven-site bits, the complete classification of the three-element subsets of $\mathbb{Z}/11\mathbb{Z}$ missing 1 in their sum of dilates, and the period-nine second-difference example. Its output is `All modular examples passed.` These small checks use direct enumeration, independently of the three main proof searches.

The archive contains a `SHA256SUMS` manifest for its files. Its own SHA-256 fingerprint is

```
bb3622739e595e923cd12ac87482963c6b69d20835618ea8af5d5303b9ce65d8
```

The fingerprint identifies the exact programs and certificate data used for the outputs above.

B.3 Five-site exhaustive verifier

```
"""Verify that every binary coloring of 0,...,122 has a monochromatic
{a + b*t + 10*t**2 : 0 <= t <= 4}.  Run with Python 3.10 or later.
"""

from collections import Counter


def minimal_edges():
    """Return the inclusion-minimal patterns as integer bit masks."""
    edges = set()
    for slope in range(-70, -9):
        offsets = {slope * t + 10 * t * t for t in range(5)}
        for start in range(-min(offsets), 123 - max(offsets)):
            edges.add(sum(1 << (start + offset) for offset in offsets))

    minimal = []
    for edge in sorted(edges, key=lambda mask: (mask.bit_count(), mask)):
        if not any(edge & other == other for other in minimal):
            minimal.append(edge)
    return minimal


def has_avoiding_coloring(edges):
    """Search both choices at each branch. Fix vertex 0 red by symmetry."""
    calls = 0

    def search(red, blue):
        nonlocal calls
        calls += 1

        while True:
            forced_red = forced_blue = 0
            open_edges = []
            for edge in edges:
                if (edge & red) and (edge & blue):
                    continue
                uncolored = edge & ~(red | blue)
                if not uncolored:
                    return False
                if uncolored.bit_count() == 1:
                    if edge & red:
                        forced_blue |= uncolored
                    else:
                        # Every minimal edge has at least two vertices.
                        forced_red |= uncolored
                open_edges.append(uncolored)

            if forced_red & forced_blue:
                return False
```

```
            if not (forced_red | forced_blue):
                break
            red |= forced_red
            blue |= forced_blue

        if not open_edges:
            return True

        score = Counter()
        for uncolored in open_edges:
            weight = 1 << (5 - uncolored.bit_count())
            while uncolored:
                bit = uncolored & -uncolored
                score[bit] += weight
                uncolored -= bit
        branch = max(score, key=score.get)
        return search(red | branch, blue) or search(red, blue | branch)

    possible = search(1, 0)
    return possible, calls


if __name__ == "__main__":
    patterns = minimal_edges()
    assert len(patterns) == 1397
    assert min(edge.bit_count() for edge in patterns) >= 2
    possible, calls = has_avoiding_coloring(patterns)
    print(f"{len(patterns)} inclusion-minimal edges")
    print(f"{calls} search calls")
    print(f"An avoiding binary coloring exists: {possible}")
    assert not possible
```

B.4 Two-six-term certificate checker

```
// C++17. Pass the directory containing the nine .rup certificates as argv[1].
#include <algorithm>
#include <array>
#include <cstdint>
#include <cstdlib>
#include <filesystem>
#include <fstream>
#include <iostream>
#include <sstream>
#include <stdexcept>
#include <string>
#include <vector>

namespace fs = std::filesystem;
using Bits = std::array<std::uint64_t, 3>;
constexpr int grid_size = 180;
constexpr std::array<int, 8> curvatures{3, 4, 10, 11, 17, 18, 24, 25};

struct Clause {
    Bits positive{};
    Bits negative{};
};

void set_bit(Bits& bits, int index) {
    bits.at(index / 64) |= std::uint64_t{1} << (index % 64);
}

std::vector<Clause> original_clauses(const std::vector<int>& coefficients) {
    std::vector<Clause> clauses;
```

```
    Clause fix_origin;
    set_bit(fix_origin.negative, 0);
    clauses.push_back(fix_origin);
    for (int c : coefficients) {
        for (int b = -grid_size - 5*c; b <= grid_size; ++b) {
            if (std::abs(5*b + 25*c) >= grid_size) continue;
            for (int a = 0; a < grid_size; ++a) {
                Clause needs_red, needs_blue;
                bool fits = true;
                for (int t = 0; t < 6; ++t) {
                    int site = a + b*t + c*t*t;
                    if (site < 0 || site >= grid_size) { fits = false; break; }
                    set_bit(needs_red.positive, site);
                    set_bit(needs_blue.negative, site);
                }
                if (fits) {
                    clauses.push_back(needs_red);
                    clauses.push_back(needs_blue);
                }
            }
        }
    }
    return clauses;
}

// A clause is justified when assuming every literal false forces a conflict
// by repeated propagation of clauses with just one unassigned literal.
bool verified_by_unit_propagation(const std::vector<Clause>& clauses, const Clause& candidate) {
    Bits yes = candidate.negative;
    Bits no = candidate.positive;
    for (int j = 0; j < 3; ++j)
        if (yes[j] & no[j]) throw std::runtime_error("tautology in certificate");
    while (true) {
        bool changed = false;
        for (const Clause& clause : clauses) {
            bool satisfied = false;
            for (int j = 0; j < 3; ++j)
                if ((yes[j] & clause.positive[j]) || (no[j] & clause.negative[j])) satisfied = true;
            if (satisfied) continue;
            std::uint64_t last_bit = 0;
            int last_word = -1;
            bool many = false;
            for (int j = 0; j < 3; ++j) {
                auto left = (clause.positive[j] | clause.negative[j]) & ~(yes[j] | no[j]);
                if (!left) continue;
                if (last_bit || (left & (left - 1))) { many = true; break; }
                last_bit = left;
                last_word = j;
            }
            if (many) continue;
            if (!last_bit) return true;
            if (clause.positive[last_word] & last_bit) yes[last_word] |= last_bit;
            else no[last_word] |= last_bit;
            changed = true;
        }
        if (!changed) return false;
    }
}

int main(int argc, char** argv) {
    std::vector<int> verified_masks;
    int learned_count = 0;
    for (const auto& file : fs::directory_iterator(argc == 2 ? argv[1] : ".")) {
        if (file.path().extension() != ".rup") continue;
        std::string name = file.path().stem().string();
        std::replace(name.begin(), name.end(), '_', ' ');
        std::istringstream input(name);
        std::vector<int> coefficients;
        int index_mask = 0, c;
```

```
        while (input >> c) {
            auto it = std::find(curvatures.begin(), curvatures.end(), c);
            if (it == curvatures.end()) throw std::runtime_error("unknown coefficient");
            coefficients.push_back(c);
            index_mask |= 1 << (it - curvatures.begin());
        }
        if (coefficients.empty()) throw std::runtime_error("no coefficients");
        auto clauses = original_clauses(coefficients);
        std::ifstream proof(file.path());
        std::string line;
        bool empty_clause = false;
        while (std::getline(proof, line)) {
            if (empty_clause) throw std::runtime_error("extraneous clause");
            std::istringstream literals(line);
            Clause next;
            int lit;
            bool terminated = false;
            empty_clause = true;
            while (literals >> lit) {
                if (lit == 0) { terminated = true; break; }
                if (std::abs(lit) > grid_size) throw std::runtime_error("literal out of range");
                set_bit(lit > 0 ? next.positive : next.negative, std::abs(lit)-1);
                empty_clause = false;
            }
            if (!terminated || !verified_by_unit_propagation(clauses, next))
                throw std::runtime_error("invalid clause in " + name);
            clauses.push_back(next);
            ++learned_count;
        }
        if (!empty_clause) throw std::runtime_error("empty clause missing");
        verified_masks.push_back(index_mask);
    }
    for (int partition = 0; partition < 256; ++partition) {
        bool covered = false;
        for (int mask : verified_masks)
            if ((partition & mask) == 0 || (partition & mask) == mask) covered = true;
        if (!covered) throw std::runtime_error("a partition is not covered");
    }
    std::cout << "Verified " << verified_masks.size() << " families, "
              << learned_count << " clauses, and all 256 partitions.\n";
}
```

B.5 The 462-point exhaustive verifier

```
// Exhaustively verify the 462-point witness for A+A-2A at missing value 42.
// Build with g++ -std=c++17 -O3 verify_kravitz_462.cpp -o verify.
// ./verify runs the full proof. ./verify r checks only the cases whose first
// (largest, ties by least residue) lane is r. The seven choices are 0 through 6.

#include <algorithm>
#include <array>
#include <cstdint>
#include <cstdlib>
#include <iostream>
#include <vector>

namespace {
using Mask = unsigned __int128;
using Lanes = std::array<Mask, 7>;

constexpr int width = 66;
constexpr int target = 127;
constexpr Mask full = (Mask(1) << width) - 1;
```

```
std::array<std::vector<Mask>, width> incident;
std::array<std::vector<Mask>, 21> patterns;
Lanes assigned{};
std::array<std::uint64_t, 8> visits{};
int first_lane = -1;

int population(Mask m) {
    return __builtin_popcountll(static_cast<std::uint64_t>(m)) +
           __builtin_popcountll(static_cast<std::uint64_t>(m >> 64));
}

int first_point(Mask m) {
    auto low = static_cast<std::uint64_t>(m);
    return low ? __builtin_ctzll(low)
               : 64 + __builtin_ctzll(static_cast<std::uint64_t>(m >> 64));
}

Mask point(int i) {
    return Mask(1) << i;
}

Mask shift(Mask m, int offset) {
    return offset < 0 ? m >> -offset : m << offset;
}

// Inside each of the seven lanes, the forbidden equation has right side 6.
void make_edges() {
    for (int x = 0; x < width; ++x) {
        for (int y = 0; y < width; ++y) {
            int z = 6 + 2*y - x;
            if (z < 0 || z >= width) continue;
            Mask edge = point(x) | point(y) | point(z);
            for (Mask t = edge; t; t &= t-1)
                incident[first_point(t)].push_back(edge);
        }
    }
    for (auto& edges : incident) {
        std::sort(edges.begin(), edges.end());
        edges.erase(std::unique(edges.begin(), edges.end()), edges.end());
    }
}

// Choose an increasing list of exactly need additional points. When all but
// one point of an edge are chosen, delete its final point from the candidates.
void list_independent(Mask chosen, Mask available, int need,
                      std::vector<Mask>& output) {
    if (need == 0) {
        output.push_back(chosen);
        return;
    }
    while (population(available) >= need) {
        int i = first_point(available);
        available &= ~point(i);
        Mask next = chosen | point(i);
        Mask allowed = available;
        bool clash = false;
        for (Mask edge : incident[i]) {
            Mask rest = edge & ~next;
            if (!rest) { clash = true; break; }
            if ((rest & (rest-1)) == 0) allowed &= ~rest;
        }
        if (!clash) list_independent(next, allowed, need-1, output);
    }
}

// Distances 3 and 6 are both forbidden in each lane, by repeated variables.
// On each residue modulo 3, selecting the earliest available point and
// excluding its next two neighbors exactly maximizes size under these two
// distance constraints alone. This gives an upper bound for the full problem.
```

```
int spacing_bound(Mask available) {
    int count = 0;
    while (available) {
        int i = first_point(available);
        ++count;
        available &= ~(Mask(73) << i); // 73 has bits 0, 3, 6.
    }
    return count;
}

// Given all selected points in distinct lanes r,s, determine every point that
// would finish an equation x+z-2y=42 with one point from each of r and s.
void impose_pair(int r, int s, Mask a, Mask b, Lanes& available) {
    const int u = (2*r - s + 14) % 7;
    const int v = (2*s - r + 14) % 7;
    const int w = (4*(r+s)) % 7;
    const int cu = (2*r-s+42-u)/7, cv = (2*s-r+42-v)/7;
    const int cw = (r+s-42-2*w)/7;

    Mask reflected_a = 0, reflected_b = 0;
    std::array<Mask, 2> half_b{};
    for (Mask t = b; t; t &= t-1) {
        int j = first_point(t);
        reflected_b |= point(65-j);
        half_b[j%2] |= point(j/2);
    }
    Mask ban_u = 0, ban_v = 0, ban_w = 0;
    for (Mask t = a; t; t &= t-1) {
        int i = first_point(t);
        reflected_a |= point(65-i);
        // All positions 2i-j+cu.
        ban_u |= shift(reflected_b, 2*i+cu-65);
        // All integral positions (i+j+cw)/2.
        int k = i+cw;
        int parity = (k%2 + 2)%2;
        ban_w |= shift(half_b[parity], (k+parity)/2);
    }
    for (Mask t = b; t; t &= t-1) {
        int j = first_point(t);
        ban_v |= shift(reflected_a, 2*j+cv-65);
    }
    available[u] &= ~ban_u;
    available[v] &= ~ban_v;
    available[w] &= ~ban_w;
}

// Assign lanes in order of decreasing final cardinality, breaking equal-size
// ties by increasing residue. available carries all constraints from pairs of
// assigned lanes. The need variable is the remaining number necessary for 127.
bool search(int used, const Lanes& available, int need, int last_size,
            int last_index, int depth) {
    ++visits[depth];
    if (need <= 0) return true;
    int free_count = 7 - depth;
    if (!free_count) return false;
    std::array<int, 7> cap{};
    int sum = 0;
    for (int i = 0; i < 7; ++i) if (!(used & (1<<i))) {
        cap[i] = std::min(last_size - (i < last_index), spacing_bound(available[i]));
        sum += cap[i];
    }
    if (sum < need) return false;
    for (int i = 0; i < 7; ++i) if (!(used & (1<<i)) && available[i] != full) {
        while (cap[i] >= 16) {
            bool fits = false;
            for (Mask p : patterns[cap[i]]) {
                if (!(p & ~available[i])) { fits = true; break; }
            }
            if (fits) break;
```

```
            --cap[i]; --sum;
            if (sum < need) return false;
        }
    }

    int average = (need + free_count - 1) / free_count;
    for (int i = 0; i < 7; ++i) if (!(used & (1<<i))) {
        if (!depth && first_lane >= 0 && i != first_lane) continue;
        int minimum = std::max(average, need - (sum-cap[i]));
        for (int n = cap[i]; n >= minimum; --n) {
            std::vector<Mask> generated;
            const std::vector<Mask>* candidates;
            if (n >= 16) candidates = &patterns[n];
            else {
                list_independent(0, available[i], n, generated);
                candidates = &generated;
            }
            for (Mask p : *candidates) {
                if (p & ~available[i]) continue;
                auto next = available;
                for (int j = 0; j < 7; ++j)
                    if (used & (1<<j)) impose_pair(i, j, p, assigned[j], next);
                assigned[i] = p;
                if (search(used | (1<<i), next, need-n, n, i, depth+1)) return true;
            }
        }
    }
    return false;
}
} // namespace

int main(int argc, char** argv) {
    if (argc > 1) {
        first_lane = std::atoi(argv[1]);
        if (first_lane < 0 || first_lane > 6) return 2;
    }
    // Check the lower witness as an ordinary integer set.
    std::array<bool, 462> example{};
    int example_size = 0;
    for (int x = 0; x < 462; ++x) {
        example[x] = (x%11 == 0 || x%11 == 4 || x%11 == 8);
        example_size += example[x];
    }
    if (example_size != 126) return 4;
    for (int x = 0; x < 462; ++x) if (example[x])
        for (int y = 0; y < 462; ++y) if (example[y]) {
            int z = 42 + 2*y - x;
            if (0 <= z && z < 462 && example[z]) return 5;
        }
    make_edges();
    std::vector<Mask> too_large;
    list_independent(0, full, 21, too_large);
    if (!too_large.empty()) return 3;
    for (int n = 16; n <= 20; ++n) list_independent(0, full, n, patterns[n]);
    Lanes available;
    available.fill(full);
    bool exists = search(0, available, target, 20, -1, 0);
    std::cout << (exists ? "FOUND 127" : "NO 127") << ". Nodes by depth:";
    for (auto count : visits) std::cout << ' ' << count;
    std::cout << '\n';
    return exists ? 1 : 0;
}
```

B.6 Independent pair-mask and local checks

```
// Independent singleton check of every cross-lane mask, plus local counts.
// The proof-search implementation is included in the same translation unit.
#define main proof_search_main
#include "verify_kravitz_462.cpp"
#undef main
#include <stdexcept>
int main() {
    unsigned checked = 0;
    for (int r=0;r<7;++r) for(int s=r+1;s<7;++s)
        for(int i=0;i<66;++i) for(int j=0;j<66;++j) {
            Lanes fast, naive; fast.fill(full); naive.fill(full);
            impose_pair(r,s,point(i),point(j),fast);
            const int x=r+7*i, y=s+7*j;
            int candidates[3]={42+2*x-y,42+2*y-x,-1};
            if ((x+y-42)%2==0) candidates[2]=(x+y-42)/2;
            for(int z:candidates) if(0<=z && z<462)
                naive[z%7] &= ~point(z/7);
            for(int a=0;a<7;++a) if(fast[a]!=naive[a])
                throw std::runtime_error("cross-lane mask mismatch");
            ++checked;
        }
    std::cout << "Cross-lane singleton checks: " << checked << " passed.\n";
    make_edges();
    for(int n=18;n<=21;++n) {
        std::vector<Mask> v;
        list_independent(0,full,n,v);
        std::cout << "Local masks of size "<<n<<": "<<v.size()<<"\n";
        for(auto m:v) for(int x=0;x<66;++x) if(m & point(x))
            for(int y=0;y<66;++y) if(m & point(y)) {
                int z=6+2*y-x;
                if(0<=z && z<66 && (m&point(z)))
                    throw std::runtime_error("invalid local mask");
            }
    }
}
```

B.7 Supplementary modular checks

```
"""Independent finite checks of the modular examples used in the report.
Run with Python 3.10 or later.  Only exact integer arithmetic is used.
"""
from itertools import combinations


def avoids_quadratic_word(word: str, length: int, coefficient: int = 1) -> None:
    q = len(word)
    for a in range(q):
        for b in range(q):
            colors = {word[(a + b*t + coefficient*t*t) % q]
                      for t in range(length)}
            assert len(colors) == 2, (word, a, b, coefficient)


# Obstructions for coefficients not divisible by 5, respectively by 2.
avoids_quadratic_word('00101', 5)
avoids_quadratic_word('00001111', 5)

# The eight-site bit for a quadratic with unit leading coefficient modulo 9.
for coefficient in (1, 2, 4, 5, 7, 8):
    for b in range(9):
        for a in range(9):
            colors = {int((a + b*t + coefficient*t*t) % 9 in (0, 1, 2))
```

```
                    for t in range(8)}
            assert len(colors) == 2
assert {(4 + 2*t*t) % 9 for t in range(-3, 4)} <= {3, 4, 6}

# The normalized seven-site bit used among three tests for integral lifts.
for b in range(9):
    for a in range(9):
        colors = {int((a + b*t + t*t) % 9 in (0, 2, 4, 6))
                  for t in range(7)}
        assert len(colors) == 2

# All three-point sets in Z/11Z missing 1 in A+A-2A.
expected = {frozenset((a + t) % 11 for t in (0, 2, 4)) for a in range(11)}
actual = set()
for k in (3, 4):
    for values in combinations(range(11), k):
        forbidden = any((x + z - 2*y) % 11 == 1
                        for x in values for y in values for z in values)
        if not forbidden:
            assert k == 3, values
            actual.add(frozenset(values))
assert actual == expected

# Integer lifts of commuting second differences need not commute.
f = (0, 0, 2, 0, 0, 1, 1, 2, 2)
def second(d: int, n: int) -> int:
    value = (f[(n+2*d) % 9] - 2*f[(n+d) % 9] + f[n % 9]) % 3
    assert value in (1, 2)
    return value
for d in (1, 3):
    for n in range(9):
        assert len({f[(n+j*d) % 9] for j in range(3)}) == 2
        second(d, n)
assert tuple(second(1, n) for n in (0, 3, 6)) == (2, 1, 2)
assert tuple(second(3, n) for n in (0, 1, 2)) == (1, 2, 2)
assert (second(1,6)-2*second(1,3)+second(1,0)
        -second(3,2)+2*second(3,1)-second(3,0)) == 3
print('All modular examples passed.')
```

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